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(54) **ERYTHROCYTE-DERIVED
EXTRACELLULAR VESICLES AND
PROTEINS ASSOCIATED WITH SUCH
VESICLES AS BIOMARKERS FOR
PARKINSON'S DISEASE**

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application No. PCT/CA2018/050150 on Feb. 9,
2018, now Pat. No. 11,662,352.

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10, 2017.

(51) **Int. Cl.**

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C12Q 1/32 (2006.01)

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CPC **G01N 33/6896** (2013.01); **C12Q 1/32**
(2013.01); **C12Q 1/34** (2013.01); **G01N 1/28**
(2013.01); **G01N 33/5076** (2013.01); **G01N**
33/573 (2013.01)

(58) **Field of Classification Search**

CPC **G01N 33/6896**; **G01N 1/28**; **G01N 33/573**;
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See application file for complete search history.

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(57) **ABSTRACT**

The present description relates to methods for clinically
assessing Parkinson's disease in a subject using protein
biomarkers of erythrocyte-derived extracellular vesicles
(EEV).

10 Claims, 17 Drawing Sheets

Specification includes a Sequence Listing.

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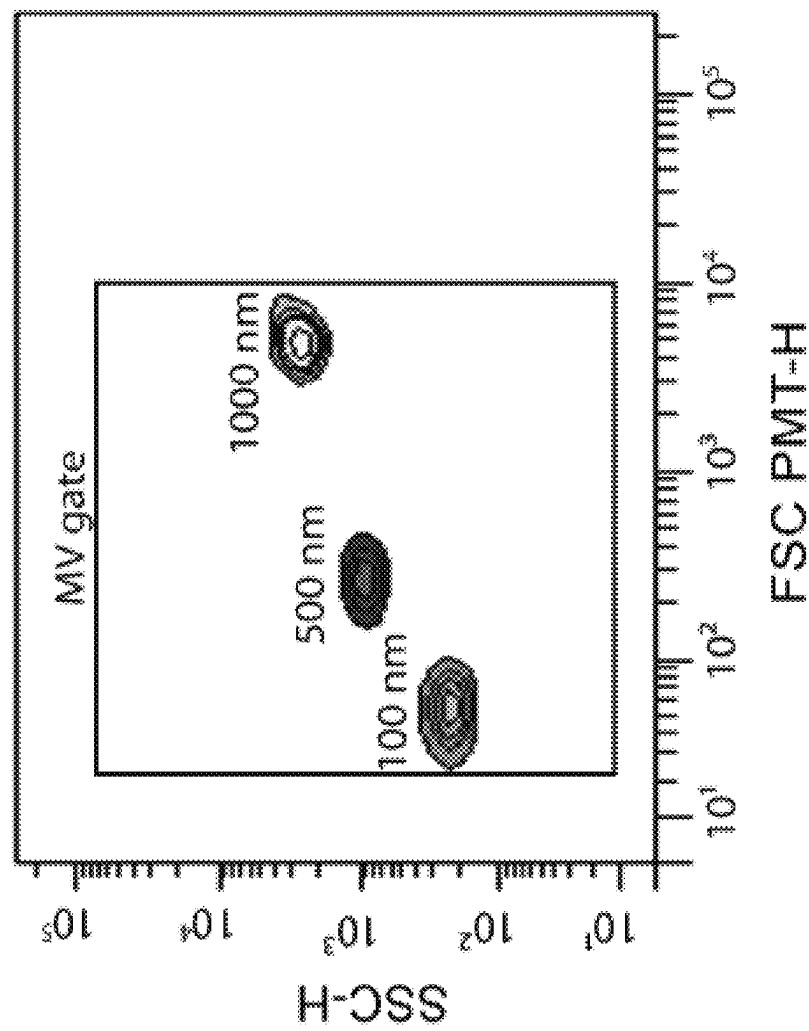


Fig. 1A

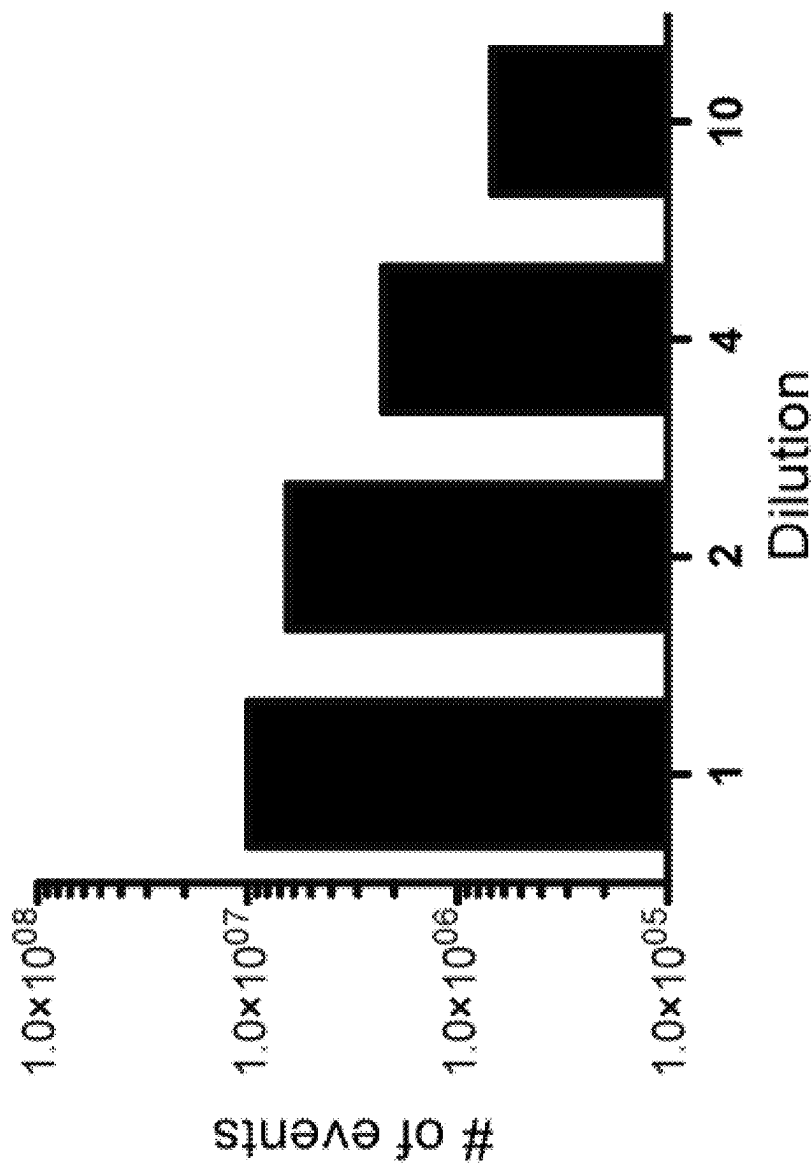
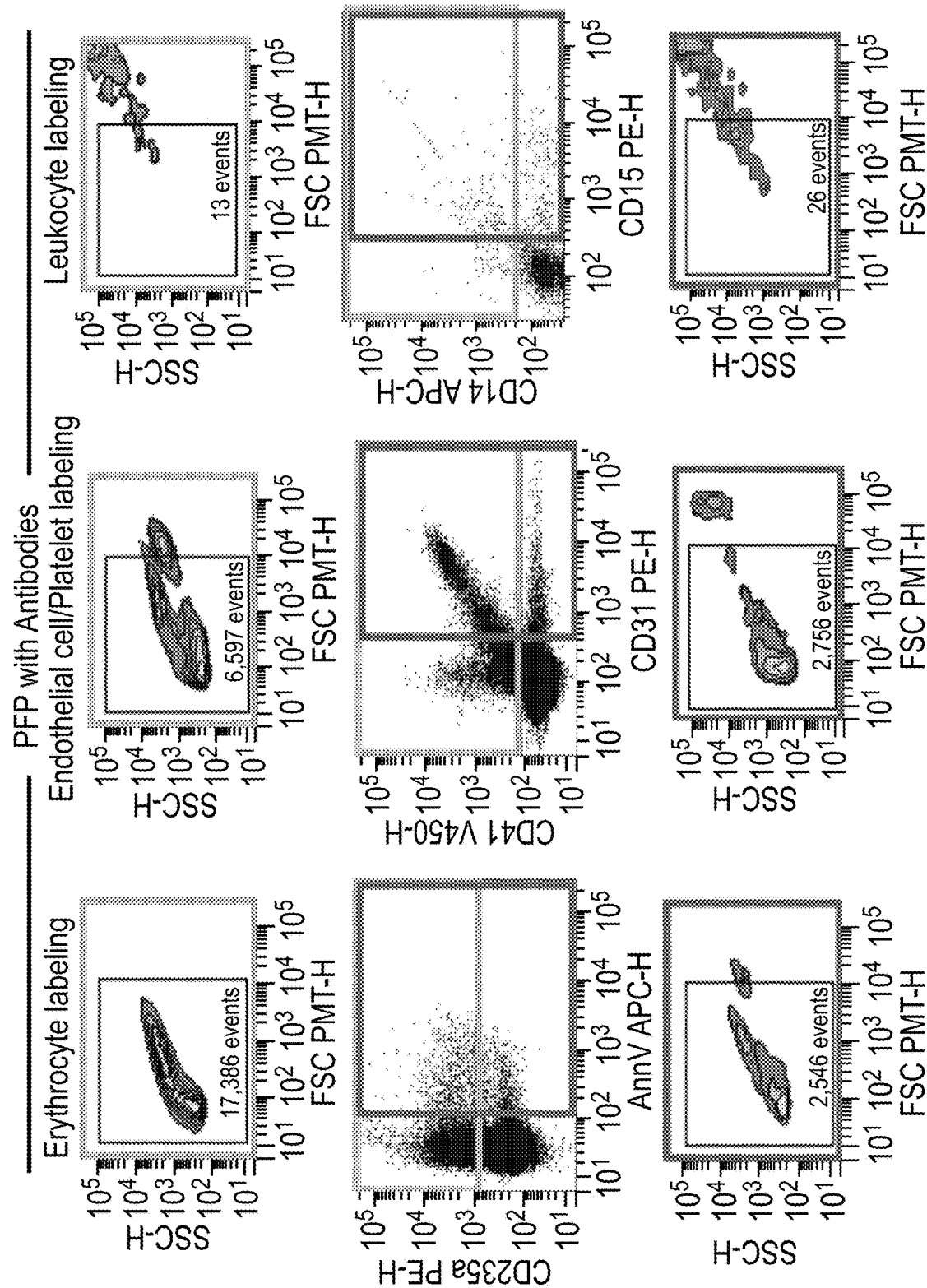


Fig. 1B

FIG. 1C



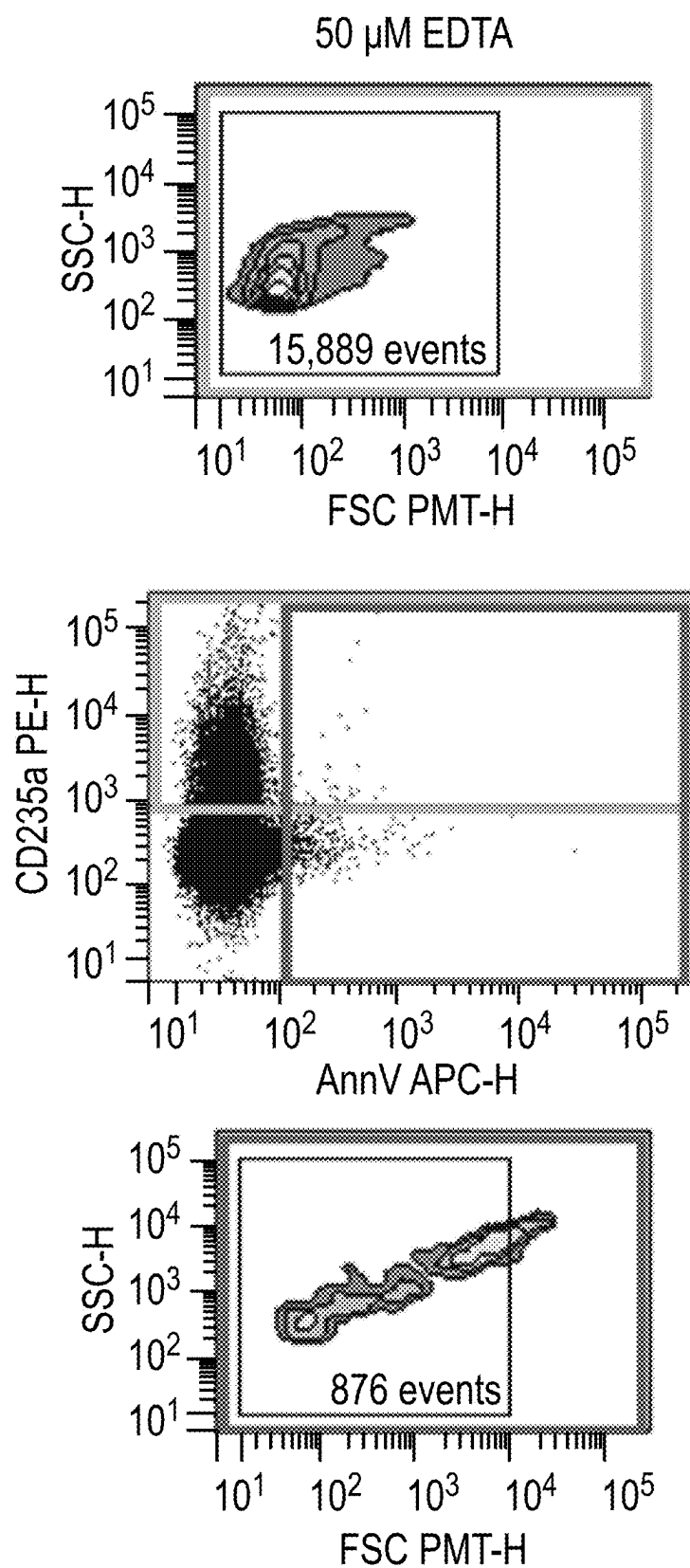


FIG. 1D

FIG. 1E

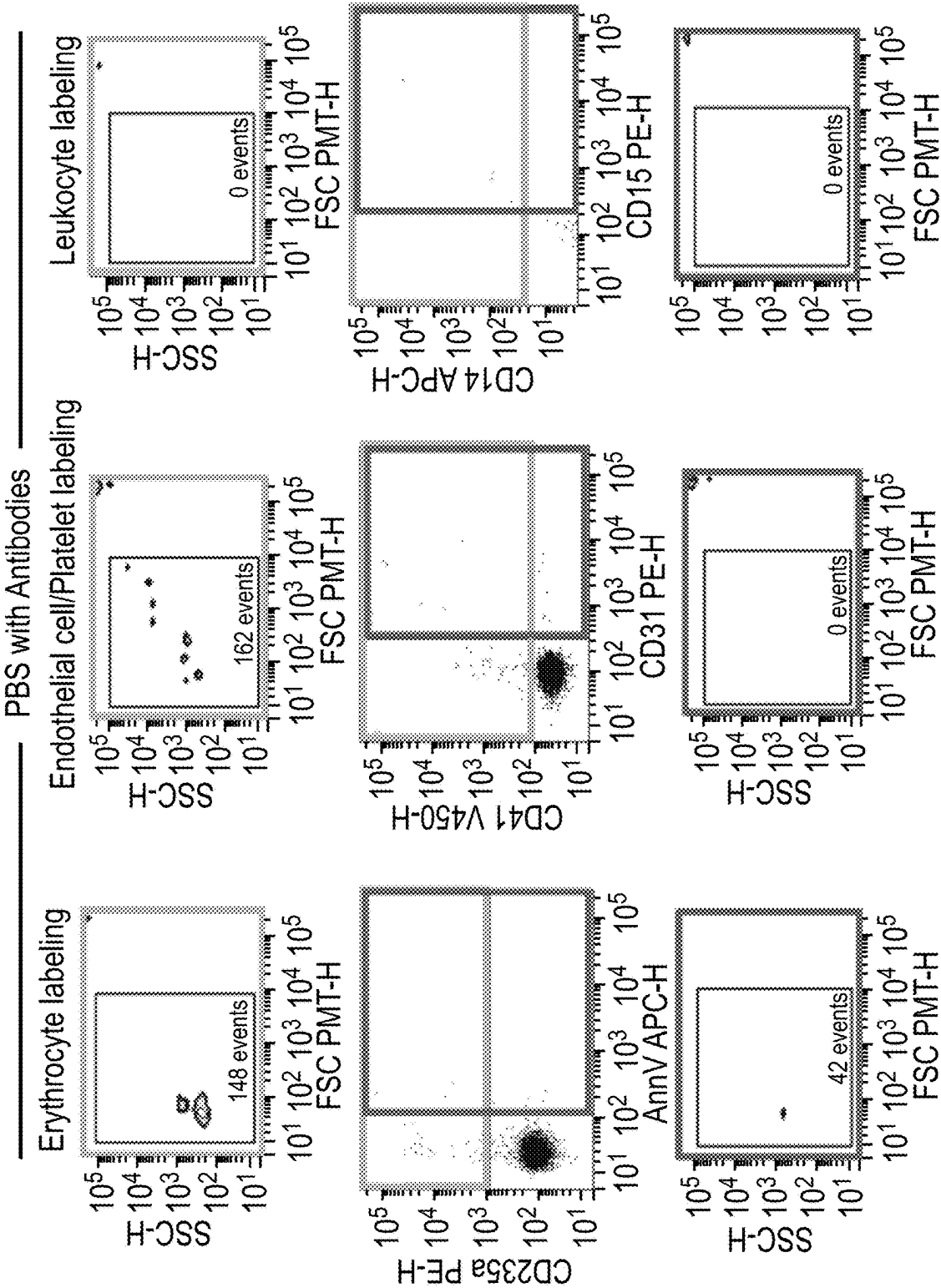
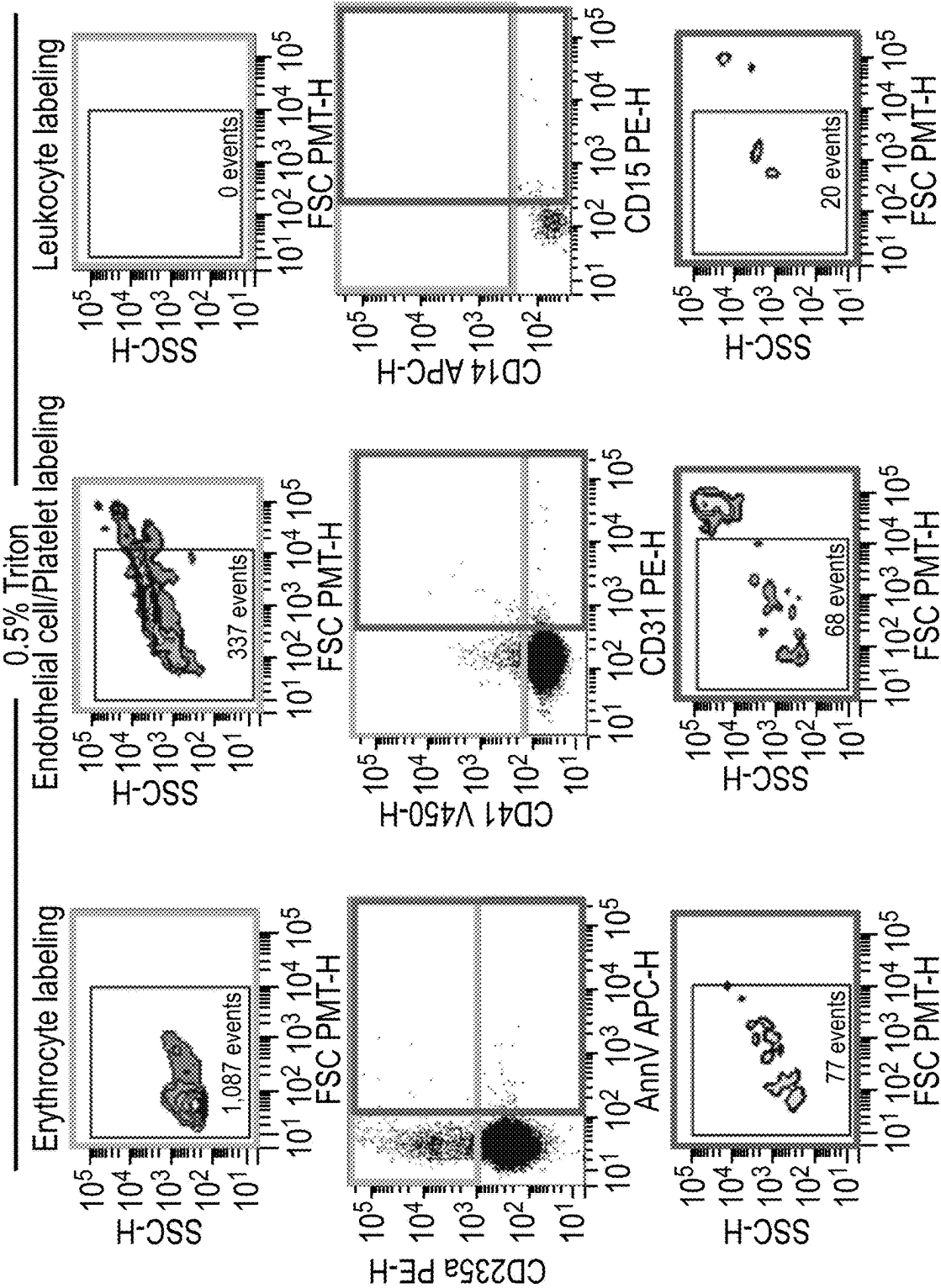
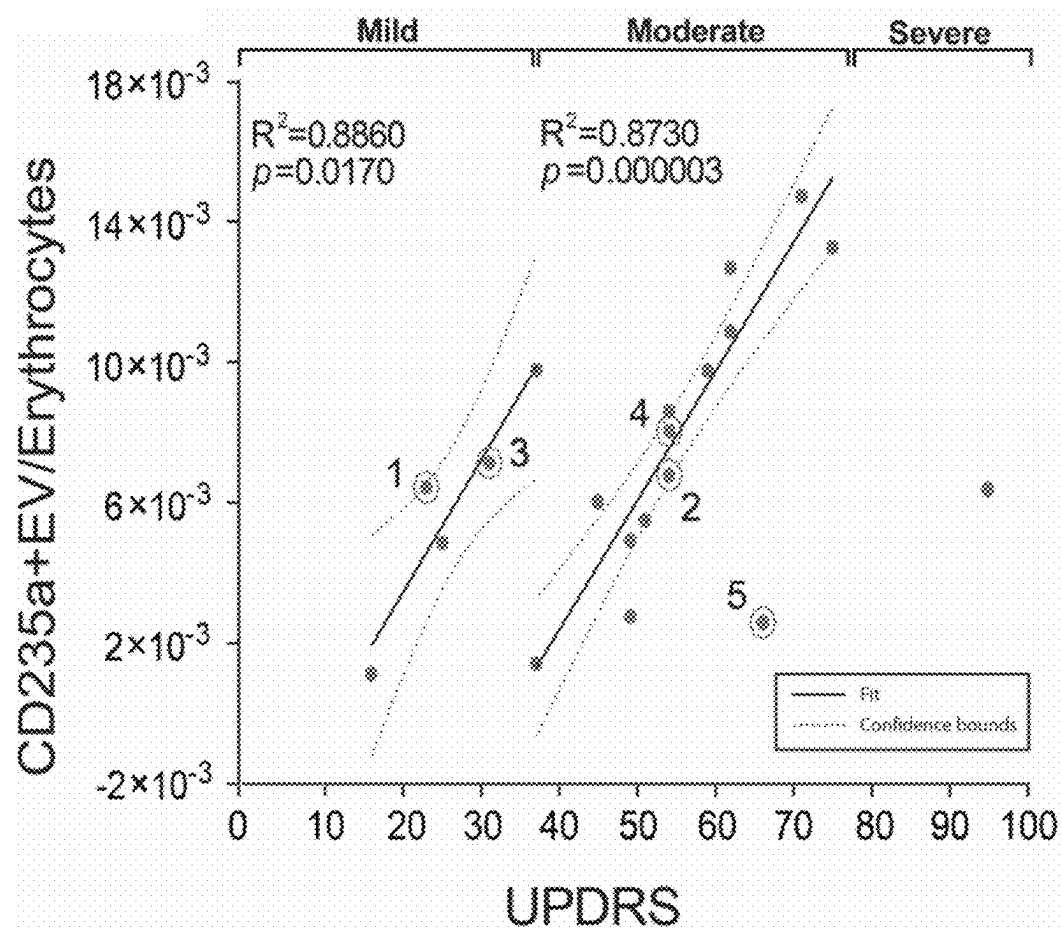


FIG. 1F





1 6.45×10^{-3} EV/erythrocyte UPDRS = 23 LEDD = 386mg VS.	3 7.14×10^{-3} EV/erythrocytes UPDRS = 31 LEDD = 705mg VS.	5 2.55×10^{-3} EV/erythrocytes UPDRS = 66 Anti-inflammatory medication
2 6.75×10^{-3} EV/erythrocytes UPDRS = 54 LEDD = 740mg	4 8.03×10^{-3} EV/erythrocytes UPDRS = 54 LEDD = 550mg	

Fig. 2A

Fig. 2B

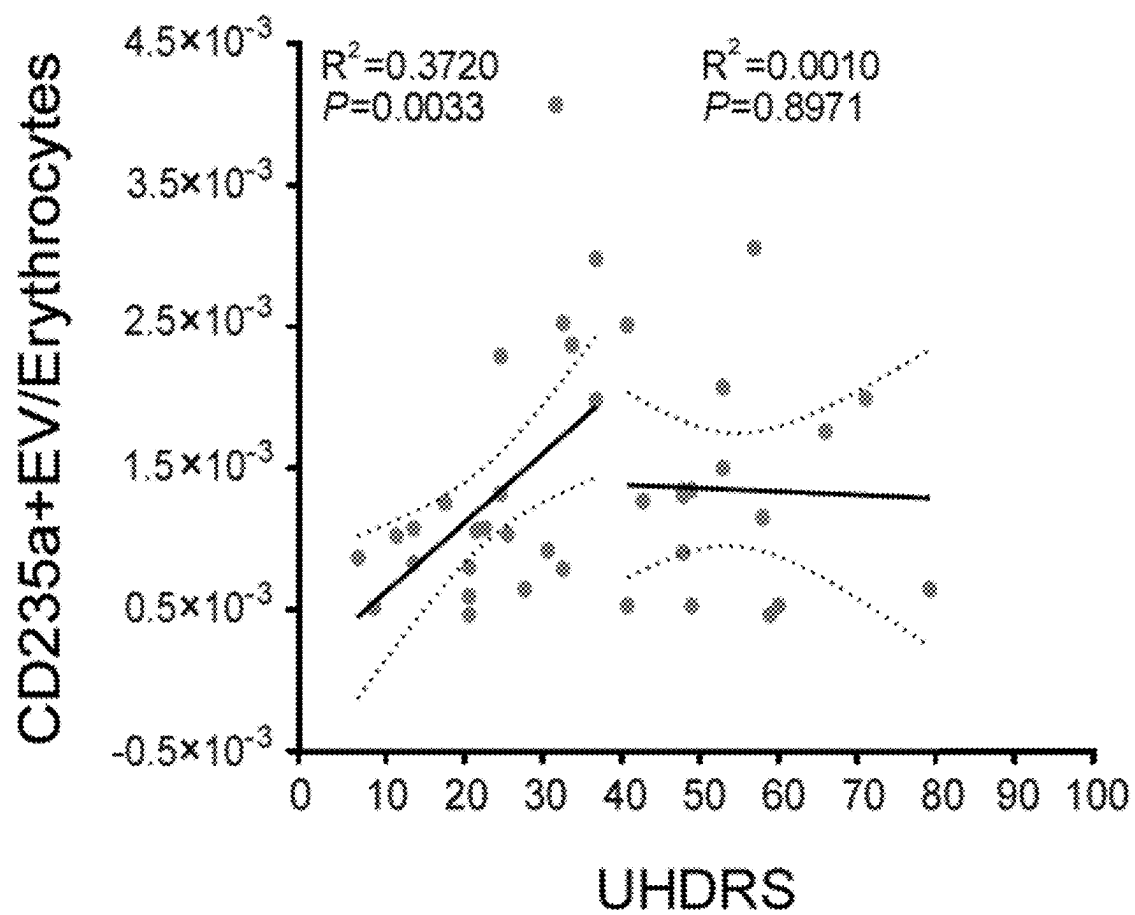


Fig. 3A

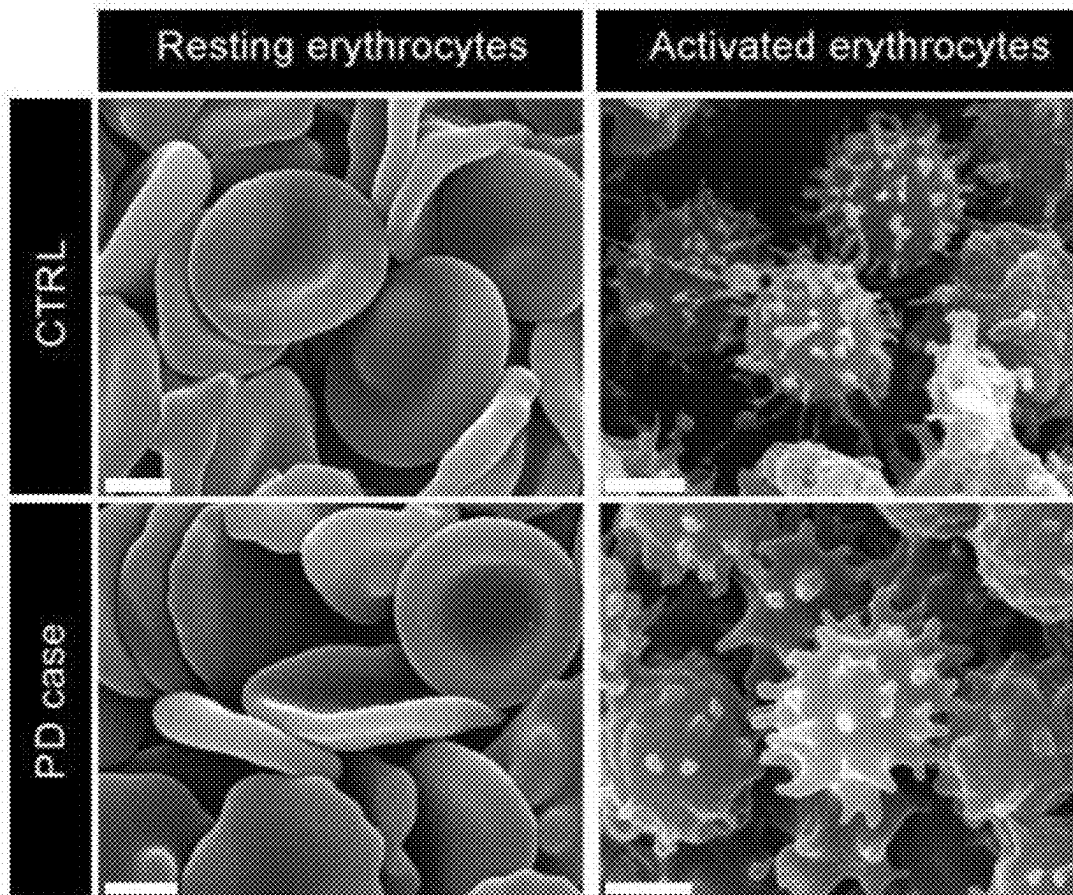


Fig. 3B

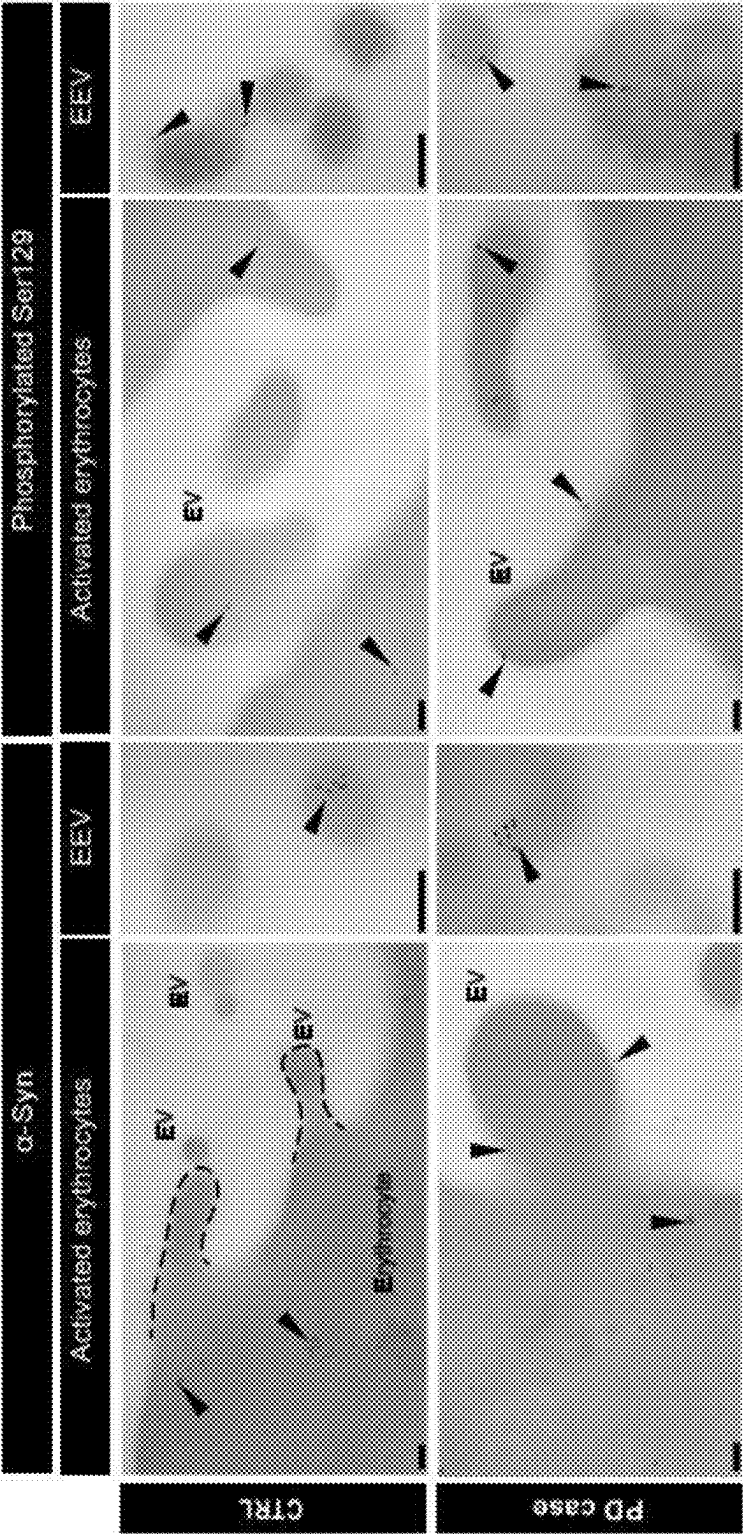


Fig. 3C

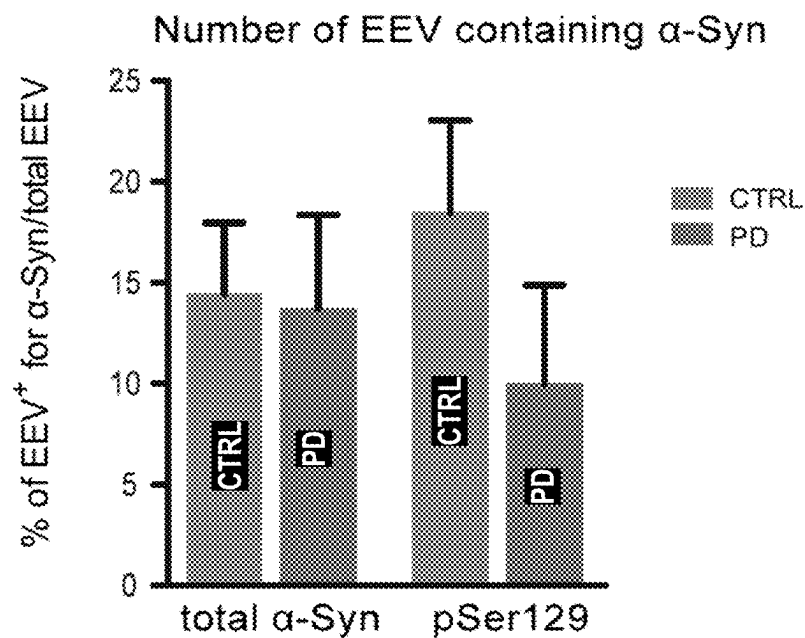


Fig. 3D

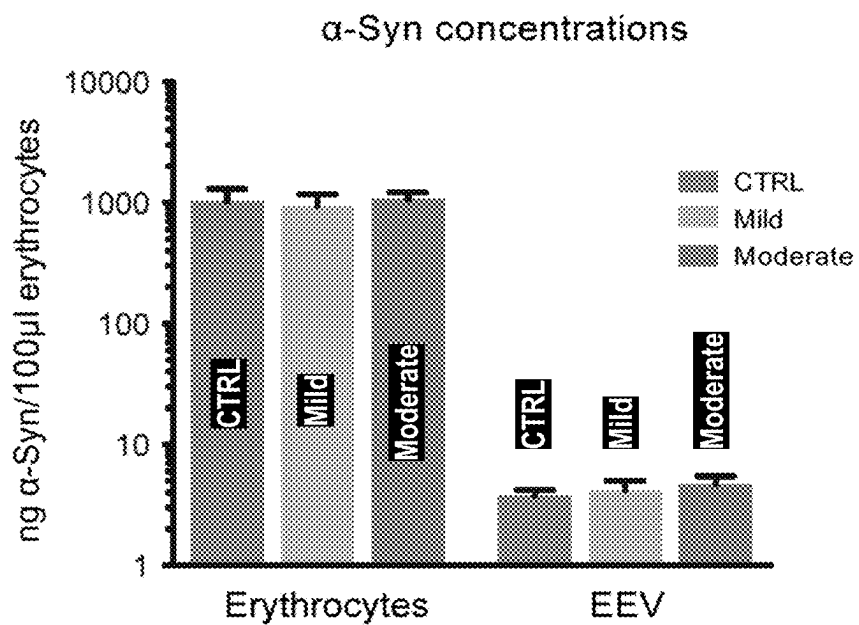


Fig. 4A

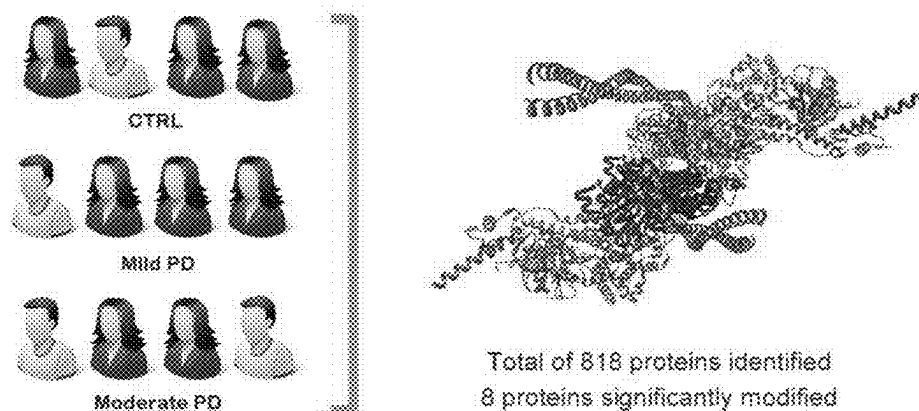
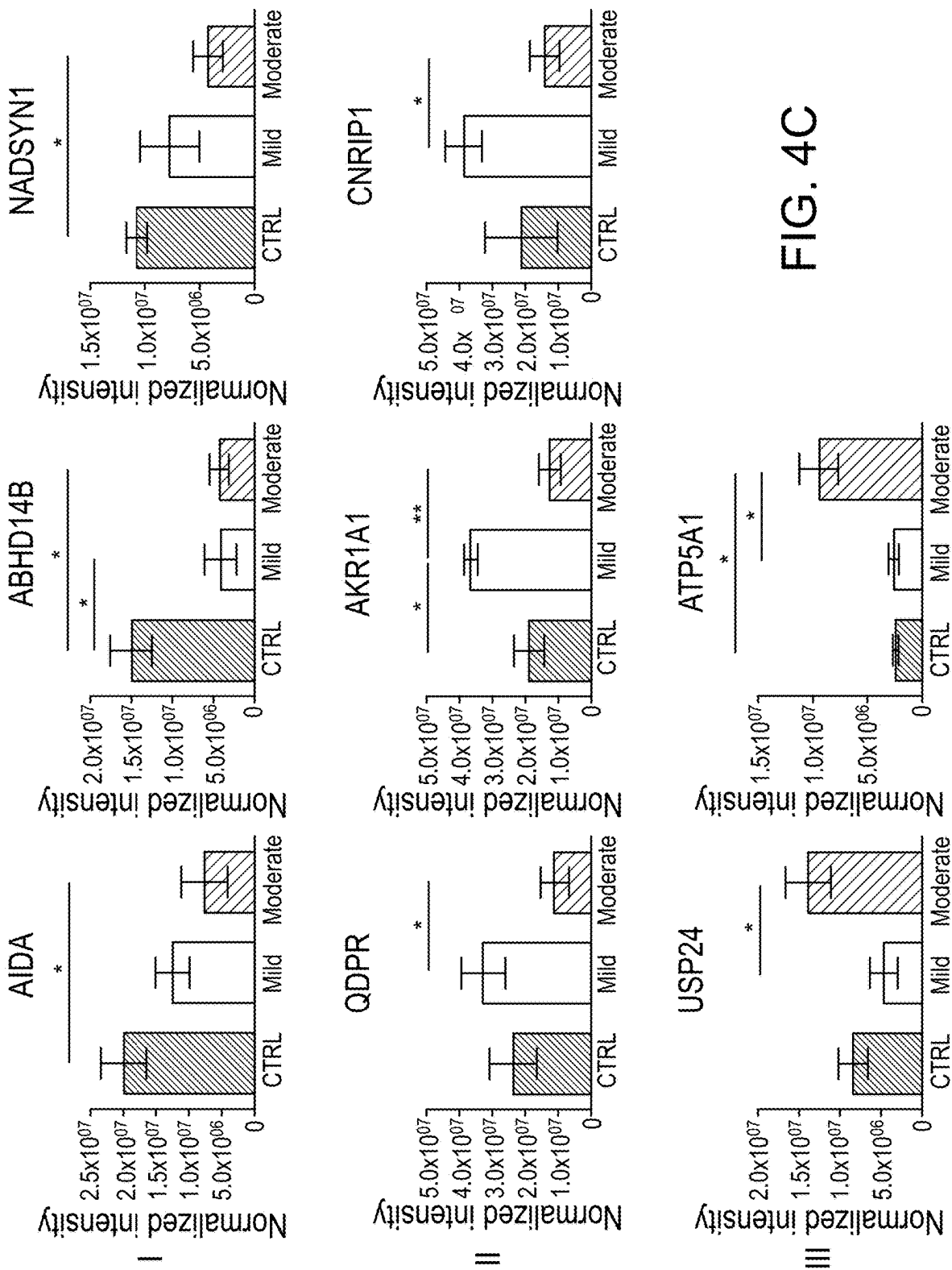


Fig. 4B

Groups	Protein biomarker	Uniprot ID	SEQ ID NO:	Corresponding gene name
I (Control)	Axin interactor, dorsalization-associated protein	Q96BJ3	1	<i>AIDA</i>
	Alpha/beta hydrolase domain-containing protein 14B	Q96IU4	2	<i>ABHD14B</i>
	Glutamine-dependent NAD(+) synthetase	E9PNF5	3	<i>NADSYN1</i>
II (mild PD)	Dihydropteridine reductase	P09417	4	<i>QDPR</i>
	Alcohol dehydrogenase [NADP(+)]	P14550	5	<i>AKR1A1</i>
	CB1 cannabinoid receptor-interacting protein 1	B8ZZB8	6	<i>CNRIP1</i>
III (moderate PD)	Ubiquitin carboxyl-terminal hydrolase 24	Q9UPU5	7	<i>USP24</i>
	ATP synthase subunit alpha, mitochondrial	K7EQH4	8	<i>ATP5A1</i>



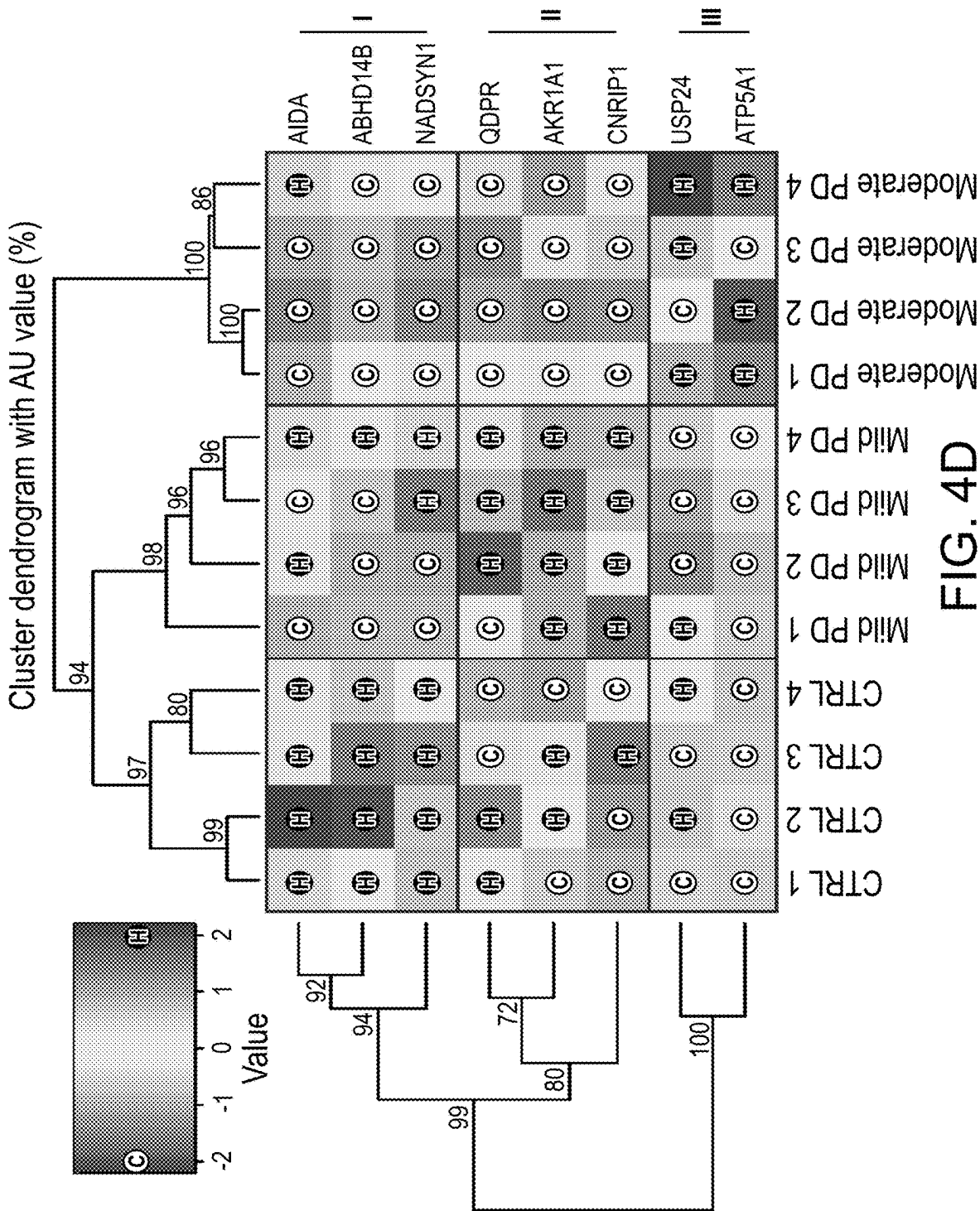


FIG. 4D

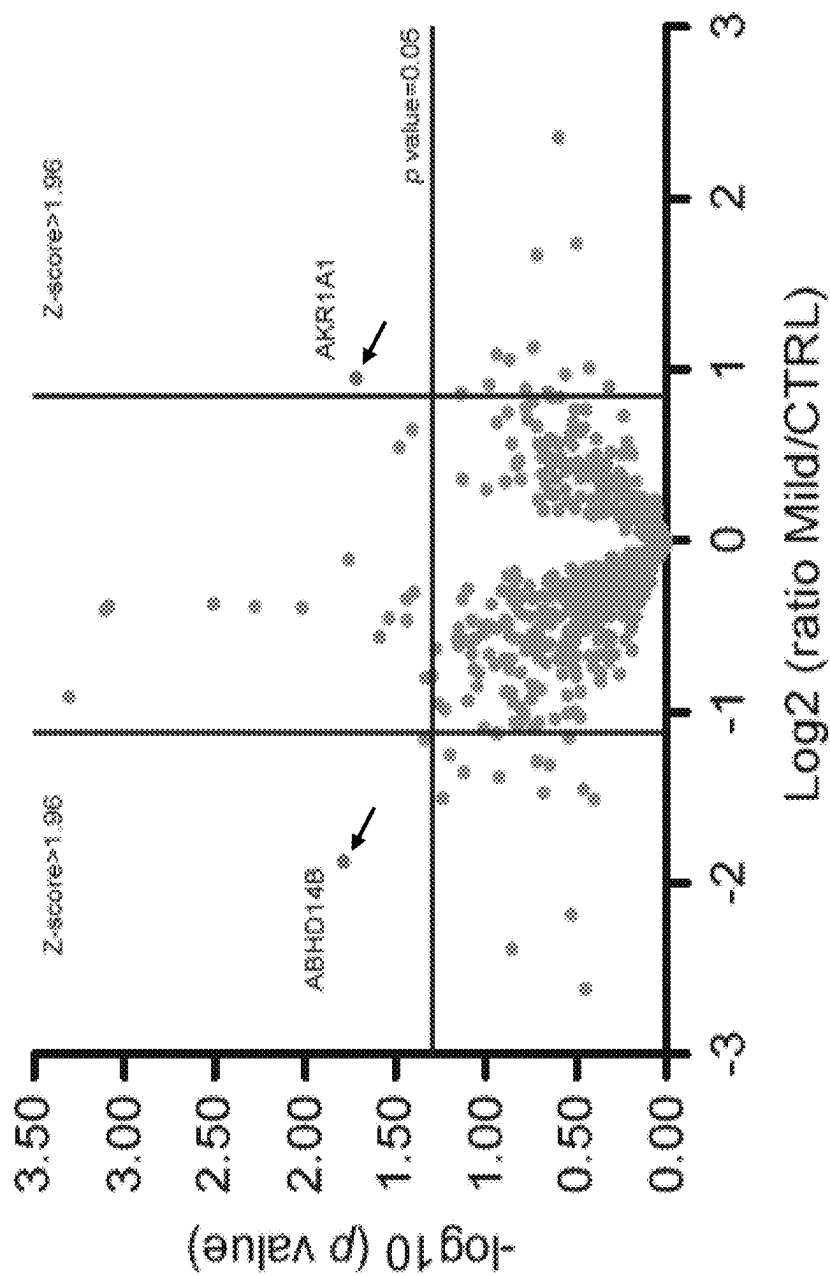


Fig. 5A

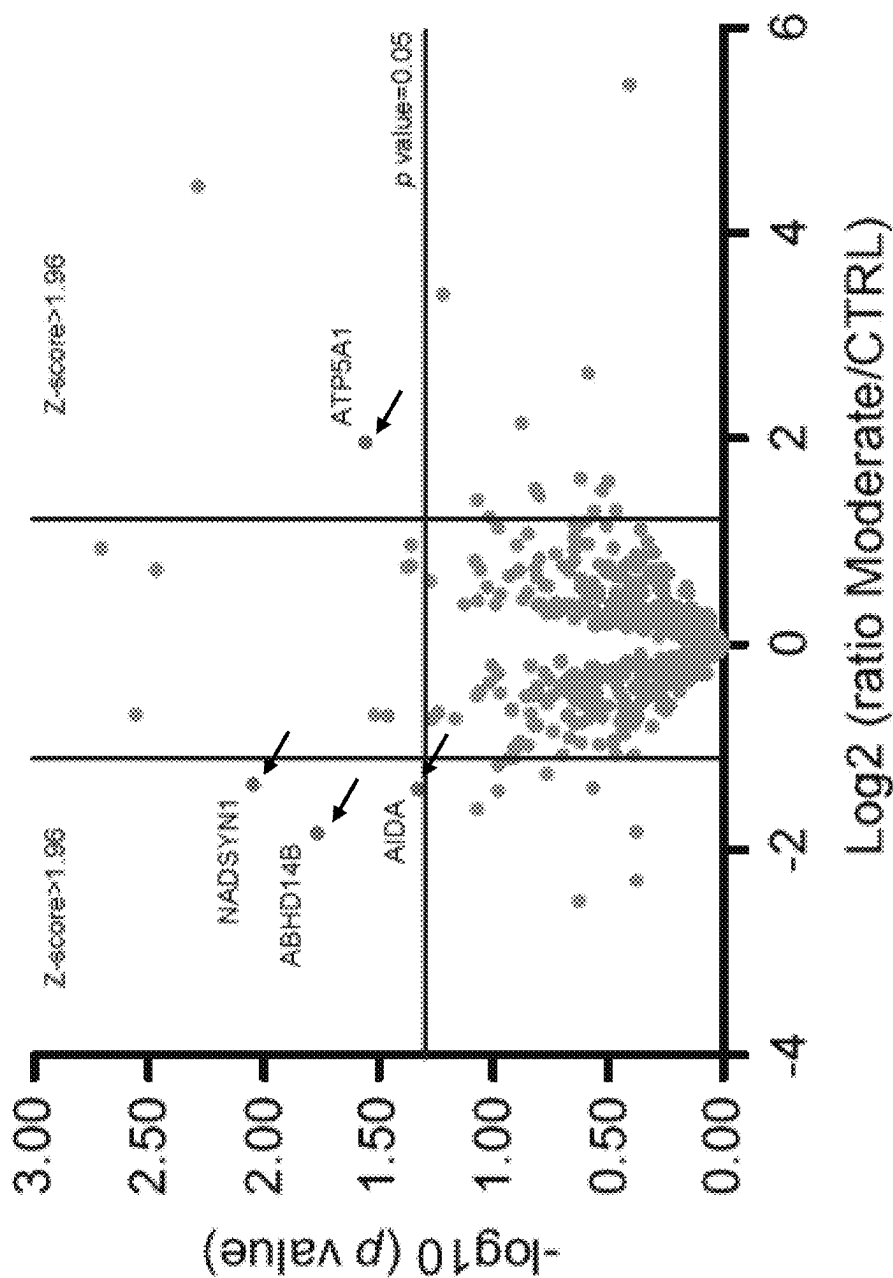


Fig. 5B

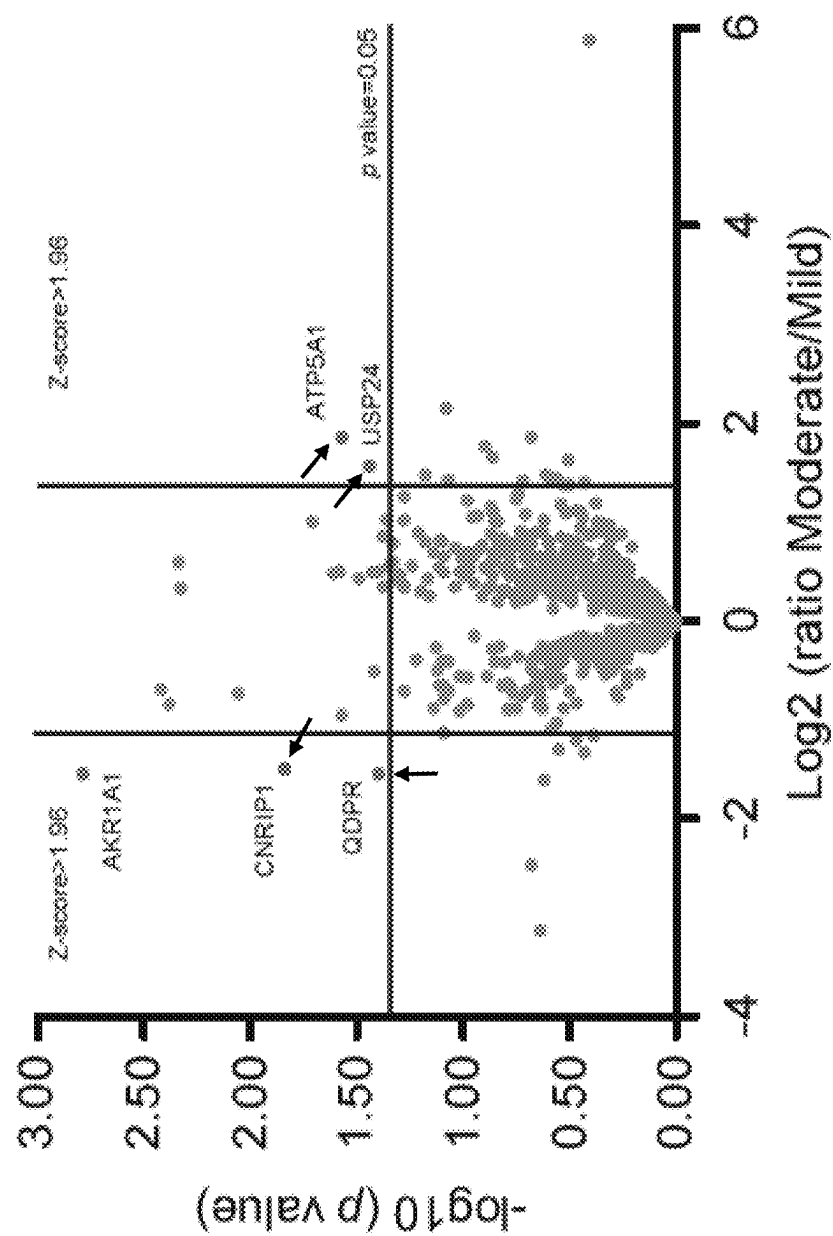


Fig. 5C

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ERYTHROCYTE-DERIVED EXTRACELLULAR VESICLES AND PROTEINS ASSOCIATED WITH SUCH VESICLES AS BIOMARKERS FOR PARKINSON'S DISEASE

RELATED APPLICATIONS

This application is a Continuation of U.S. patent application Ser. No. 16/484,073, which is the U.S. National Stage of International Application No. PCT/CA2018/050150, filed Feb. 9, 2018, which designates the U.S., published in English, and claims the benefit of U.S. Provisional Application No. 62/457,350, filed Feb. 10, 2017. The entire teachings of the above applications are incorporated herein by reference.

INCORPORATION BY REFERENCE OF MATERIAL IN XML

This application incorporates by reference the Sequence Listing contained in the following eXtensible Markup Language (XML) file being submitted concurrently herewith:

- a) File name: 54931003003_Sequence_Listing.xml; created Apr. 25, 2023, 15,256 Bytes in size. The computer readable form is incorporated herein by reference.

SEQ ID NO:	Protein sequence of:	Uniprot ID
1	Axin interactor, dorsalization-associated protein	Q96BJ3
2	Alpha/beta hydrolase domain-containing protein 14B	Q96IU4
3	Glutamine-dependent NAD(+) synthetase	E9PNF5
4	Dihydropteridine reductase	P09417
5	Alcohol dehydrogenase [NADP(+)]	P14550
6	CB1 cannabinoid receptor-interacting protein 1	B8ZZB8
7	Ubiquitin carboxyl-terminal hydrolase 24	Q9UPU5
8	ATP synthase subunit alpha, mitochondrial	K7EQH4

BACKGROUND

Parkinson's disease (PD) is one of the most common neurodegenerative disorders affecting millions of people worldwide. Definite diagnosis for PD can only be made postmortem, for instance, by the characteristic accumulation of the protein alpha-synuclein into Lewy body inclusions observed within neurons. Currently, the diagnosis of PD is based on fitting observed symptoms and their severity into clinical rating scales such as the Unified Parkinson's Disease Rating Scale (UPDRS) or the Hoehn & Yahr scale. Current clinical assessments are subjective, however, and would benefit from improved methods of clinically assessing PD, particularly at early stages of the disease when therapeutic options are likely to be most efficient.

SUMMARY

The present description relates to Parkinson's disease. More particularly, the present description relates to extracellular vesicles originating from erythrocytes, and protein biomarkers associated erythrocyte-derived extracellular vesicles, for use in clinically assessing Parkinson's disease in a subject.

The present description relates to the proteomic analysis of erythrocyte-derived extracellular vesicles (EEV) from healthy control subjects, subjects with mild Parkinson's

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disease, and subjects with moderate Parkinson's disease, as well as the identification of protein biomarkers for which expression within EEV was found to be significantly modified therebetween. Accordingly, the present description generally relates to methods of clinically assessing Parkinson's disease based on the detection and/or quantification of the expression level of one or more protein biomarker(s) within EEV.

In some aspects, the present description may relate to one or more of the following items:

1. An in vitro method for clinically assessing Parkinson's disease, the method comprising obtaining a preparation of isolated erythrocyte-derived extracellular vesicles (EEV) from a blood sample of a subject having or suspected of having Parkinson's disease; and determining the expression level(s) of one or more protein biomarker(s) in the preparation of isolated EEV, wherein the one or more protein biomarker(s) comprises: (i) Alpha/beta hydrolase domain-containing protein 14B, if the subject has or is suspected of having mild or moderate Parkinson's disease; (ii) Alcohol dehydrogenase [NADP(+)], if the subject has or is suspected of having mild Parkinson's disease; (iii) ATP synthase subunit alpha, mitochondrial, if the subject has or is suspected of having moderate Parkinson's disease; or (iv) any combination of (i) to (iii), wherein the expression level of one or more of the protein biomarker(s) is indicative of, or correlates with, the subject's Parkinson's disease state.
2. An in vitro method for clinically assessing Parkinson's disease, the method comprising: obtaining a preparation of isolated erythrocyte-derived extracellular vesicles (EEV) from a blood sample of a subject having or suspected of having Parkinson's disease; and determining the expression level(s) of one or more protein biomarker(s) in the preparation of isolated EEV, wherein the expression level of one or more of the protein biomarker(s) is indicative of, or correlates with, the subject's Parkinson's disease state.
3. The method of item 2, wherein the one or more protein biomarker(s) comprises at least one of: (a) Axin interactor, dorsalization-associated protein; (b) Alpha/beta hydrolase domain-containing protein 14B; (c) Glutamine-dependent NAD(+) synthetase; (d) Dihydropteridine reductase; (e) Alcohol dehydrogenase [NADP(+)]]; (f) CB1 cannabinoid receptor-interacting protein 1; (g) Ubiquitin carboxyl-terminal hydrolase 24; and (h) ATP synthase subunit alpha, mitochondrial.
4. The method of item 3, wherein the one or more protein biomarkers comprises at least two of (a) to (h).
5. The method of item 3, wherein the one or more protein biomarkers comprises at least three of (a) to (h).
6. The method of item 3, wherein the one or more protein biomarkers comprises at least four of (a) to (h).
7. The method of item 3, wherein the one or more protein biomarkers comprises at least five of (a) to (h).
8. The method of item 3, wherein the one or more protein biomarkers comprises at least six of (a) to (h).
9. The method of item 3, wherein the one or more protein biomarkers comprises at least seven of (a) to (h).
10. The method of any one of items 2 to 9, wherein the one or more protein biomarkers comprises (a).
11. The method of any one of items 2 to 10, wherein the one or more protein biomarkers comprises (b).
12. The method of any one of items 2 to 11, wherein the one or more protein biomarkers comprises (c).

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13. The method of any one of items 2 to 12, wherein the one or more protein biomarkers comprises (d).
14. The method of any one of items 2 to 13, wherein the one or more protein biomarkers comprises (e).
15. The method of any one of items 2 to 14, wherein the one or more protein biomarkers comprises (f).
16. The method of any one of items 2 to 15, wherein the one or more protein biomarkers comprises (g).
17. The method of any one of items 2 to 16, wherein the one or more protein biomarkers comprises (h).
18. The method of any one of items 2 to 17, further comprising determining the expression level(s) of one or more of the EEV protein(s) listed in Table 4, wherein the one or more EEV protein(s) does not comprise (a) to (h).
19. The method of any one of items 2 to 18, further comprising normalizing the expression level(s) of the one or more protein biomarker(s) to one or more of the EEV protein(s) as defined in item 18.
20. The method of any one of items 1 to 19, wherein the preparation of isolated EEV is obtained after inducing the calcium-dependent production of EEV from activated erythrocytes in the blood sample of the subject.
21. The method of any one of items 1 to 20, wherein the preparation of isolated erythrocyte-derived extracellular vesicles (EEV) is obtained by separating the EEV by flow cytometry, differential centrifugation, nanomembrane ultrafiltration, immunoabsorbent capture, size-exclusion chromatography, ultracentrifugation, magnetic activated cell sorting (MACS), nanoparticle tracking analysis, light scattering, electrophoretic light scattering, dynamic light scattering, electron microscopy, or any combination thereof.
22. The method of any one of items 1 to 21, wherein the EEV are CD235a+ extracellular vesicles.
23. The method of any one of items 1 to 22, wherein the EEV are TSG101+, Rabs+, CD9+, CD63+, CD81+, or any combination thereof.
24. The method of any one of items 1 to 23, wherein the EEV are between about 20 nm and about 1000 nm in diameter.
25. The method of any one of items 1 to 24, wherein said EEV are greater than about 100 nm in diameter.
26. The method of any one of items 1 to 25, further comprising removing hemoglobin from the preparation of isolated EEV prior to determining the expression level(s) of the one or more protein biomarker(s).
27. The method of any one of items 1 to 26, wherein determining the expression level(s) of one or more of the protein biomarker(s) in the preparation of isolated EEV comprises contacting the protein biomarker with an antibody directed against the protein biomarker.
28. The method of any one of items 1 to 27, wherein determining the expression level(s) of one or more of the protein biomarker(s) in the preparation of isolated EEV comprises mass spectrometry.
29. The method of item 28, wherein determining the expression level(s) of one or more of the protein biomarker(s) in the preparation of isolated EEV comprises nano liquid chromatography tandem mass spectrometry (nanoLC-MS/MS).
30. The method of any one of items 1 to 29, wherein clinically assessing Parkinson's disease comprises diagnosing Parkinson's disease.
31. The method of any one of items 1 to 30, wherein clinically assessing Parkinson's disease in the subject comprises staging Parkinson's disease.

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32. The method of any one of items 1 to 31, wherein clinically assessing Parkinson's disease comprises monitoring the progression of Parkinson's disease.
33. The method of any one of items 1 to 32, wherein clinically assessing Parkinson's disease comprises monitoring the effectiveness of treatment of a Parkinson's disease subject.
34. The method of item 32 or 33, comprising determining the expression level(s) of one or more of the protein biomarker(s) in a preparation of isolated EEV from a further blood sample of the subject obtained at a later point of time.
35. A method for analyzing proteins of erythrocyte-derived extracellular vesicles (EEV), the method comprising: (i) isolating erythrocyte-derived extracellular vesicles (EEV) from a blood sample of a subject and forming an EEV preparation therefrom; (ii) extracting proteins from the EEV preparation; (iii) removing hemoglobin from the EEV preparation to obtain a hemoglobin-free EEV preparation; and (iv) detecting EEV proteins present in the hemoglobin-free EEV preparation.
36. The method of item 35, wherein: step (iii) comprises removing a hemoglobin-containing fraction of proteins from the EEV preparation to obtain a hemoglobin-depleted EEV preparation and a hemoglobin-rich fraction; and step (iv) comprises detecting EEV proteins present in the hemoglobin-free EEV preparation and EEV proteins present in the hemoglobin-rich fraction.
37. The method of item 35 or 36, wherein the EEV proteins detected comprise the protein biomarkers as defined in any one of items 3 to 17.
38. The method of any one of items 35 to 37, wherein the preparation of isolated EEV is obtained after inducing the calcium-dependent production of EEV from activated erythrocytes in the blood sample of the subject.
39. The method of any one of items 35 to 38, wherein the preparation of isolated erythrocyte-derived extracellular vesicles (EEV) is obtained as defined in item 21.
40. The method of any one of items 35 to 39, wherein the EEV are as defined in any one of items 21 to 24.
41. The method of any one of items 35 to 40, wherein the EEV proteins are detected as defined in any one of items 26 to 28.
42. Use of one or more erythrocyte-derived extracellular vesicle (EEV) protein biomarker(s) as defined in any one of items 3 to 17 for clinically assessing Parkinson's disease in a subject.
43. The use of item 42, wherein the clinical assessment is as defined in any one of items 30 to 33.
44. A method of treating a subject with Parkinson's disease, the method comprising: (i) clinically assessing the subject according to the method of any one of items 1 to 41; and (ii) beginning or modifying the subject's Parkinson's disease treatment based on the clinical assessment in (i).

General Definitions

Headings, and other identifiers, e.g., (a), (b), (i), (ii), etc., are presented merely for ease of reading the specification and claims. The use of headings or other identifiers in the specification or claims does not necessarily require the steps or elements be performed in alphabetical or numerical order or the order in which they are presented.

The use of the word "a" or "an", when used in conjunction with the term "comprising" in the claims and/or the speci-

fication may mean “one” but it is also consistent with the meaning of “one or more”, “at least one”, and “one or more than one”.

The term “about” is used to indicate that a value includes the standard deviation of error for the device or method being employed to determine the value. In general, the terminology “about” is meant to designate a possible variation of up to 10%. Therefore, a variation of 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10% of a value is included in the term “about”. Unless indicated otherwise, use of the term “about” before a range applies to both ends of the range.

As used in this specification and claim(s), the words “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”) or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, un-recited elements or method steps.

Other objects, advantages and features of the present description will become more apparent upon reading of the following non-restrictive description of specific embodiments thereof, given by way of example only with reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

In the appended drawings:

FIGS. 1A-1F. Optimization of EV detection: controls for flow cytometry. FIG. 1A: To properly set the EV gate, fluorescent silica beads of 100 nm (Red), 500 nm (Blue) and 1000 nm (Yellow) were acquired on a flow cytometer Canto II modified with a FSC-PMT small particles option. The EV gate was used throughout the experiments. FIG. 1B: Serial dilutions (1, 2, 4 and 10) of erythrocyte-derived EV (EEV) to confirm the linearity of the quantification. FIG. 1C: FSC-PMT/SSC gates of platelet-free plasma (PFP) stained with annexin V and respective fluorochrome-conjugated antibodies directed against erythrocyte (CD235a+), endothelial (CD31+/CD41-)/platelets (CD41+) and leukocytes (CD14+CD45+, monocytes; CD15+CD45+, granulocytes)-derived EV. Controls for EV labeling. FIG. 1D: Treatment with the ion chelator EDTA inhibited the binding of annexin V to phosphatidylserine. FIG. 1E: Minimal background was observed using antibodies in absence of PFP. This background was subtracted from all subsequent EV quantifications. FIG. 1F: EV sensitivity to 0.5% Triton™ was assessed. Abbreviations: AnnV, annexin V; FSC PMT-H, forward scatter photomultiplier; PBS, phosphate buffered saline; PFP, platelet free plasma; SSC-H, side scatter.

FIGS. 2A and 2B. EEV: A biomarker of PD state. FIG. 2A shows the correlations between the number of erythrocyte-derived extracellular vesicles (EEV; expressed as CD235a+ EV/total number of erythrocytes) and the Unified Parkinson's Disease Rating Scale (UPDRS) of subjects (n=20). Robust correlations between the number of EEV/total number of erythrocytes and UPDRS scores (PD, n=20), displayed a clear split between mild and moderate patients, supporting the potential of EEV as a biomarker for disease state. Additional information on five patients (identified as 1-5) derived from each correlation is provided (number of EEV/total number of erythrocytes and levodopa dose equivalent) to illustrate that levodopa dosing cannot account for differences in EEV counts. Note that patient no. 5, which falls outside the confidence boundary, is the only patient on a regimen of anti-inflammatory drugs to manage arthritis. FIG. 2B shows the results of a similar analysis as in FIG. 2A,

but performed on Huntington's disease subjects (n=42) using the Unified Huntington's Disease Rating Scale (UHDRS). The numbers of EEV are expressed as CD235a+ EV/total number of erythrocytes. No statistically significant correlations were found between the number of EEV/total number of erythrocytes and UHDRS scores. This argues in favor of EEV being a specific biomarker of PD. Distributions were determined using unpaired t-test with Welch's correction (PD) or one-way ANOVA (HD). Correlations were determined using Pearson's correlation, *p<0.05. Abbreviations: CD235a, glycophorin A; CTRL, Controls; EEV, erythrocyte-derived extracellular vesicle; EV, extracellular vesicle; HD, Huntington's disease; LEDD, Levodopa equivalent daily dose; PD, Parkinson's disease; Pre-HD, Pre-manifest; UHDRS, Unified Huntington's Disease Rating Scale; UPDRS, Unified Parkinson's Disease Rating Scale.

FIGS. 3A and 3B. Detection of normal and phosphorylated α -Syn in EEV. FIG. 3A: Representative scanning electron microscopy observations of resting and activated erythrocytes (treated with calcium ionophore A23187 to generate EEV) in both PD patients and healthy sex- and age-matched CTRL. Scale bar: 2 μ m. FIG. 3B: Representative transmission electron microscopy images of immunogold labeling for α -Syn and α -Syn pS129 in activated erythrocytes and EEV (some examples delineated by dotted lines). Arrowheads point to positive immunolabeling for either α -Syn or α -Syn pS129. Scale bar: 100 nm. FIG. 3C: Quantification of α -Syn in EEV as detected by transmission electron microscopy and expressed as the percentage of EEVs positive for α -Syn/total number of EEV in healthy sex- and age-matched CTRL and PD patients (n=100 erythrocytes sampled in n=3 CTRL and n=3 PD). FIG. 3D: Quantification of α -Syn in EEV by ELISA assay in healthy sex- and age-matched CTRL, mild and moderate stage patients selected according to their H&Y stage (n=4 erythrocytes per group; n=13 EEV per group) revealing the absence of measurable changes in α -Syn levels between PD and healthy sex- and age-matched CTRL. Statistical analyses were performed using a Mann-Whitney U test (FIG. 3C) or a Kruskal-Wallis ANOVA (FIG. 3D). Abbreviations: α -Syn, α -synuclein; α -Syn pS129, α -synuclein phosphorylated Serine 129; CTRL, Control; EEV, erythrocyte-derived extracellular vesicle; H&Y, Hoehn and Yahr; PD, Parkinson's disease.

FIGS. 4A-4D. Specific protein signature of EEV in PD patients. FIG. 4A: NanoLC-MS/MS Label-free analysis of EEV in PD patients and healthy age-matched CTRL (PD, n=4; CTRL, n=4) revealed a total of 818 proteins, with 8 of which the expression was significantly modified as a function of PD states. FIG. 4B: The 8 differentially expressed proteins are referenced according to the gene to which they are associated, and further separated into 3 groups in relation to their expression variations in comparison to CTRL (Group I), mild PD (Group II) or moderate PD (Group III).

FIG. 4C: Normalized expression (intensity) of the proteins associated with the Group I genes AIDA, ABHD14B, and NADSYN1; the Group II genes QDPR, AKR1A1, and CNRIP1; and the Group III genes USP24 and ATP5A1. FIG. 4D: Heatmap establishing correlations between disease states and the abundance of the variable proteins. Cold (C) and hot (H) colors represent low and high correlation levels, respectively. The AU p value is indicated for each node. Protein modulation was determined by unpaired t-tests with Welch's correction using the criteria of a p value under 0.05 and a minimum of 2-fold change between groups, *p<0.05,

**p<0.01. Abbreviations: AU, Approximately Unbiased; CTRL Control; PD, Parkinson's disease.

FIGS. 5A-5C. Confirmation of EEV proteins selectively modified in PD patients by Volcano plots. The protein ratios (log 2(ratio)) of the three comparison (FIG. 5A: mild PD/CTRL, FIG. 5B: moderate PD/CTRL, and FIG. 5C: moderate PD/mild PD) were plotted over the corresponding Welch's test p value (-log 10(p-value)). The graphs display a V shape, as expected, and only the proteins falling outside the limits of a p value<0.05 and absolute value of z-score>1.96 (identified by black lines) were considered as variant proteins (arrows). Two variant proteins were excluded given that they were quantified using only one peptide. Abbreviations: CTRL: Control; PD, Parkinson's disease.

DETAILED DESCRIPTION

The present description relates to the proteomic analysis of erythrocyte-derived extracellular vesicles (EEV) and the identification of protein biomarkers whose expression levels within EEV correlate with different Parkinson's disease (PD) states.

In one aspect, the present description relates to a method for clinically assessing Parkinson's disease in a human subject, based on the expression level of one or more protein biomarker(s) that may correlate with or be indicative of the subject's Parkinson's disease state. As used herein, the expression "clinically assessing" or "clinical assessment" in the context of PD refers to an evaluation of a subject's PD state, which may or may not occur in a clinical setting, and which may or may not be performed by a health care professional. For example, clinically assessing may comprise screening and/or diagnosing PD in a subject having or suspected of having PD, staging a subject's PD, monitoring the progression of PD in a subject, monitoring the effect of PD medication or treatment (e.g., over time), or any combination thereof. The use of the methods described herein with other methods for clinically assessing PD subjects is also envisaged.

In some aspects, the methods described herein may comprise obtaining a preparation of isolated erythrocyte-derived extracellular vesicles (EEV) from a blood sample of a subject having or suspected of having Parkinson's disease. As used herein, the expression "extracellular vesicles" (EV) refers to subcellular membrane vesicles found in the extracellular environment (e.g., bodily fluids) that originate from cells, and which range in size from about 20 nm to about 1000 nm. EV may comprise exosomes, microvesicles (MV), multivesicular endosomes (MVE), or vesicles produced by apoptotic bodies, or any combination thereof, as well as other types of extracellular vesicles. Whereas the majority of the circulating EV that are detected by flow cytometric assays are likely to be MV, we do not completely exclude the potential contribution of larger exosomes or vesicles produced by apoptotic bodies. In some embodiments, the EV of the present description comprise vesicles between about 30, 40, 50, 60, 70, 80, 90, or 100 nm to about 500, 600, 700, 800, 900, or 1000 nm in size. In some embodiments, the EV of the present description comprise vesicles from 100 nm to 1000 nm in size. In some embodiments, the EV of the present description comprise vesicles between 150 nm to 1000 nm in size. All EV are composed of membrane proteins and lipids, as well as cytoplasmic components of the cell from which they originate, such as mRNA and miRNA, organelles or infectious particles (e.g., prions, virus). A variety of methods may be used to determine the origin of

EV. For example, cell surface markers (e.g., with immunolabeling and/or flow cytometry techniques) may be used to identify, enrich/purify/isolate, and/or quantify EV according to their cell of origin. Examples of such markers include: CD235a+ (erythrocytes), CD31+/CD41-(endothelial cells), CD41+ (platelets), CD45+ (leukocytes), CD45+ CD14+ (monocytes), and CD45+ CD15+(granulocytes). Of particular interest for the present description are markers that are present in (or specific for) EEV that may be used to identify, enrich/purify/isolate, and/or quantify EEV from other types of EV. Examples of such EEV markers include endosome or membrane-bonding proteins such as TSG101 and Rabs (enriched in exosomes), tetraspanins such as CD9, CD63 and CD81 (enriched in exosomes), golgi and mitochondrial proteins (enriched in MVs and absent in exosomes) (Lotvall et al., 2014). In some embodiments, the EEVs of the present description may comprise one or more of the EEV protein(s) listed in Table 4.

As used herein, the expression "[marker]+EV" or "[marker]-positive" in relation to extracellular vesicles refers to the presence or detectability of that marker in an EV population of interest, regardless of whether that marker is actually detected (e.g., using an immunolabel). Conversely, the expression "[marker]-EV" or "[marker]-negative EV" refers to the absence or lack of detectability of that marker in an EV population of interest, regardless of whether that marker is actually detected (e.g., using an immunolabel). For example, the expression "CD235+EV" or "CD235a-positive EV" means EV that comprise the marker CD235a (Glycophorin A).

As used herein, the term "protein biomarker" refers to a molecular indicator that is a polypeptide or protein that is associated with a particular pathological or physiological state (e.g., PD disease state). For example, the expression "Parkinson's disease biomarker" or "PD biomarker" refers to a molecular indicator that is associated with the presence, stage, and/or progression of PD in a subject. Furthermore, the term "EEV protein biomarker" refers to a protein biomarker that is expressed in EEV, whose level of expression is associated with a particular pathological or physiological state (e.g., PD disease state).

In some embodiments, obtaining a preparation of isolated EEV from a blood sample of a subject (e.g., a subject having or suspected of having Parkinson's disease) may involve identifying, enriching/purifying/isolating, and/or quantifying EEV in a blood sample from the subject. In some embodiments, the blood samples may be processed to obtain platelet-free plasma (PFP), and the preparation of isolated EEV may be prepared from PFP. As used herein, the terms "enriched", "purified", "isolated" and the like, refer to either removing contaminants from a biological sample and/or increasing the concentration of an analyte of interest (e.g., EEV) in the sample, to an extent that is not found in nature. In some embodiments, identifying, enriching/purifying/isolating, and/or quantifying EEV may involve flow cytometry, differential centrifugation, nanomembrane ultrafiltration, immunoabsorbent capture, size-exclusion chromatography, ultracentrifugation, magnetic activated cell sorting (MACS), nanoparticle tracking analysis, light scattering, electrophoretic light scattering, dynamic light scattering, electron microscopy or any combination thereof, or using other techniques that can separate vesicles based on their size and/or surface protein expression. Quantifying EEVs may also be performed by methods such as nanoparticle tracking (NTA), biochemical approaches and semi-quantitative electron microscopy approaches. In some embodiments, the methods described herein may further comprise quantifying

the level of EEV in a blood sample from a subject. The quantification of EEV may be expressed as a relative value by normalizing the number of EEV (e.g., in terms of the total number of erythrocytes).

In some embodiments, preparations of isolated EEV described herein may comprise at least 70%, 75%, 80%, 85%, 90%, 95%, or 99% (e.g., by weight or number) of EEV, and/or less than 30%, 25%, 20%, 15%, 10%, 5%, or 1% (e.g., by weight or number) of non-EEV.

In some embodiments, the methods described herein may comprise obtaining a blood sample from a subject that is less than 5 mL, less than 4 mL, less than 3 mL, less than 2 mL, less than 1 mL, less than 900 μ L, less than 800 μ L, less than 700 μ L, less than 600 μ L, less than 500 μ L, less than 400 μ L, less than 300 μ L, less than 200 μ L, less than 100 μ L, less than 50 μ L, less than 40 μ L, less than 30 μ L, less than 25 μ L, less than 20 μ L, or less than 10 μ L.

In some embodiments, the methods described herein may comprise detecting or determining the expression level of an EEV protein biomarker that is differentially expressed in controls (e.g., non-PD subjects) and/or in different PD states (e.g., mild, moderate, or severe), for example based on the unified Parkinson's disease rating scale (UPDRS). As used herein, the expression "mild PD patients" are defined as patients characterized by a UPDRS score of lower than 37; "moderate PD patients" are defined as patients characterized by a UPDRS score of between 37 and 44; "severe PD patients" are defined as patients characterized by a UPDRS score of greater than 44. In some embodiments, the EEV protein biomarkers described herein may be used, for example, to distinguish between mild, moderate and severe PD patients. In some embodiments, the EEV protein biomarkers described herein may be used to distinguish between patients characterized by UPDRS scores within different ranges from those mentioned above. As used herein, "control subjects" or "controls" refer to non-PD subjects (e.g., healthy subjects).

In some embodiments, the methods described herein comprise determining the expression level(s) of one or more protein biomarker(s) in a preparation of isolated EEV, wherein the protein biomarker(s) is/are one or more of the proteins defined in the Table below.

Groups	Protein biomarker	SEQ ID NO:	Related accession no.	Related UniProt ID	Corresponding gene name
I (Control)	Axin interactor, dorsalization-associated protein	1	NP_073742.2	Q96BJ3	AIDA
	Alpha/beta hydrolase domain-containing protein 14B	2	NP_116139.1	Q96IU4	ABHD14B
	Glutamine-dependent NAD(+) synthetase	3	EAW74792.1	E9PNF5	NADSYN1
II (mild PD)	Dihydropteridine reductase	4	NP_001293069.1	QDPR	QDPR
	Alcohol dehydrogenase [NADP(+)]	5	AAP36383.1	P14550	AKR1A1
	CB1 cannabinoid receptor-interacting protein 1	6	NP_056278.1	B8ZZB8	CNRP1
III (moderate PD)	Ubiquitin carboxyl-terminal hydrolase	7	NP_001104571.1	Q9UPU5	USP24
	ATP synthase subunit alpha, mitochondrial	8	NP_056121.2	K7EQH4	ATP5A1

In some embodiments, the protein biomarkers identified herein as belonging to Group I may be differentially expressed in the EEV of healthy or non-PD control subjects, as compared to those of mild and/or moderate PD subjects. In some embodiments, the protein biomarkers identified herein as belonging to Group II may be differentially expressed in the EEV of mild PD subjects, as compared to

those of healthy or non-PD control subjects and/or moderate PD subjects. In some embodiments, the protein biomarkers identified herein as belonging to Group III may be differentially expressed in the EEV of moderate PD subjects, as compared to those of healthy or non-PD control subjects and/or mild PD subjects.

In some embodiments, the methods described herein may comprise determining the expression of one or more of the protein biomarkers described herein, based on the PD state which is to be clinically assessed. For example, different protein biomarkers or combinations of protein biomarkers may be selected based on, for example, the differential expression patterns reported herein (e.g., in Table 3.1 and in FIG. 4C). In some embodiments, the methods described herein may comprise determining the expression of: (i) Alpha/beta hydrolase domain-containing protein 14B, if the subject has or is suspected of having mild or moderate Parkinson's disease; (ii) Alcohol dehydrogenase [NADP(+)], if the subject has or is suspected of having mild Parkinson's disease; (iii) ATP synthase subunit alpha, mitochondrial, if the subject has or is suspected of having moderate Parkinson's disease; or (iv) any combination of (i) to (iii). In some embodiments, the subject is determined to have mild or moderate PD when the expression level of alpha/beta hydrolase domain-containing protein 14B in the preparation of isolated EEV from a blood sample of the subject is lower than that corresponding to a control subject (i.e., not having PD). In some embodiments, the subject is determined to have mild PD when the expression level of alcohol dehydrogenase [NADP(+)] in the preparation of isolated EEV from a blood sample of the subject is higher than that corresponding to a control subject (i.e., not having PD). In some embodiments, the subject is determined to have moderate PD when the expression level of ATP synthase subunit alpha, mitochondrial in the preparation of isolated EEV from a blood sample of the subject is higher than that corresponding to a control subject (i.e., not having PD). In some embodiments, the terms "higher" or "lower" may refer to a difference in expression from the reference value of at least 1.5-fold, 2-fold, 2.5-fold, 3-fold, or 3.5-fold. In some embodiments, methods described herein may comprise determining the expression level(s) of at least one, at

least two, at least three, at least four, at least five, at least six, at least seven, or all eight of the protein biomarker(s):

- Axin interactor, dorsalization-associated protein (represented by SEQ ID NO: 1);
- Alpha/beta hydrolase domain-containing protein 14B (represented by SEQ ID NO: 2);

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- (c) Glutamine-dependent NAD(+) synthetase (represented by SEQ ID NO: 3);
- (d) Dihydropteridine reductase (represented by SEQ ID NO: 4);
- (e) Alcohol dehydrogenase [NADP(+)] (represented by SEQ ID NO: 5);
- (f) CB1 cannabinoid receptor-interacting protein 1 (represented by SEQ ID NO: 6);
- (g) Ubiquitin carboxyl-terminal hydrolase 24 (represented by SEQ ID NO: 7); and
- (h) ATP synthase subunit alpha, mitochondrial (represented by SEQ ID NO: 8).

In some embodiments, the expression level of one or more of the protein biomarker(s) described herein may be determined by detecting and/or quantifying the presence of a polypeptide fragment of any one of the polypeptides of SEQ ID NOs: 1-8. Such fragments may be comprise or consist of at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 45, 50, 55, 60, 65, 70, 75, 80, 90, 95, 100, 125, 150, 175, 200, or more contiguous amino acids of any one of SEQ ID NOs: 1-8. In some embodiments, the polypeptides of any one of SEQ ID NOs: 1-8 may be present in the EEV (or a preparation of isolated EEV) as a fragment of SEQ ID NOs: 1-8. Detection of such fragments are considered within the scope of the present description. In some embodiments, a protein biomarker described herein may comprise one or more fragments of any one of SEQ ID NOs: 1-8, for example a fragment comprising or consisting of: residues 127-192 of SEQ ID NO: 1; residues 188-200 of SEQ ID NO: 2; residues 112-260 or residues 349-446 of SEQ ID NO: 3; residues 1-213 of SEQ ID NO: 4; residues 1-325 of SEQ ID NO: 5; residues 1-110 or residues 84-110 of SEQ ID NO: 6; residues 2570-2620 of SEQ ID NO: 7; or residues 51-161, residues 111-137, or residues 53-152 of SEQ ID NO: 8.

In some embodiments, methods described herein may comprise determining the expression level(s) of at least one, at least two, at least three, at least four, at least five, at least six, at least seven, or all eight of the protein biomarker(s) encoded by the human genes AIDA, ABHD148, NADSYN1, QDPR, AKR1A1, CNRIP1, USP24, and ATP5A1.

In some embodiments, methods described herein may comprise determining the expression level(s) of at least one, at least two, at least three, at least four, at least five, at least six, at least seven, or all eight of the protein biomarker(s) defined by accession numbers: NP_073742.2, NP_116139.1, EAW74792.1, NP_001293069.1, AAP36383.1, NP_056278.1, NP_001104571.1, NP_056121.2, and EAX01470.1.

In some embodiments, protein biomarker (a) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 1. In some embodiments, protein biomarker (b) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 2. In some embodiments, protein biomarker (c) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 3. In some embodiments, protein biomarker (d) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 4. In some embodiments, protein biomarker (e) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 5. In some embodiments, protein biomarker (f) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO:

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6. In some embodiments, protein biomarker (g) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 7. In some embodiments, protein biomarker (h) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 8.

In some embodiments, the methods described herein may comprise determining the expression levels of an EEV protein signature comprising any combination of the protein biomarkers (a) to (h). In some embodiments, the methods described herein may comprise determining the expression levels of at least one of protein biomarkers (a), (b), and (c) [Group I]; at least one of protein biomarkers (d), (e), and (f) [Group II]; and/or at least one of protein biomarkers (g) and (h) [Group III].

In some embodiments, the methods described herein may further comprise comparing the expression level of the protein biomarker(s) so detected and comparing the expression level to a reference value corresponding to that of a control group, non-PD group, mild PD group, moderate PD group, or severe PD group.

In some embodiments, the EEV protein biomarkers described herein do not comprise alpha-synuclein; and/or the methods described herein do not comprise detecting and/or quantifying alpha-synuclein.

In some embodiments, the expression level(s) of one or more of the protein biomarker(s) in the preparation of isolated EEV may comprise contacting the protein biomarker with an antibody directed against the protein biomarker (e.g., an immunoassay). In some embodiments, the antibody specifically binds to the polypeptide of any one of SEQ ID NOs: 1-8. Antibodies against the proteins of SEQ ID NOs: 1-8 are commercially available, for example from Novus Biologicals (e.g., catalog numbers NBP1-88323, NBP2-26122, H00055191-M01, H00005860-M02, NBP2-02164, NBP1-86800, NB100-40830, and NBP2-38525, respectively). In some embodiments, antibodies described herein may bind to, or be raised against, a fragment of any one of SEQ ID NOs: 1-8, for example a fragment comprising or consisting of: residues 127-192 of SEQ ID NO: 1; residues 188-200 of SEQ ID NO: 2; residues 112-260 or residues 349-446 of SEQ ID NO: 3; residues 1-213 of SEQ ID NO: 4; residues 1-325 of SEQ ID NO: 5; residues 84-110 of SEQ ID NO: 6; residues 2570-2620 of SEQ ID NO: 7; or residues 53-152 of SEQ ID NO: 8. In some embodiments, antibodies described herein may bind to an epitope comprising or consisting of at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 contiguous residues of any one of SEQ ID NOs: 1-8.

As used herein, the term "antibody" may encompass any type of antibody, including but not limited to monoclonal antibodies, polyclonal antibodies, "antigen-binding fragments" (or portion), such as Fab, Fab', F(ab')₂, Fd, Fv, Fc, etc., of intact antibodies that retain the ability to specifically bind to a given antigen (e.g., an EEV protein described herein), an isolated complementarity determining region (CDR), bispecific antibodies, heteroconjugate antibodies, mutants thereof, fusion proteins having an antibody, or antigen-binding fragment thereof, (e.g., a domain antibody), single chain (ScFv) and single domain antibodies (e.g., shark and camelid antibodies), maxibodies, minibodies, intrabodies, diabodies, triabodies, tetrabodies, v-NAR and bis-scFv, humanized antibodies, chimeric antibodies and any other modified configuration of the immunoglobulin molecule that includes an antigen recognition site of the required specificity, including glycosylation variants of antibodies, amino acid sequence variants of antibodies, and

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covalently modified antibodies. The antibodies may be murine, rat, human, or any other origin (including chimeric or humanized antibodies).

In some embodiments, the expression level(s) of one or more of the protein biomarker(s) in the preparation of isolated EEV may comprise mass spectrometry (e.g., nano liquid chromatography tandem mass spectrometry (nanoLC MS/MS)).

In some embodiments, the methods described herein may comprise combining the expression of one or more of the protein biomarker(s) in the preparation of isolated EEV with the quantity of EEV to increase the power of a biomarker described herein.

In some embodiments, the methods described herein may further comprise determining the expression level(s) of one or more of the EEV protein(s) listed in Table 4 (Proteins identified in the EEV proteome), wherein the one or more EEV protein(s) does not comprise (a) to (h). The expression level of one or more EEV proteins that are not differentially expressed between PD states may be useful for example as normalization parameters in conjunction with the one or more of the protein biomarkers (a) to (h).

In some embodiments, the methods described herein may comprise determining the expression level(s) of one or more of the EEV protein(s) defined by UniProt ID: P02549, P11277, P16157, P02730, P55072, P16452, P11171, P04040, Q13228, B4DT77, Q8WUM4, P35612, P68871, Q00610, P69905, B4DVE7, J3QLD9, P08758, P09525, O75955, P11142, P32119, P00491, P27105, P00918, P23634, C9JIF9, P00915, P63261, Q5VU58, P30041, E7EU23, Q00013, E7EV01, J3KPS3, Q08495, P23276, P69892, P60174, P62258, O75326, O75340, E7EV99, Q5VZU9, P53396, C9JOK6, P04406, P07738, B7Z319, Q86X55, P07195, P23526, Q32Q12, B7Z7A9, P11166, P22303, Q9NP58, P40925, P00352, F2Z2V0, K7EMC9, F5H7S3, A6NN80, P30043, H7BXD5, P04083, P62937, P37837, Q06830, O75131, P00390, E7EQB2, P07384, P02042, P30086, P35613, Q9H0U4, P63092, P48506, P06702, Q9UBV8, P17931, P28066, P07451, E7EQ12, P50895, P28074, G3V5Z7, P25786, G3VID3, P49247, Q5T9B7, P25789, B4E022, J3QS39, HOY7A7, P28070, Q9H4G4, Q9BY43, P48426, P28289, P07911, Q9GZP4, P78417, P25788, O14818, P08107, H0YD13, P61225, P05109, P23528, Q99808, P84077, P31946, C9JIS1, P53990, Q99497, F5H7U0, B7Z7E9, P62834, P04899, P25325, Q9NP79, P00492, Q16531, P22314, P00441, H7BY58, P10768, P09543, P06733, P26038, O75368, K7EQ48, P25787, P49721, F5H8J2, C9J9P4, P09211, B5MDF5, P07900, K7EQ02, Q13630, F5H0T1, P50502, P20618, P62805, P51148, H7C2G2, J3KQ18, H3BPK3, B4DIT7, O43633, B4DQH4, Q9UN37, I3L397, Q9Y5Z4, Q9UKV8, F5H442, H3BLV0, P02008, E7EPV7, P63104, P02724, U3KQE2, Q9NP59, Q5QPM9, C9J8U2, P61981, Q9UQ80, E5RJR5, Q9NRV9, Q9H444, P10599, Q5VSJ9, Q9UK41, Q9Y311, P61026, Q14974, P27797, P18669, P54725, P30613, P63000, P05164, F5GWY2, P48637, F5H5V4, G5E9R5, P46976, P28072, P26447, F5GXXQ0, P08754, Q99436, P62942, U3KQK0, J3QKR3, P01116, P13489, Q08722, Q5T123, Q8WYQ7, O75695, P00167, Q9Y4D1, P11021, H7C1D4, P07737, M0R389, A6NJA2, P10644, Q9BS40, G5EA52, P53004, Q04656, H9KV70, O00299, F8WF69, G3V2F7, F8WDS9, P60891, K7ESE8, H0YNE3, P16930, F8VSD4, P07203, P62328, E5RIW3, M0R0Y2, P15374, P04921, H0YDI1, B4E220, C9JEN3, F5H2R5, Q53TN4, Q9NZD4, Q8NHG7, Q5JYX0, Q71RC9, E9PNW4, P09105, R4GN98, O75531, Q5T6W5, F5H4Q5, J3QK90, H3BV85, Q9NRX4, H3BS66, E7ESC6,

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In some embodiments, the methods described herein may comprise determining the expression level(s) of one or more of the EEV protein(s) defined by protein accession number: NP_000017.1, NP_000022.3, NP_000028.3, NP_000034.1, NP_000036.2, NP_000043.4, NP_000058.1, NP_000110.2, NP_000128.1, NP_000134.2, NP_000149.3, NP_000166.2, NP_000169.1, NP_000175.1, NP_000185.1, NP_000230.1, NP_000241.1, NP_000260.1, NP_000261.2, NP_000276.2, NP_000280.1, NP_000282.1, NP_000289.1, NP_000311.2, NP_000333.1, NP_000338.3, NP_000356.1, NP_000365.3, NP_000366.1, NP_000382.3, NP_000389.1, NP_000393.4, NP_000410.2, NP_000411.1, NP_000421.1, NP_000445.1, NP_000471.1, NP_000507.1, NP_000508.1, NP_000509.1, NP_000510.1, NP_000549.1, NP_000550.2, NP_000572.2, NP_000623.2, NP_000628.2, NP_000656.1, NP_000678.1, NP_000680.2, NP_000687.3, NP_000691.1, NP_000692.2, NP_000703.2, NP_000704.1, NP_000792.1, NP_000843.1, NP_000886.1, NP_001001323.1, NP_001001396.1, NP_001001521.1, NP_001002021.2, NP_001002857.1, NP_001002858.1, NP_001002860.2, NP_001003938.1, NP_001003945.1, NP_001005360.1, NP_001005361.1, NP_001005362.1, NP_001005386.1, NP_001005753.1, NP_001007068.1, NP_001007069.1, NP_001007070.1, NP_001007071.1, NP_001008390.1, NP_001008800.1, NP_001009185.1, NP_001009186.1, NP_001009570.1, NP_001010935.1, NP_001010942.1, NP_001013275.1, NP_001013454.1, NP_001017963.2, NP_001019397.1, NP_001019398.1, NP_001019399.1, NP_001019820.1, NP_001020029.1, NP_001020276.1, NP_001020560.1,

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 XP_016869038.1, XP_016869039.1, XP_016869040.1, 15
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 XP_016869050.1, XP_016869051.1, XP_016869299.1,
 XP_016869300.1, XP_016869434.1, XP_016869469.1,
 XP_016869596.1, XP_016871328.1, XP_016871329.1,
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 XP_016874004.1, XP_016874335.1, XP_016874336.1,
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 XP_016882662.1, XP_016883197.1, XP_016883198.1,
 XP_016883528.1, XP_016884292.1, XP_016884293.1,
 XP_016884294.1, XP_016884295.1, XP_016885269.1, 40
 XP_016885270.1, or any combination thereof.

In some embodiments, the preparation of isolated EEV described herein may be obtained after inducing the production of EEV from activated erythrocytes in the blood sample of the subject. Erythrocytes may be activated to induce EEV production, for example, using a calcium ionophore (e.g., A23187), cold-storage, or ATP depletion (Prudent et al., 2015).

In some embodiments, the methods described herein may comprise a step of removing or depleting hemoglobin from a preparation of EEV prior to determining the level of expression of EEV proteins. Given the significant amounts of hemoglobin within erythrocytes that could mask the true nature of the protein signature in EEV, this step may improve the detection of non-hemoglobin EEV proteins. In some embodiments, hemoglobin-depleted and hemoglobin-rich fractions may be prepared in parallel and analyzed separated for expression of EEV proteins (e.g., in mass spectrometry-based proteomic detection techniques).

In some aspects, the methods described herein may further comprise comparing the level of an EEV protein biomarker to a suitable reference value indicative of the presence, stage and/or progression of Parkinson's disease, thereby clinically assessing Parkinson's disease in the subject.

As used herein, the expression "reference value" means a control value or range of values corresponding to a known

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level or range of an EEV protein biomarker associated with the presence, stage and/or progression of Parkinson's disease. In some embodiments, for example where the expression level of an EEV protein biomarker has previously been measured in a blood sample from a subject, the reference value may be a value corresponding to the same subject's previous reading (e.g., a baseline). The term "suitable" in the expression "suitable reference value" reflects the observations reported herein that the number of EEV (and/or the protein expressed therein) in blood samples from PD subjects may vary depending on, for example, factors which may also affect the EV and/or EEV levels. For example, it is reported herein that a subject's EEV levels may be affected by whether or not the subject is being treated for their PD symptoms, whether the subject has or previously had cancer, whether the subject has or previously had diabetes, or whether the subject is taking anti-inflammatory medication.

In some embodiments, the present description relates to a method for treating a subject with PD, the method comprising clinically assessing Parkinson's disease in the subject by a method described herein, and commencing, administering, and/or modifying PD treatment based on the clinical assessment.

The scope of the claims should not be limited by the particular embodiments set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole

EXAMPLES

Example 1—Methods

1.1 Participant Recruitment and Ethic Statement

Human blood was obtained from two cohorts of participants. The first cohort was composed of Parkinson's disease (PD) patients and healthy age- and sex-matched Controls, and the second cohort was composed of Huntington's disease (HD) patients and healthy age- and sex-matched Controls. The demographics for both cohorts are shown in Table 1. For the two cohorts, the Controls were recruited amongst the caregivers, spouses, family and friends of the patients. Institutional review boards approved this study (CHU de Québec-Université Laval, #A13-2-1096; CHUM, #14.228; Cambridge Central Regional Ethics Committee, REC #03/303 & #08/H0306/26; and Cambridge University Hospitals Foundation Trust Research and Development department, R&D #A085170 & A091246) in accordance with the Declaration of Helsinki, and written informed consent was obtained from all participants.

In the case of PD patients, the clinical evaluation included measures on the Unified Parkinson Disease Rating Scale (UPDRS), the Hoehn and Yahr (H&Y) stage, the Mini Mental State Examination (MMSE), the Addenbrooke's Cognitive Examination (ACE), and the Beck Depression Inventory (BDI). In the case of the HD patients, we collected their scores on the Unified Huntington Disease Rating Scale (UPDRS), Total Functional capacity (TFC) and calculated values for burden of disease (BDS). All the clinical evaluations were conducted within 9 months of the blood drive. Participants were further asked to fill out a questionnaire related to health issues and medication and their full blood count performed on the day of blood sampling.

Of note, participants excluded from the present EEV-related analyses included those with diabetes and those suffering or having suffered from cancer, because we observed a significant PD-independent increase in EEV

concentration in the platelet-free plasma of these participants. Furthermore, PFP samples with elevated free hemoglobin (>45 000 ng/mL), potentially due to hemolysis at blood sampling, were also excluded from EEV-related analyses, which explains the discrepancies between the total number of participants initially recruited and those contained in each analysis.

TABLE 1

Participant demographics								
Table 1: Participant clinical information.								
Parkinson's disease (PD) cohort								
PD Patients - Stages of disease								
	Ctrl	Unknown	Mild	Moderate	Severe		P value	
n	37	7	12	33	8			
Age	66.8	69.8	66.7	71.1	75.0*		0.04	
Gender F (M)	18 (19)	1 (6)	6 (6)	16 (17)	0 (8)		0.05	
Disease severity								
Hoehn & Yahr (n)			1 ± 0.3 (12)	2 ± 0.2 (33)	3 ± 0.5 (8)		<0.0001	
UPDRS (n)			38 ± 11 (6)	52 ± 19 (17)	73 ± 20 (6)		0.02	
ACE (n)			96 ± 4 (6)	92 ± 7 (17)	84 ± 14 (6)		0.13	
MMSE (n)			29 ± 2 (7)	29 ± 1 (19)	26 ± 3 (6)		0.01	
BDI (n)			3 ± 2 (6)	4 ± 2 (17)	13 ± 7 (4)		0.03	
Comorbidities								
Asthma	3	1	1	5	0		0.71	
Hypertension	10	1	2	10	3		0.76	
Diabetes	2	0	0	1	2		0.10	
Cancer	5	0	3	4	1		0.64	
Allergies	2	0	2	6	2		0.28	
Depression	3	1	2	1	2		0.29	
Hypercholesterolemia	5	0	1	6	1		0.73	
Huntington's disease (HD) cohort								
HD Patients - Stages of disease								
	Ctrl	Pre-HD	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	P value
n	55	11	15	13	12	10	2	
Age	55.0	37.5	53.1	54.2	58.3	58.1	55.5	0.02
Gender F (M)	31 (22)	6 (5)	5 (10)	4 (9)	8 (4)	7 (3)	1 (1)	0.26
Disease severity								
UHDRS (n)		2.7 (11)	15.7 (14)	34.5 (11)	42.9 (12)	55.9 (10)	67.5 (2)	<0.001
TFC (n)	13 (16)	13 (11)	12.5 (15)	7.8 (13)	4.3 (12)	1.6 (10)	0 (2)	<0.001
CAG (n)	28.3 (3)	41.1 (10)	42.3 (13)	42.6 (12)	43.7 (7)	44.3 (7)		<0.001
BDS (n)		206 (10)	337 (13)	356 (12)	442 (7)	465 (7)		<0.001
Comorbidities								
Asthma	0	0	1	0	0	0	0	0.65
Hypertension	4	1	2	1	1	2	0	0.92
Diabetes	3	1	1	1	1	1	0	0.99
Cancer	0	0	0	0	0	0	0	
Allergies	3	0	2	2	0	0	0	0.33
Depression	8	1	1	3	6	4	1	0.0497
Hypercholesterolemia	8	1	1	0	0	1	0	0.32

(PD cohort) Disease severity levels in relation to the H&Y scale (score): Mild (1-1.5); Moderate (2-2.5); Severe (3-3.5).

*p < 0.05 vs. CTRL. Statistical analyses were performed using a Welch ANOVA followed by Dunnett's multiple comparison test. Disease severity was evaluated within 6 months of blood sampling. Comorbidities were determined from medical information reported by the participant or caregiver. Cancer refers to participant having suffered from cancer in the past. (HD cohort) Disease severity levels in relation to the TFC scale: Stage 1 (11-13); Stage 2 (7-10); Stage 3 (3-6); Stage 4 (1-2); Stage 5 (0). Disease severity was evaluated within 6 months of blood sampling. Comorbidities were determined from medical information reported by the participant or caregiver. Abbreviations: ACE, Addenbrooke's cognitive examination; BDI, Beck depression inventory; BDS, Burden of Disease Score; CAG, Trinucleotide repeat; MMSE, Mini-Mental State Examination; UHDRS, Unified Huntington's Disease Rating Scale; TFC, Total Function Capacity.

1.2 Preparation of Platelet-Free Plasma (PFP) and Extracellular Vesicle (EV) Labeling

Citrated blood was centrifuged twice for 15 minutes at 2500 g at room temperature. Platelet-free plasma (PFP) was harvested and stored at -80° C. within 2 hours of collection following guidelines suggested by Lacroix and colleagues (Lacroix et al., 2012).

For all experiments, diluted annexin-V buffer (BD Pharmingen, Mississauga, ON, Canada) and phosphate buffered saline (PBS) were filtered on 0.22 µm pore size membranes. To quantify the EV according to their cell of origin, the following surface markers were used: CD235a+ (erythrocytes) (5 µL), CD31+/CD41- (endothelial cells) (1 µL), CD41+ (platelets) (5 µL), CD45+ (leukocytes) (3 µL),

CD45+CD14+ (monocytes) (10 µL), and CD45+CD15+ (granulocytes) (2 µL), with or without annexin-V staining (5 µL). PFP (5 µL) was incubated with Phenylalanyl-prolyl-arginyl Chloromethyl Ketone (PPACK) (Calbiochem, Eto-bicoke, ON, Canada) for 5 minutes, followed by a 30-minute incubation with antibodies and annexin-V in a final PBS volume of 100 µL, all at room temperature. Finally, the samples were diluted to a final volume of 2 mL prior to

FACS analysis. The following antibodies were purchased at BD Pharmingen and used throughout the experiments: FITC-conjugated mouse anti-human CD235a (clone GA-R2 (HIR2), 1/20), PE-conjugated mouse anti-human CD31 (clone WM59, 1/100), V450-conjugated mouse anti-human CD41a (clone HIPS, 1/20), APC mouse anti-human CD14 (clone M5E2, 1/10), PE-conjugated mouse anti-human CD15 (clone H198, 1/50), V450-conjugated mouse anti-human CD45 (clone H130, 1/33), V450- and PerCP-CyTM5.5-conjugated annexin-V (1/33 and 1/10, respectively).

1.3 Flow Cytometry Quantification

For EV quantification, we used a FACS Canto II Special Order Research Product equipped with a forward scatter (FSC) coupled to a photomultiplier tube (FSC-PMT). Flow cytometer performance tracking was carried out daily using the BD cytometer setup and tracking beads (BD Biosciences, San Jose, CA, USA). The size of the EV was determined using fluorescent silicone beads of 100, 500 and 1000 nm. Controls and optimization of the detection method are presented in FIGS. 1A-1F. The settings for the EV detection were determined as described previously (Rousseau et al., 2015) using a threshold of 200 for SSC. Between PD and HD analyses, the blue laser had to be replaced for maintenance issues and therefore laser settings were reassessed. For FSC-PMT, the assigned voltage was 363 (PD) and 160 (HD) Volts. For SSC, the assigned voltage was 407 (PD) and 300 (HD) Volts. All other parameters were set between 450 and 500 Volts. The acquisition of EV was performed at low speed with an approximate rate of 10 μ L/min. To determine background noise level, antibody mixes were incubated in absence of PFP sample and unlabeled PFP was used as a negative control.

1.4 Statistical Analyses

All statistical analyses pertaining to Examples 2.1 and 2.2 were performed using "The Statistics and Machine Learning Toolbox" provided by MathWorksTM under the MATLABTM platform. The version used was MATLAB[®] R2015a. The analysis included the scatter plot, the classical least-squares linear regression model, the R-squared and p values, as well as Pearson's goodness-of-fit model. Interval cut-off values were determined using a loop program developed in MATLABTM. Model diagnostics, including residual behaviour and homoscedasticity, were also obtained with the same Toolbox.

1.5 Production and Purification of EEV

Blood was collected in heparin tubes and centrifuged for 10 minutes at 282 g at room temperature. Blood cells were washed first in PBS-2% FBS, then with 0.9% sodium chloride solution and centrifuged for 10 minutes at 750 g. To avoid leukocyte and/or platelet contamination, the buffy coat and the upper fraction of erythrocytes were removed. To preserve erythrocytes, two volumes of glycerolyte 57 solution (57% glycerol, 142 mM sodium lactate, 1 mM KCl, 25 mM sodium phosphate pH 6.8) were added to the pellet and stored at -80° C.

For the production of EEV, red bloods cells were thawed and EV production was induced as previously described (Minetti et al., 2004). Briefly, the erythrocyte pellet was activated with 3 volumes of calcium ionophore solution (150 mM NaCl; 10 mM Tris-HCl; 1 mM CaCl_2 ; 5 μ M ionophore A23187 (Sigma, St Louis, MO)) for 30 minutes at 37° C. The activation was stopped by the addition of 5 mM EDTA. Remaining erythrocytes were pelleted at 15 000 g for 20 minutes. The EEV were centrifuged at 20 000 g for 90

minutes and washed once in PBS. The EEV pellet was resuspended in PBS and frozen at -80° C. until further analyses.

1.6 C-Reactive Protein, Free Hemoglobin and α -Synuclein Quantification

The concentrations of C-reactive protein (CRP) and free hemoglobin were determined in the PFP of all donors using the RayBioTM Human CRP ELISA Kit (RayBiotech, Norcross, GA, USA) and the Hemoglobin Human ELISA kit (Abcam, Toronto, ON, Canada). To quantify α -synuclein (α -Syn) in erythrocytes and EEV, we used the human α -Syn ELISA kit (ThermoFisher Scientific, Waltham, MA, USA). Absorbance values were measured at 450 nm using a multi-detection microplate reader (Synergy HT; BioTek; Winooski, VT, USA). All ELISA tests were performed according to the manufacturer's instructions.

1.7 Scanning Electron Microscopy

Preparations of erythrocytes (5 μ L) were fixed in 2% paraformaldehyde and 2.5% glutaraldehyde in PBS buffer at least 24 hours before standard dehydration. Samples were washed 3 times for 10 minutes with sodium cacodylate buffer (0.1 M, pH 7.3) and fixed with 1% osmium tetroxide in sodium cacodylate buffer for 90 minutes. Subsequently, samples were washed and processed in 50%, 70%, 90% and 100% ethanol for dehydration (10 minutes/step). Finally, samples were soaked in two subsequent baths of 100% ethanol, for 40 minutes and 10 minutes, air-dried overnight and coated with palladium. Observations were completed using a JEOL 6360LV scanning electron microscope (JEOL, Peabody, MA, USA).

1.8 Transmission Electron Microscopy

Preparations of EEV (30 μ L) and activated erythrocytes (5 μ L) were fixed in 2% paraformaldehyde at least 24 hours before being dehydrated and sealed in LR white resin. Slices of LR white resin were placed on a Formvar/carbon-coated grid and processed for immunolabeling. The tissues mounted on grids were blocked in 0.5% BSA-c (Aurion, Wageningen, The Netherlands) in HBSS and incubated for 120 minutes with rabbit anti- α -Syn antibody (Abcam, Toronto, ON, Canada) or rabbit anti- α -Syn (phospho S129) antibody (Abcam, Toronto, ON, Canada), both diluted at 1:250 in HBSS and washed several times with distilled water. Finally, the grids were incubated for 60 minutes with an anti-rabbit IgG conjugated to 6 nm gold particles (EMS, Hatfield, PA, USA) diluted at 1:200 and washed several times with distilled water to ultimately be fixed in 2.5% glutaraldehyde (EMS, Hatfield, PA, USA) in HBSS for 15 minutes. For this last step, the grids were treated with 3% uranyl acetate-0.075 M oxalate (pH 7.0) (EMS, Hatfield, PA, USA) for 1 minute, which was followed by several washes in distilled water. Observations were completed with a TECNAI Spirit G2 transmission electron microscope at 80 kV (FEI, Hillsboro, OR, USA).

1.9 Mass Spectrometry Analysis and Label Free Protein Quantification

For proteomic analyses, EEV from 4 individuals per group (Control, mild PD and moderate PD) were prepared as described above. For each individual, 25 μ g of protein sample, according to Bradford protein assay, were migrated onto an electrophoresis gel 4-12% Bis-Tris to separate hemoglobin from higher proteins. Following gel staining using Sypro Ruby (Thermo Fischer Scientific), the 12 kDa band corresponding to the hemoglobin size was cut out and the remaining part of the gel further fractioned into 7 slices, exposed to trypsin digestion and peptide extraction on a MassPrepTM liquid handling robot (Waters, Milford, USA) according to the manufacturer's specifications and to the

protocol of Shevchenko et al., 1996, with the modifications suggested by Havlis et al., 2003. The extracted peptides from the 7 slices of the same individual were pooled and analyzed by nanoLC-MS/MS. The excised hemoglobin gel slices were also analyzed in the same conditions. One μg of each individual sample was injected on a Dionex Ulti-Mate™ 3000 nanoRSLC system (Thermo Scientific) equipped with a nanoviper Acclaim Pepmap100™, C18, 3 μm , 75 μm ×column (Thermo Scientific) connected to the nanoelectrospray source of an Orbitrap Fusion™ mass spectrometer (Thermo Scientific). The peptides were eluted at 300 nL/min using an acetonitrile gradient of 90 minutes and the mass spectrometer was operating in Data Dependent Acquisition mode. Peptide masses were measured in MS spectra detected in the orbitrap at 120K resolution. MSMS fragmentation spectra of peptides were generated by Higher energy Collisional Dissociation (HCD) and detected in the ion trap. Spectra were searched against a human protein database (Uniprot Complete Proteome, taxonomy *Homo sapiens*—83512 sequences) using Andromeda™ search engine included in MaxQuant™ software version 1.5.5.1 (Cox et al., 2008). MaxQuant™ was also used to validate proteins and peptides at 1% False Discovery Rate using a target/decoy database search and to perform Label Free Quantification of the identified proteins using the ‘match between runs’ option.

1.10 Further Statistical Analyses

For FIGS. 3C and 3D, data were first tested for normality using the D’Agostino & Pearson normality test. Comparisons between groups were obtained by Mann-Whitney U test or Kruskal-Wallis ANOVA and performed using Prism 6.0 (GraphPad Software, LaJolla, CA). For analyses displayed in FIG. 4C, the ‘Intensity values’ contained in the output ‘protein_group.txt’ file of MaxQuant™ were used to quantify each identified protein in each individual sample. The values were normalized by the median of each column (all intensity values of proteins for one sample). The missing values were imputed with a noise value corresponding to the 1-percentile of each sample column. For each comparison between two groups (Control, mild PD or moderate PD), proteins with too many imputed values were considered not quantifiable (a minimum of three not-imputed values in one of the 2 groups are required). A protein ratio was calculated between the two groups using the average of intensity values in each group. Finally, a statistical Welch’s test was performed between the two groups. The protein ratios were transformed into $\log_2(\text{ratio})$ then centered by calculation of a z-score ($z\text{-score} = (x - \mu) / \sigma$). A protein was considered as variant if it fulfilled the following criteria: minimum of 2 peptides quantified, Welch’s test p value < 0.05 and absolute value of z-score > 1.96 (corresponding to values outside of

the 95% confidence interval). The Gene Ontology enrichment analysis on the identified proteins (FIG. 4D) was performed on the Cytoscape™ platform (v. 3.4.0) using the BinGO™ software version 3.0.3 (Maere et al., 20005) against all human genes with GO annotation (Uniprot-GOA generated 2015-06-22). Enrichment was calculated by hypergeometric test and Bonferroni Family-Wise Error Rate (FWER) was used to correct for multiple testing. The data for the resulting 8 proteins was standardized, hierarchically clustered and visualized as a heatmap by using the statistical framework R (R Core Team, 2016). The robustness of the nodes was evaluated by computing Approximately Unbiased (AU) p values using the R package pvclust (10000 bootstraps, average method and correlation-based dissimilarity matrix) (Suzuki et al., 2006).

Example 2—Results

The cohorts studied here included Parkinson’s disease (PD) (n=60) and Huntington’s disease (HD) patients (n=63) of all stages (see Example 1.1), as well as their respective age- and sex-matched healthy controls (n=37; n=55, respectively). The demographics for both cohorts are shown in Table 1. Full blood counts (erythrocytes, lymphocytes, platelets, leukocytes, monocytes, neutrophils) and C-reactive protein (indicative of an inflammatory response) quantification were obtained for all participants, but they did not reveal any significant differences between groups (data not shown). Similarly, the hematocrit, the mean corpuscular hemoglobin, as well as the mean corpuscular volume values were similar between PD and control groups (data not shown).

2.1 PD Patients Exhibit a Disease-Specific Increase in Erythrocyte-Derived EV

Platelet-free plasma (PFP) and extracellular vesicles (EV) were labeled and quantified according to their cell of origin for all participants, as described in Examples 1.2 and 1.3. Results are summarized in Table 2A (PD patients and controls) and Table 2B (HD patients and controls).

As shown in Table 2A, no significant differences between PD patient and control samples were observed in the number of EV originating from platelets, endothelial cells, monocytes, granulocytes, and leukocytes. Similarly, as shown in Table 2B, no significant differences between HD patient and control samples were observed in the concentrations of EV originating from these same cell types.

Interestingly, a significant increase in erythrocyte-derived EV in patients with PD was observed, as compared to the control group (see values highlighted in black in Table 2A). This increase in erythrocyte-derived EV in patients with PD was disease-specific, as the same effect was not observed in erythrocyte-derived EV in patients with HD (Table 2B).

TABLE 2A

Quantification of extracellular vesicles (EV) derived from different cell types of PD patients and controls									
Cell type	Markers	Units	CTRL			PD			P value
			n	Mean	SEM	n	Mean	SEM	
Platelets	CD41+PS−	$\times 10^3 / \mu\text{L}$	37	7.88	1.68	59	10.3	1.33	0.27
	CD41+PS+		37	15.2	3.20	59	17.9	2.53	0.51
	CD41+CD31+		37	1.51	0.69	59	1.99	0.54	0.59
	CD41+ total		37	23.1	4.62	59	28.2	3.66	0.38
	EV CD41+/platelet		35	0.106	0.021	57	0.125	0.016	0.49
Endothelial cells	CD31+CD41−PS−	$\times 10^3 / \mu\text{L}$	37	15.8	8.04	59	11.7	6.37	0.75
	CD31+CD41−PS+		37	0.91	0.13	59	0.92	0.10	0.96
	CD31+CD41− total		37	16.7	8.03	59	12.6	6.36	0.75

TABLE 2A-continued

Quantification of extracellular vesicles (EV) derived from different cell types of PD patients and controls									
Cell type	Markers	Units	CTRL			PD			P
			n	Mean	SEM	n	Mean	SEM	
Monocytes	CD45-CD14+ PS-	$\times 10^3/\mu\text{L}$	37	1.70	0.30	59	1.62	0.24	0.85
	CD45-CD14+ PS+		37	1.20	4.00	59	5.84	3.17	0.50
	CD45+CD14+ PS-		37	0.16	0.04	59	0.14	0.03	0.74
	CD45+CD14+PS+		37	0.60	0.79	59	1.47	0.63	0.59
	CD14+ total		37	3.66	4.88	59	9.06	3.87	0.60
Granulocytes	EV CD14+/monocyte	$\times 10^3/\mu\text{L}$	35	7.08	1.99	57	9.16	1.56	0.41
	CD45-CD15+ PS-		37	12.3	7.96	59	16.7	6.30	0.92
	CD45-CD15+ PS+		37	2.21	0.77	59	1.39	0.61	0.47
	CD45+CD15+ PS-		37	0.55	0.36	59	1.15	0.29	0.20
	CD45+CD15+PS+		37	1.01	0.30	59	1.25	0.24	0.56
Leukocytes	CD15+ total	$\times 10^3/\mu\text{L}$	37	16.0	8.83	59	20.6	6.99	0.91
	EV CD15+/granulocyte		35	3.70	0.64	57	3.16	0.50	0.53
	CD45+ total		37	10.4	2.21	59	13.8	1.75	0.26
	CD235a+PS-		36	18.2	46.5	59	32.0	36.3	0.04
	CD235a+PS+		36	0.22	0.07	59	0.29	0.05	0.70
Erythrocytes	CD235a+ total	$\times 10^3/\mu\text{L}$	36	18.4	47.0	59	32.3	36.7	0.04
	EV CD235a+/erythrocyte		34	0.0039	0.011	57	0.0069	0.008	0.04

Abbreviations: CD235a, glycophorin A; EV, extracellular vesicle; PD, Parkinson's disease; PS, phosphatidylserine.

TABLE 2B

Quantification of extracellular vesicles (EV) derived from different cell types of HD patients and controls												
Cell type	Markers	Units	CTRL			HD pre-manifest			HD			P
			n	Mean	SEM	n	Mean	SEM	n	Mean	SEM	
Platelets	CD41+PS-	$\times 10^3/\mu\text{L}$	54	9.2	2.2	10	4.3	1.3	50	6.1	1.0	0.78
	CD41+PS+		54	19.3	4.8	10	7.1	2.0	50	12.4	2.4	0.74
	CD41+ total		54	28.4	6.9	10	11.4	3.2	50	18.6	3.4	0.70
	EV CD41+/platelet		53	0.12	0.03	10	0.05	0.02	48	0.08	0.01	0.34
	CD31+CD41-PS-		54	1.4	0.3	10	0.6	0.2	50	1.2	0.2	0.31
Endothelial cells	CD31+CD41-PS+	$\times 10^3/\mu\text{L}$	54	0.68	0.16	10	0.25	0.06	50	0.46	0.09	0.59
	CD31+CD41- total		54	2.1	0.4	10	0.8	0.2	50	1.7	0.3	0.26
	CD45-CD14+ PS-		54	3.4	1.1	10	1.6	0.2	51	1.6	0.1	0.91
	CD45-CD14+ PS+		54	1.8	0.3	10	0.8	0.3	51	1.5	0.2	0.14
	CD45+CD14+ PS-		54	0.18	0.07	10	0.069	0.016	51	0.056	0.008	0.34
Monocytes	CD45+CD14+PS+	$\times 10^3/\mu\text{L}$	54	0.62	0.12	10	0.24	0.06	51	0.55	0.14	0.12
	CD14+ total		54	6.0	1.3	10	2.6	0.4	51	3.7	0.4	0.08
	EV CD14+/monocyte		53	12.3	2.5	10	5.7	0.6	48	8.0	1.0	0.13
	CD45-CD15+ PS-		54	1.2	0.1	10	1.2	0.3	51	1.5	0.2	0.33
	CD45-CD15+ PS+		54	0.12	0.04	10	0.18	0.08	51	0.22	0.11	0.33
Granulocytes	CD45+CD15+ PS-	$\times 10^3/\mu\text{L}$	54	0.20	0.05	10	0.07	0.02	51	0.15	0.04	0.64
	CD45+CD15+PS+		54	0.25	0.05	10	0.13	0.06	51	0.20	0.04	0.39
	CD15+ total		54	1.7	0.2	10	1.6	0.4	51	0.20	0.3	0.67
	EV CD15+/granulocyte		53	0.41	0.04	10	0.42	0.13	48	0.50	0.08	0.75
	CD45+ total		54	33.4	2.7	10	31.6	5.3	51	31.7	2.4	0.88
Leukocytes	CD235a+PS-	$\times 10^3/\mu\text{L}$	54	15.2	2.0	10	10.3	3.5	51	14.1	1.4	0.16
	CD235a+PS+		54	1.1	0.2	10	0.4	0.2	51	1.1	0.1	0.04
	CD235a+ total		54	16.4	2.0	10	10.7	3.5	51	15.3	1.5	0.09
	EV		54	0.0035	0.0005	10	0.0023	0.0008	50	0.0033	0.0003	0.11
	CD235a+/erythrocyte											

Abbreviations: CD235a, glycophorin A; EV, extracellular vesicle; HD, Huntington's disease; PS, phosphatidylserine.

2.2 Increase in Erythrocyte-Derived EV in PD Patient Samples Correlates with PD Progression and PD Treatment

To evaluate its suitability as a potential biomarker for monitoring PD progression, we examined correlations between the number of erythrocyte-derived EV (EEV) and the Unified Parkinson's Disease Rating Scale (UPDRS) PD staging system, because of its greater sensitivity and the recent publications validating this approach (Martinez-Martin et al., 2015). Strikingly, statistical linear regression analysis revealed strong correlations between the number of erythrocyte-derived EV and PD stage/progression. As

shown in FIG. 2A, strong correlations (correlations exceeding were observed between the number of erythrocyte-derived EV (expressed as CD135a+ EV/total number of erythrocytes) and patient UPDRS score, and thus PD stages. The R^2 values obtained demonstrated that in both "mild" and "moderate" PD patient groups, at least 87% of the variation in the total number of EEV/erythrocytes is due to the variation of the UPDRS. Moreover, the results are significant with respect to the p values obtained for each fit, since they fall below the 5% confidence level. Hence, the statistical tests on EEV counts uncovered a clear cut-off point

between mild and moderate PD patients, which could not be accounted for by daily levodopa doses administered to the patient (FIG. 2A, see details of the 5 patients pinpointed).

The above correlations observed with respect to the number of EEV in PD patients was found to be disease-specific, since a similar analysis performed in HD patients failed to reveal the same strong correlations (see FIG. 2B). In the HD cohort, the total number of EEV between pre-manifest and manifest HD was similar to their age- and sex-matched healthy Controls. In contrast to PD, correlation analyses failed to reveal an association between the number of EEV and HD stage using the United Huntington's Disease Rating Scale (UHDRS) score (FIG. 2B). Based on this, all our subsequent analyses focused only on the PD cohort. It is should be re-emphasized that total blood counts did not indicate any significant differences in the number of endothelial cell-, platelet-, monocyte- and granulocyte-derived EV in the platelet-free plasma (PFP) in PD (Table 2A) and HD patients (Table 2B), when compared to their respective control cohorts.

We have thus identified at least two distinct groups of PD patients with highly significant correlations to the number of EV derived from erythrocytes, which relates to PD stage and/or PD treatment (FIG. 2A). Strikingly, these correlations appear to be specific to PD, as similar correlations were not observed in the cohort of HD patients (of varying degrees of severity) in which we performed identical analyses (FIG. 2B).

2.3 α -Syn is not Differentially Expressed in Normal Vs. Diseased PD Conditions

Having established that EEV counts correlate with disease state in PD, we aimed to assess whether α -Syn—which is not only the main component of Lewy bodies but is highly expressed in most blood cells—was differentially expressed in normal vs. diseased conditions. For this, we opted to use scanning electron microscopy, but this did not reveal any morphological changes between resting and activated erythrocytes in either condition (FIG. 3A). We further used transmission electron microscopy to quantify the number of EEV containing α -Syn and phosphorylated (serine 129) forms of the protein but again no significant differences between PD patients and age- and sex-matched healthy Controls (FIGS. 3B and 3C) were observed. Quantified α -Syn levels in EEV from PD patients and Controls using commercial ELISA kits corroborated these results (FIG. 3D).

2.4 Analysis of the EEV Proteome Revealed 8 Differentially Expressed Proteins

Since our combined quantifications (ELISA and transmission electron microscopy) suggested that α -Syn levels could not be used as a blood marker of disease, we sought to obtain the specific protein signature of EEV from mild and moderate PD patients (with respect to the UPDRS scores) and their age-matched Controls. Given the significant amounts of hemoglobin within erythrocytes that could mask the true nature of the protein signature in EEV, we performed a label free quantitative proteomic analysis by nanoLC/MSMS (Wither et al., 2016) using two distinct approaches: with and without hemoglobin.

By removing the hemoglobin, we identified a total of 818 proteins in comparison with 356 when we did not perform this methodological step (refer to Table 4 for complete list of proteins), which clearly, provides a much more thorough evaluation of the protein content of EEV. Additionally, a Gene Ontology enrichment analysis on the 'Cellular Component' ontology performed on the two sets of identified proteins in comparison with the whole human proteome, revealed that our samples are enriched with elements associated to 'vesicles' and 'hemoglobin complex' which show the efficiency of our EEV production and purification protocol (data not shown).

Out of the 818 proteins identified in the proteome of EEV, 8 had their expression significantly modified according to the different stages of PD (FIGS. 4A and 4B). Hierarchical clustering, coupled to a heatmap (FIG. 4D), allowed us to group individuals according to stages of disease (Control, mild PD and moderate PD) and provided compelling evidence that the 8 proteins identified could also be grouped into three categories. Proteins of group I were highly and predominantly expressed in Controls, proteins of group II were highly and predominantly expressed in mild PD patients, and proteins belonging to group III were highly and predominantly expressed in moderate PD patients (FIGS. 4B and 4C; Table 3). This data set was further confirmed by volcano plots (FIG. 5). Of note, two proteins of group I are associated with the regulation system of the cell (ABHD148, NADSYN1) and one protein significantly expressed in moderate PD patients (ATP5A1) is involved in the regulation of mitochondrial ATP.

TABLE 3

EEV proteins differentially expressed according to different stages of PD		
Groups	Protein	Gene
I (Control)	Axin interactor, dorsalization-associated protein	AIDA
	Alpha/beta hydrolase domain-containing protein 14B	ABHD14B
	Glutamine-dependent NAD(+) synthetase	NADSYN1
II (mild PD)	Dihydropteridine reductase	QDPR
	Alcohol dehydrogenase [NADP(+)]	AKR1A1
	CB1 cannabinoid receptor-interacting protein 1	CNRIP1
III (moderate PD)	Ubiquitin carboxyl-terminal hydrolase 24	USP24
	ATP synthase subunit alpha, mitochondrial	ATP5A1

Fold changes and the results of statistical analyses of the proteins of Table 3 are shown in Table 3.1, in which the proteins are listed in order of their "fold" difference for each group comparison. For example, a "fold" of 1.931 indicates that the AKR1A1 protein was detected in the isolated EEV preparations almost two times higher in the mild group, as compared to the control group.

TABLE 3.1

Fold changes and statistics of EEV proteins of Table 3				
Protein	Gene	Welch's test	Fold change	Z-score
Mild (Group II) v. Controls (Group I)				
Alcohol dehydrogenase [NADP(+)]	AKR1A1	0.019	1.931	2.25
Alpha/beta hydrolase domain-containing protein 14B	ABHD14B	0.016	0.274	-3.57
Moderate (Group III) v. Controls (Group I)				
ATP synthase subunit alpha, mitochondrial	ATP5A1	0.028	3.906	3.25
Glutamine-dependent NAD(+) synthetase	NADSYN1	0.009	0.391	-2.31
Axin interactor, dorsalization-associated protein	AIDA	0.046	0.379	-2.38
Alpha/beta hydrolase domain-containing protein 14B	ABHD14B	0.017	0.282	-3.09
Moderate (Group III) v. Mild (Group II)				
ATP synthase subunit alpha, mitochondrial	ATP5A1	0.027	3.629	2.77
Ubiquitin carboxyl-terminal hydrolase 24	USP24	0.036	2.971	2.29
CB1 cannabinoid receptor-interacting protein 1	CNRIP1	0.014	0.352	-2.80
Alcohol dehydrogenase [NADP(+)]	AKR1A1	0.002	0.342	-2.87
Dihydropteridine reductase	QDPR	0.040	0.341	-2.88

Proteins were considered as significantly differentially expressed if they respected two conditions: Z-score >1.96 or <-1.96, and Welch's test p-value < 0.05.

As seen in Table 3.1 and in FIG. 4C, the ABHD14B protein (alpha/beta hydrolase domain-containing protein 14B) was detected over 3.5-fold lower both in the mild v. controls and in the moderate v. controls groups, suggesting that this EEV protein may be a useful biomarker for clinically assessing symptomatic PD subjects. The results in Table 3.1 and in FIG. 4C suggest that the AKR1A1 protein (alcohol dehydrogenase [NADP(+)] may be a useful biomarker for clinically assessing mild PD subjects, as this EEV protein was detected almost 2-fold higher in mild v. control PD subjects, but was not differentially expressed in moderate v. control PD subjects. Finally, the results in Table 3.1 and in FIG. 4C suggest that the ATP5A1 protein may be a useful biomarker for clinically assessing moderate PD subjects, since this protein was detected over 3.6-fold higher in both moderate v. controls and in moderate v. mild PD subjects.

Example 3—Discussion

We have identified biomarkers correlating to different states of PD based on the quantification of EV shed from erythrocytes and UPDRS scores. In particular, we have shown a strong correlation between the number of EEV and the clinical expression/stages of PD. Finding this level of correlation in a 60-patient sample size in such a heterogeneous disorder highlights the robustness of the biomarkers identified herein.

Although staging of PD is often done using the H & Y clinical scale, we sought to use the UPDRS given its greater sensitivity and the recent publications validating this approach (Martinez-Martin et al., 2015). Using these scores, we found that mild PD patients—with a UPDRS score lower than 37—are characterized by an increased number of EV during the mild disease stage (correlations=0.886); and that the same pattern repeated itself with patients who had UPDRS scores between 37 and 75 (correlations=0.873). When comparing individual patients from both correlation curves, the dose of levodopa taken daily did not seem to account for the differences, as there was no correlation between the levodopa dose and the number of EEV.

Finally, we herein report an improved method to perform proteomic analysis of EEV in blood samples by removing hemoglobin, a large protein that can easily mask other ones within a protein signature. Indeed, the high dynamic range of protein concentrations in erythrocytes and therefore in EEV, due to the high abundance of hemoglobin, decreases the capacity of the mass spectrometer to detect signals corresponding to low abundance proteins. Analyzing the hemoglobin separately from the other proteins of other molecular weights allowed us to go deeper in the EEV proteome by identifying 129% more proteins than in the initial analysis.

TABLE 4

Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome <i>Homo sapiens</i> database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search.			
Protein ID	Protein names	Gene names	Razor + unique peptides
(A). UNIPROT COMPLETE PROTEOME <i>HOMO SAPIENS</i> DATABASE FOR THE NON-DEPLETED SAMPLE			
P02549	Spectrin alpha chain, erythrocytic 1	SPTA1	136
P11277	Spectrin beta chain, erythrocytic	SPTB	123
P16157	Ankyrin-1	ANK1	70
P02730	Band 3 anion transport protein	SLC4A1	38
P55072	Transitional endoplasmic reticulum ATPase	VCP	33
P16452	Erythrocyte membrane protein band 4.2	EPB42	33

TABLE 4-continued

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Protein ID	Protein names	Gene names	Razor + unique peptides
P11171	Protein 4.1	EPB41	33
P04040	Catalase	CAT	32
Q13228	Selenium-binding protein 1	SELENBP1	29
B4DT77	Annexin; Annexin A7	ANXA7	27
Q8WUM4	Programmed cell death 6-interacting protein	PDCD6IP	27
P35612	Beta-adducin	ADD2	26
P68871	Hemoglobin subunit beta; LW-hemorphin-7; Spinorphin	HBB	25
Q00610	Clathrin heavy chain 1	CLTC	25
P69905	Hemoglobin subunit alpha	HBA1	22
B4DVE7	Annexin A11	ANXA11	20
J3QLD9	Flotillin-2	FLOT2	20
P08758	Annexin A5; Annexin	ANXA5	19
P09525	Annexin A4; Annexin	ANXA4	18
O75955;	Flotillin-1	FLOT1	18
P11142	Heat shock cognate 71 kDa protein	HSPA8	18
P32119	Peroxiredoxin-2	PRDX2	17
P00491	Purine nucleoside phosphorylase	PNP	17
P27105	Erythrocyte band 7 integral membrane protein	STOM	17
P00918	Carbonic anhydrase 2	CA2	17
P23634	Plasma membrane calcium-transporting ATPase 4	ATP2B4	17
C9JIF9	Acylamino-acid-releasing enzyme	APEH	17
P00915	Carbonic anhydrase 1	CA1	16
P63261	Actin, cytoplasmic 2	ACTG1	16
Q5VU58	Tropomyosin alpha-3 chain	TPM3	16
P30041	Peroxiredoxin-6	PRDX6	16
E7EU23	Rab GDP dissociation inhibitor beta	GDI2	16
Q00013	55 kDa erythrocyte membrane protein	MPP1	16
E7EV01	Calpain-5	CAPN5	16
J3KPS3	Fructose-bisphosphate aldolase A	ALDOA	15
Q08495	Dematin	DMTN	15
P23276	Kell blood group glycoprotein	KEL	15
P69892	Hemoglobin subunit gamma-2	HBG2	14
P60174	Triosephosphate isomerase	TPI1	14
P62258	14-3-3 protein epsilon	YWHAE	14
O75326	Semaphorin-7A	SEMA7A	14
O75340	Programmed cell death protein 6	PDCD6	13
E7EV99	Alpha-adducin	ADD1	13
Q5VZU9	Tripeptidyl-peptidase 2	TPP2	13
P53396	ATP-citrate synthase	ACLY	13
C9J0K6	Sorcin	SRI	12
P04406	Glyceraldehyde-3-phosphate dehydrogenase	GAPDH	12
P07738	Bisphosphoglycerate mutase	BPGM	12
B7Z319	Delta-aminolevulinic acid dehydratase	ALAD	12
Q86X55	Histone-arginine methyltransferase CARM1	CARM1	12
P07195	L-lactate dehydrogenase B chain; L-lactate dehydrogenase	LDHB	12
P23526	Adenosylhomocysteinase	AHCY	12
Q32Q12	Nucleoside diphosphate kinase	NME1-NME2	12
B7Z7A9	Phosphoglycerate kinase 1	PGK1	11
P11166	Solute carrier family 2, facilitated glucose transporter member 1	SLC2A1	11
P22303	Acetylcholinesterase	ACHE	11
Q9NP58	ATP-binding cassette sub-family B member 6, mitochondrial	ABCB6	11
P40925	Malate dehydrogenase, cytoplasmic; Malate dehydrogenase	MDH1	11
P00352	Retinal dehydrogenase 1	ALDH1A1	11
F2Z2V0	Copine-1	CPNE1	11
K7EMC9	WW domain-binding protein 2	WBP2	10
F5H7S3	Tropomyosin alpha-1 chain	TPM1	10
A6NN80	Annexin A6; Annexin	ANXA6	10
P30043	Flavin reductase (NADPH)	BLVRB	9
H7BXD5	Grancalcin	GCA	9
P04083	Annexin A1; Annexin	ANXA1	9
P62937	Peptidyl-prolyl cis-trans isomerase	PPIA	9
P37837	Transaldolase	TALDO1	9
Q06830	Peroxiredoxin-1	PRDX1	9
O75131	Copine-3	CPNE3	9
P00390	Glutathione reductase, mitochondrial	GSR	9
E7EQB2	Lactotransferrin	LTF	9
P07384	Calpain-1 catalytic subunit	CAPN1	9
P02042	Hemoglobin subunit delta	HBD	8
P30086	Phosphatidylethanolamine-binding protein 1; Hippocampal cholinergic neurostimulating peptide	PEBP1	8

TABLE 4-continued

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Protein ID	Protein names	Gene names	Razor + unique peptides
P35613	Basigin	BSG	8
Q9H0U4	Ras-related protein Rab-1B; Putative Ras-related protein Rab-1C	RAB1B; RAB1C	8
P63092	Guanine nucleotide-binding protein G(s) subunit alpha isoforms	GNAS	8
P48506	Glutamate-cysteine ligase catalytic subunit	GCLC	8
P06702	Protein S100-A9	S100A9	7
Q9UBV8	Peflin	PEF1	7
P17931	Galectin-3; Galectin	LGALS3	7
P28066	Proteasome subunit alpha type-5	PSMA5	7
P07451	Carbonic anhydrase 3	CA3	7
E7EQ12	Calpastatin	CAST	7
P50895	Basal cell adhesion molecule	BCAM	7
P28074	Proteasome subunit beta type-5	PSMB5	7
G3V5Z7	Proteasome subunit alpha type; Proteasome subunit alpha type-6	PSMA6	7
P25786	Proteasome subunit alpha type-1	PSMA1	7
G3V1D3	Dipeptidyl peptidase 3	DPP3	7
P49247	Ribose-5-phosphate isomerase	RPIA	7
Q5T9B7	Adenylate kinase isoenzyme 1	AK1	7
P25789	Proteasome subunit alpha type-4; Proteasome subunit beta type	PSMA4	7
B4E022	Transketolase	TKT	7
J3QS39	Ubiquitin-60S ribosomal protein L40; Ubiquitin; 60S ribosomal protein L40; Ubiquitin-40S ribosomal protein S27a; Ubiquitin; 40S ribosomal protein S27a; Polyubiquitin-B; Ubiquitin; Polyubiquitin-C; Ubiquitin	UBB; RPS27A; UBC; UBA52; UBBP4	6
H0Y7A7	Calmodulin	CALM2	6
P28070	Proteasome subunit beta type-4	PSMB4	6
Q9H4G4	Golgi-associated plant pathogenesis-related protein 1	GLIPR2	6
Q9BY43	Charged multivesicular body protein 4a	CHMP4A	6
P48426	Phosphatidylinositol 5-phosphate 4-kinase type-2 alpha	PIP4K2A	6
P28289	Tropomodulin-1	TMOD1	6
P07911	Uromodulin; Uromodulin, secreted form	UMOD	6
Q9GZP4	PITH domain-containing protein 1	PITHD1	6
P78417	Glutathione S-transferase omega-1	GSTO1	6
P25788	Proteasome subunit alpha type-3	PSMA3	6
O14818	Proteasome subunit alpha type-7; Proteasome subunit alpha type-7-like	PSMA7; PSMA8	6
P08107	Heat shock 70 kDa protein 1A/1B	HSPA1A	6
H0YD13	CD44 antigen	CD44	6
P61225	Ras-related protein Rap-2b; Ras-related protein Rap-2c; Ras-related protein Rap-2a	RAP2B; RAP2A; RAP2C	6
P05109	Protein S100-A8; Protein S100-A8, N-terminally processed	S100A8	6
P23528	Cofilin-1	CFL1	6
Q99808	Equilibrative nucleoside transporter 1	SLC29A1	6
P84077	ADP-ribosylation factor 1; ADP-ribosylation factor 3; ADP-ribosylation factor 5; ADP-ribosylation factor 4	ARF1; ARF3; ARF5; ARF4	6
P31946	14-3-3 protein beta/alpha; 14-3-3 protein beta/alpha, N-terminally processed	YWHA8	6
C9JIS1	Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-2; Guanine nucleotide-binding protein subunit beta-4	GNB2; GNB4	6
P53990	IST1 homolog	IST1	6
Q99497	Protein DJ-1	PARK7	6
F5H7U0	6-phosphogluconate dehydrogenase, decarboxylating	PGD	6
B7Z7E9	Aspartate aminotransferase, cytoplasmic	GOT1	6
P62834	Ras-related protein Rap-1A; Ras-related protein Rap-1b; Ras-related protein Rap-1b-like protein	RAP1A; RAP1B	6
P04899	Guanine nucleotide-binding protein G(i) subunit alpha-2	GNAI2	6
P25325	3-mercaptopyruvate sulfurtransferase; Sulfurtransferase	MPST	6
Q9NP79	Vacuolar protein sorting-associated protein VTA1 homolog	VTA1	6
P00492	Hypoxanthine-guanine phosphoribosyltransferase	HPRT1	6
Q16531	DNA damage-binding protein 1	DDB1	6
P22314	Ubiquitin-like modifier-activating enzyme 1	UBA1	6
P00441	Superoxide dismutase [Cu—Zn]	SOD1	5
H7BY58	Protein-L-isoaspartate O-methyltransferase; Protein-L-isoaspartate(D-aspartate) O-methyltransferase	PCMT1	5
P10768	S-formylglutathione hydrolase	ESD	5
P09543	2,3-cyclic-nucleotide 3-phosphodiesterase	CNP	5
P06733	Alpha-enolase; Enolase	ENO1	5
P26038	Moesin	MSN	5
O75368	SH3 domain-binding glutamic acid-rich-like protein	SH3BGRL	5
K7EQ48	Glucose-6-phosphate isomerase	GPI	5
P25787	Proteasome subunit alpha type-2	PSMA2	5
P49721	Proteasome subunit beta type-2	PSMB2	5
F5H8J2	Protein disulfide-isomerase	P4HB	5
C9J9P4	Phospholipid scramblase 1	PLSCR1	5
P09211	Glutathione S-transferase P	GSTP1	5

TABLE 4-continued

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Protein ID	Protein names	Gene names	Razor + unique peptides
B5MDF5	GTP-binding nuclear protein Ran	RAN	5
P07900	Heat shock protein HSP 90-alpha	HSP90AA1	5
K7EQ02	DAZ-associated protein 1	DAZAP1	5
Q13630	GDP-L-fucose synthase	TSTA3	5
F5H0T1	Stress-induced-phosphoprotein 1	STIP1	5
P50502	Hsc70-interacting protein; Putative protein FAM10A5; Putative protein FAM10A4	ST13; ST13P5; ST13P4	5
P20618	Proteasome subunit beta type-1	PSMB1	5
P62805	Histone H4	HIST1H4A	5
P51148	Ras-related protein Rab-5C	RAB5C	5
H7C2G2	Ecto-ADP-ribosyltransferase 4	ART4	5
J3KQ18	D-dopachrome decarboxylase; D-dopachrome decarboxylase-like protein	DDT; DDTL	5
H3BPK3	Hydroxyacylglutathione hydrolase, mitochondrial	HAGH	5
B4DIT7	Protein-glutamine gamma-glutamyltransferase 2	TGM2	5
O43633	Charged multivesicular body protein 2a	CHMP2A	5
B4DQH4	T-complex protein 1 subunit theta	CCT8	5
Q9UN37	Vacuolar protein sorting-associated protein 4A	VPS4A	5
I3L397	Eukaryotic translation initiation factor 5A-1; Eukaryotic translation initiation factor 5A-1-like	EIF5A; EIF5AL1	5
Q9Y5Z4	Heme-binding protein 2	HEBP2	5
Q9UKV8	Protein argonaute-2	AGO2	5
F5H442	Tumor susceptibility gene 101 protein	TSG101	5
H3BLV0	Complement decay-accelerating factor	CD55	5
P02008	Hemoglobin subunit zeta	HBZ	5
E7EPV7	Alpha-synuclein	SNCA	4
P63104	14-3-3 protein zeta/delta	YWHAZ	4
P02724	Glycophorin-A	GYPA; GP-Erik	4
U3KQE2	Calpain small subunit 1	CAPNS1	4
Q9NP59	Solute carrier family 40 member 1	SLC40A1	4
Q5QPM9	Proteasome inhibitor PI31 subunit	PSMF1	4
C9J8U2	Nicotinate phosphoribosyltransferase	NAPRT	4
P61981	14-3-3 protein gamma; 14-3-3 protein gamma, N-terminally processed	YWHAH	4
Q9UQ80	Proliferation-associated protein 2G4	PA2G4	4
E5RJR5	S-phase kinase-associated protein 1	SKP1	4
Q9NRV9	Heme-binding protein 1	HEBP1	4
Q9H444	Charged multivesicular body protein 4b	CHMP4B	4
P10599	Thioredoxin	TXN	4
Q5VSJ9	Blood group Rh(CE) polypeptide; Blood group Rh(D) polypeptide	RHCE; RHD	4
Q9UK41	Vacuolar protein sorting-associated protein 28 homolog	VPS28	4
Q9Y3I1	F-box only protein 7	FBXO7	4
P61026	Ras-related protein Rab-10	RAB10	4
Q14974	Importin subunit beta-1	KPNB1	4
P27797	Calreticulin	CALR	4
P18669	Phosphoglycerate mutase 1; Probable phosphoglycerate mutase 4	PGAM1; PGAM4	4
P54725	UV excision repair protein RAD23 homolog A	RAD23A	4
P30613	Pyruvate kinase PKLR	PKLR	4
P63000	Ras-related C3 botulinum toxin substrate 1; Ras-related C3 botulinum toxin substrate 3; Ras-related C3 botulinum toxin substrate 2	RAC1; RAC3; RAC2	4
P05164	Myeloperoxidase	MPO	4
F5GWY2	Bifunctional purine biosynthesis protein PURH; Phosphoribosylaminoimidazolecarboxamide formyltransferase; IMP cyclohydrolase	ATIC	4
P48637	Glutathione synthetase	GSS	4
F5H5V4	26S proteasome non-ATPase regulatory subunit 9	PSMD9	4
G5E9R5	Low molecular weight phosphotyrosine protein phosphatase	ACP1	3
P46976	Glycogenin-1	GYG1	3
P28072	Proteasome subunit beta type-6	PSMB6	3
P26447	Protein S100-A4	S100A4	3
F5GXQ0	BRO1 domain-containing protein BROX	BROX	3
P08754	Guanine nucleotide-binding protein G(k) subunit alpha	GNAI3	3
Q99436	Proteasome subunit beta type-7	PSMB7	3
P62942	Peptidyl-prolyl cis-trans isomerase	FKBP1A	3
U3KQK0	Histone H2B	HIST1H2B	3
J3QKR3	Proteasome subunit beta type-3	PSMB3	3
P01116	GTPase KRas; GTPase KRas, N-terminally processed; GTPase HRas; GTPase HRas, N-terminally processed; GTPase NRas	KRAS; HRAS; NRAS	3
P13489	Ribonuclease inhibitor	RNH1	3
Q08722	Leukocyte surface antigen CD47	CD47	3
Q5T123	SH3 domain-binding glutamic acid-rich-like protein 3	SH3BGRL3	3
Q8WYQ7	Galectin; Galectin-9	LGALS9	3
O75695	Protein XRP2	RP2	3
P00167	Cytochrome b5	CYB5A	3

TABLE 4-continued

Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome <i>Homo sapiens</i> database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search.			
Protein ID	Protein names	Gene names	Razor + unique peptides
Q9Y4D1	Disheveled-associated activator of morphogenesis 1	DAAM1	3
P11021	78 kDa glucose-regulated protein	HSPA5	3
H7C1D4	Translin	TSN	3
P07737	Profilin-1	PFN1	3
M0R389	Platelet-activating factor acetylhydrolase IB subunit gamma	PAFAH1B3	3
A6NJA2	Ubiquitin carboxyl-terminal hydrolase 14	USP14	3
P10644	cAMP-dependent protein kinase type I-alpha regulatory subunit	PRKAR1A	3
Q9BS40	Latexin	LXN	3
G5EA52	Protein disulfide-isomerase A3	PDIA3	3
P53004	Biliverdin reductase A	BLVRA	3
Q04656	Copper-transporting ATPase 1	ATP7A	3
H9KV70	Neutrophil gelatinase-associated lipocalin	LCN2	3
O00299	Chloride intracellular channel protein 1	CLIC1	3
F8WF69	Clathrin light chain A	CLTA	3
G3V2F7	Ubiquitin-conjugating enzyme E2 variant 1; Ubiquitin-conjugating enzyme E2 variant 2	UBE2V1; UBE2V2	3
F8WDS9	LanC-like protein 1	LANCL1	3
P60891	Ribose-phosphate pyrophosphokinase 1	PRPS1	3
K7ESE8	Bleomycin hydrolase	BLMH	3
H0YNE3	Proteasome activator complex subunit 1	PSME1	3
P16930	Fumarylacetoacetase	FAH	3
F8VSD4	Ubiquitin-conjugating enzyme E2 N	UBE2N	3
P07203	Glutathione peroxidase 1	GPX1	3
P62328	Thymosin beta-4; Hematopoietic system regulatory peptide	TMSB4X; TMSB4XP4	3
E5RIW3	Tubulin-specific chaperone A	TBCA	3
M0R0Y2	Alpha-soluble NSF attachment protein	NAPA	3
P15374	Ubiquitin carboxyl-terminal hydrolase isozyme L3	UCHL3	3
P04921	Glycophorin-C	GYPC	2
H0YDI1	Lymphocyte function-associated antigen 3	CD58	2
B4E220	Aquaporin-1	AQP1	2
C9JEN3	Protein lifeguard 3	TMBIM1	2
F5H2R5	Rho GDP-dissociation inhibitor 2	ARHGDIB	2
Q53TN4	Cytochrome b reductase 1	CYBRD1	2
Q9NZD4	Alpha-hemoglobin-stabilizing protein	AHSP	2
Q8NHG7	Small VCP/p97-interacting protein	SVIP	2
Q5JYX0	Cell division control protein 42 homolog	CDC42	2
Q71RC9	Small integral membrane protein 5	SMIM5	2
E9PNW4	CD59 glycoprotein	CD59	2
P09105	Hemoglobin subunit theta-1	HBQ1	2
R4GN98	Protein S100; Protein S100-A6	S100A6	2
O75531	Barrier-to-autointegration factor; Barrier-to-autointegration factor, N-terminally processed	BANF1	2
Q5T6W5	Heterogeneous nuclear ribonucleoprotein K	HNRNPK	2
F5H4Q5	Vacuolar protein sorting-associated protein 37C	VPS37C	2
J3QK90	NSFL1 cofactor p47	NSFL1C	2
H3BV85	BolA-like protein 2	BOLA2B; BOLA2	2
Q9NRX4	14 kDa phosphohistidine phosphatase	PHPT1	2
H3BS66	Small integral membrane protein 1	SMIM1	2
E7ESC6	Exportin-7	XPO7	2
P68402	Platelet-activating factor acetyl hydrolase IB subunit beta	PAFAH1B2	2
Q9BRF8	Serine/threonine-protein phosphatase CPPED1	CPPED1	2
P08246	Neutrophil elastase	ELANE	2
E9PN50	26S protease regulatory subunit 6A	PSMC3	2
E7EUC7	UTP-glucose-1-phosphate uridylyltransferase	UGP2	2
B8ZZB8	CB1 cannabinoid receptor-interacting protein 1	CNRIP1	2
E9PCS3	26S proteasome non-ATPase regulatory subunit 2	PSMD2	2
P59666	Neutrophil defensin 3; HP 3-56; Neutrophil defensin 2; Neutrophil defensin 1; HP 1-56; Neutrophil defensin 2	DEFA3; DEFA1	2
O15400	Syntaxin-7	STX7	2
P00338	L-lactate dehydrogenase A chain	LDHA	2
P61970	Nuclear transport factor 2	NUTF2	2
E7EMV0	Protein diaphanous homolog 1	DIAPH1	2
F5GY90	Porphobilinogen deaminase	HMBS	2
P61020	Ras-related protein Rab-5B	RAB5B	2
Q99828	Calcium and integrin-binding protein 1	CIB1	2
B4DUA0	Plastin-2	LCP1	2
C9JTY3	Protein TFG	TFG	2
P27348	14-3-3 protein theta	YWHAQ	2
H0YKZ7	Annexin; Annexin A2; Putative annexin A2-like protein	ANXA2; ANXA2P2	2

TABLE 4-continued

Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome <i>Homo sapiens</i> database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search.			
Protein ID	Protein names	Gene names	Razor + unique peptides
P08238	Heat shock protein HSP 90-beta	HSP90AB1	2
J3KQP6	Ras-related protein Rab-11B; Ras-related protein Rab-11A	RAB11A; RAB11B	2
A6NMU3	Signal transducing adapter molecule 1	STAM	2
P53985	Monocarboxylate transporter 1	SLC16A1	2
F6USW4	F-actin-capping protein subunit beta	CAPZB	2
O14964	Hepatocyte growth factor-regulated tyrosine kinase substrate	HGS	2
P20020	Plasma membrane calcium-transporting ATPase 1; Calcium-transporting ATPase	ATP2B1	2
P36959	GMP reductase 1	GMPT	2
Q9Y376	Calcium-binding protein 39	CAB39	2
Q9Y6M5	Zinc transporter 1	SLC30A1	2
Q8IZ83	Aldehyde dehydrogenase family 16 member A1	ALDH16A1	2
Q99459	Cell division cycle 5-like protein	CDC5L	2
P06132	Uroporphyrinogen decarboxylase	UROD	2
J3KNT0	Fascin	FSCN1	2
P49189	4-trimethylaminobutyraldehyde dehydrogenase	ALDH9A1	2
H3BNT7	26S proteasome non-ATPase regulatory subunit 7	PSMD7	2
P05023	Sodium/potassium-transporting ATPase subunit alpha-1; Sodium/potassium-transporting ATPase subunit alpha-3; Sodium/potassium-transporting ATPase subunit alpha-2; Sodium/potassium-transporting ATPase subunit alpha-4; Potassium-transporting ATPase alpha chain 1; Potassium-transporting ATPase alpha chain 2	ATP1A1; ATP1A2; ATP1A3; ATP1A4; ATP4A; ATP12A	2
P34932	Heat shock 70 kDa protein 4	HSPA4	2
K7EMV3	Histone H3	H3F3B	2
Q8IU68	Transmembrane channel-like protein 8	TMC8	2
E7ENZ3	T-complex protein 1 subunit epsilon	CCT5	2
Q5TZA2	Rootletin	CROCC	2
Q9P203	BTB/POZ domain-containing protein 7	BTBD7	2
Q7LBR1	Charged multivesicular body protein 1b	CHMP1B	2
U3KQ56	Glyoxylate reductase/hydroxypyruvate reductase	GRHPR	2
HOYJ11	Alpha-actinin-1; Alpha-actinin-2; Alpha-actinin-4	ACTN1; ACTN4; ACTN2	2
Q9UDT6	CAP-Gly domain-containing linker protein 2	CLIP2	2
P09960	Leukotriene A-4 hydrolase	LTA4H	2
Q5HY54	Filamin-A	FLNA	2
E9PJL5	Uncharacterized protein C12orf55; Putative uncharacterized protein C12orf63	C12orf55; C12orf63	2
G3V2U7	Acylphosphatase; Acylphosphatase-1	FBN3; ACYP1	2
E9PQN4	Complement receptor type 1	CR1	2
Q9BSL1	Ubiquitin-associated domain-containing protein 1	UBAC1	2
Q04917	14-3-3 protein eta	YWHAH	2
B7ZBP9	Serine/threonine-protein phosphatase 2A activator	PPP2R4; DKF Zp781M17165	2
Q8NDC0	MAPK-interacting and spindle-stabilizing protein-like	MAPK1IP1L	1
P68133	Actin, alpha skeletal muscle; Actin, alpha cardiac muscle 1; Actin, gamma-enteric smooth muscle; Actin, aortic smooth muscle	ACTA1; ACTC1; ACTG2; ACTA2	1
P69891	Hemoglobin subunit gamma-1	HBG1	1
S4R3Y4	Protein AM BP; Alpha-1-microglobulin; Inter-alpha-trypsin inhibitor light chain; Trypstatin	AMBP	1
I3L3E4	Charged multivesicular body protein 6	CHMP6	1
Q16570	Atypical chemokine receptor 1	ACKR1	1
Q5VY30	Retinol-binding protein 4; Plasma retinol-binding protein(1-182); Plasma retinol-binding protein(1-181); Plasma retinol-binding protein(1-179); Plasma retinol-binding protein(1-176)	RBP4	1
E7END7	Ras-related protein Rab-1A	RAB1A	1
Q5VU59		TPM3	1
P17066	Heat shock 70 kDa protein 6; Putative heat shock 70 kDa protein 7	HSPA6; HSPA7	1
Q04760	Lactoylglutathione lyase	GLO1	1
D6RD66	WD repeat-containing protein 1	WDR1	1
K7EM02	Katanin p60 ATPase-containing subunit A-like 2	KATNAL2	1
P14209	CD99 antigen	CD99	1
E9PIR7	Thioredoxin reductase 1, cytoplasmic	GML; TXNRD1	1
K7EMQ9		EIF3K	1
P15531	Nucleoside diphosphate kinase A	NME1	1
H7BZT4	Small ubiquitin-related modifier 4; Small ubiquitin-related modifier 2; Small ubiquitin-related modifier 3	SUMO2; SUMO3; SUMO4	1
O00560	Syntenin-1	SDCBP	1
Q9BVM4	Gamma-glutamylaminocyclotransferase	GGACT	1
K7EKH5	Fructose-bisphosphate aldolase C	ALDOC	1
P49773	Histidine triad nucleotide-binding protein 1	HINT1	1
HOYBY6	Disks large-associated protein 2	DLGAP2	1

TABLE 4-continued

Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome <i>Homo sapiens</i> database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search.			
Protein ID	Protein names	Gene names	Razor + unique peptides
Q9Y624	Junctional adhesion molecule A	F11R	1
B1AKQ8	Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-1; Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-3	GNB1; GNB3	1
K7EKN6	Urea transporter 1	SLC14A1	1
I3L0K2	Thioredoxin domain-containing protein 17	TXNDC17	1
A8MX0	Syntaxin-4	STX4	1
O14773	Tripeptidyl-peptidase 1	TPP1	1
E9PNW0	Nucleosome assembly protein 1-like 1; Nucleosome assembly protein 1-like 4	NAP1L4; NAP1L1	1
Q5TDH0	Protein DDI1 homolog 2	DDI2	1
Q96JM4	Leucine-rich repeat and IQ domain-containing protein 1	LRRIQ1	1
F5GWT9	Phosphoribosylformylglycinamide synthase	PFAS	1
F2Z3J2	26S proteasome non-ATPase regulatory subunit 5	PSMD5	1
J3QL74	Zinc finger and BTB domain-containing protein 14	ZBTB14	1
E9PJ7	CD82 antigen	CD82	1
Q9H936	Mitochondrial glutamate carrier 1	SLC25A22	1
D6RD63	COP9 signalosome complex subunit 4	COPS4	1
Q6B0K9	Hemoglobin subunit mu	HBM	1
Q31611	HLA class I histocompatibility antigen, alpha chain G	HLA-G	1
H7BY04	Laminin subunit gamma-3	LAMC3	1
Q9UL25	Ras-related protein Rab-21	RAB21	1
H7C3P7	Ras-related protein Ral-A	RALA	1
P08311	Cathepsin G	CTSG	1
E9PE37	Ras-related protein Rab-2A; Ras-related protein Rab-2B	RAB2B; RAB2A	1
G3V1N2		HBA2	1
P00387	NADH-cytochrome b5 reductase 3; NADH-cytochrome b5 reductase 3 membrane-bound form; NADH-cytochrome b5 reductase 3 soluble form	CYB5R3	1
O75339	Cartilage intermediate layer protein 1; Cartilage intermediate layer protein 1 C1; Cartilage intermediate layer protein 1 C2	CILP	1
P14324	Farnesyl pyrophosphate synthase	FDPS	1
K7EKG2	Thioredoxin-like protein 1	TXNL1	1
(B). UNIPROT COMPLETE PROTEOME <i>HOMO SAPIENS</i> DATABASE WHEN THE HEMOGLOBIN WAS ANALYZED SEPARATELY			
P02549	Spectrin alpha chain, erythrocytic 1	SPTA1	197
P11277	Spectrin beta chain, erythrocytic	SPTB	177
P16157	Ankyrin-1	ANK1	94
P55072	Transitional endoplasmic reticulum ATPase	VCP	53
P111714	Protein 4.1	EPB41	52
P35579	Myosin-9	MYH9	44
Q8WUM4	Programmed cell death 6-interacting protein	PDCD6IP	43
P02730	Band 3 anion transport protein	SLC4A1	43
P16452	Erythrocyte membrane protein band 4.2	EPB42	42
P04040	Catalase	CAT	40
A0A087WVQ6	Clathrin heavy chain; Clathrin heavy chain 1	CLTC	38
P35612	Beta-adducin	ADD2	37
P16157	Ankyrin-1	ANK1	36
Q14254	Flotillin-2	FLOT2	33
P20073	Annexin A7	ANXA7	31
O75955	Flotillin-1	FLOT1	30
P53396	ATP-citrate synthase	ACLY	30
P06753		TPM3	29
P49368	T-complex protein 1 subunit gamma	CCT3	29
P236344	Plasma membrane calcium-transporting ATPase 4	ATP2B4	28
P11142	Heat shock cognate 71 kDa protein	HSPA8	28
P60709	Actin, cytoplasmic 1; Actin, cytoplasmic 1, N-terminally processed	ACTB	28
Q5T4S7	E3 ubiquitin-protein ligase UBR4	UBR4	27
P78371	T-complex protein 1 subunit beta	CCT2	26
P28289	Tropomodulin-1	TMOD1	26
P50395	Rab GDP dissociation inhibitor beta	GDI2	25
P27105	Erythrocyte band 7 integral membrane protein	STOM	25
P68871	Hemoglobin subunit beta; LW-hemophrin-7; Spinorhin	HBB	25
P02730	Band 3 anion transport protein	SLC4A1	25
Q00013	55 kDa erythrocyte membrane protein	MPP1	24
J3KPS3	Fructose-bisphosphate aldolase; Fructose-bisphosphate aldolase A	ALDOA	24
P00352	Retinal dehydrogenase 1	ALDH1A1	24
Q86VP6	Cullin-associated NEDD8-dissociated protein 1	CAND1	24
P49327	Fatty acid synthase	FASN	24
Q13228	Selenium-binding protein 1	SELENBP1	22
P00915	Carbonic anhydrase 1	CA1	22
P50991	T-complex protein 1 subunit delta	CCT4	22

TABLE 4-continued

Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome <i>Homo sapiens</i> database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search.			
Protein ID	Protein names	Gene names	Razor + unique peptides
P50990	T-complex protein 1 subunit theta	CCT8	22
P32119	Peroxiredoxin-2	PRDX2	21
P50995	Annexin A11	ANXA11	21
P69905	Hemoglobin subunit alpha	HBA1	21
P00558	Phosphoglycerate kinase 1	PGK1	20
H7BXX9	ATP-binding cassette sub-family B member 6, mitochondrial	ABCB6	20
P08758	Annexin A5; Annexin	ANXA5	20
P09525	Annexin A4; Annexin	ANXA4	20
P07900	Heat shock protein HSP 90-alpha	HSP90AA1	20
P48643	T-complex protein 1 subunit epsilon	CCT5	20
P07384	Calpain-1 catalytic subunit	CAPN1	20
P22314	Ubiquitin-like modifier-activating enzyme 1	UBA1	19
P04406	Glyceraldehyde-3-phosphate dehydrogenase	GAPDH	19
Q08495	Dematin	DMTN	19
Q99832	T-complex protein 1 subunit eta	CCT7	19
P29144	Tripeptidyl-peptidase 2	TPP2	19
P30041	Peroxiredoxin-6	PRDX6	18
E9PM69	26S protease regulatory subunit 6A	PSMC3	18
P40227	T-complex protein 1 subunit zeta	CCT6A	18
P50570	Dynamin-2	DNM2	18
E7EQB2	Lactotransferrin; Lactoferricin-H; Kaliocin-1; Lactoferronin-A; Lactoferronin-B; Lactoferronin-C	LTF	18
P31948	Stress-induced-phosphoprotein 1	STIP1	18
E7ESC6	Exportin-7	XPO7	18
E7EV99	Alpha-adducin	ADD1	17
P62258	14-3-3 protein epsilon	YWHAE	17
P30613	Pyruvate kinase PKLR	PKLR	17
A0A0G2JIW1	Heat shock 70 kDa protein 1B; Heat shock 70 kDa protein 1A	HSPA1B; HSPA1A	17
Q16531	DNA damage-binding protein 1	DDB1	17
P11021	78 kDa glucose-regulated protein	HSPA5	17
F5H2F4	C-1-tetrahydrofolate synthase, cytoplasmic; Methylenetetrahydrofolate dehydrogenase	MTFHD1	17
P07195	L-lactate dehydrogenase B chain; L-lactate dehydrogenase	LDHB	16
P45974	Ubiquitin carboxyl-terminal hydrolase 5	USP5	16
Q43242	26S proteasome non-ATPase regulatory subunit 3	PSMD3	16
C9J0K6	Sorcin	SRI	16
P08133	Annexin A6; Annexin	ANXA6	16
Q13200	26S proteasome non-ATPase regulatory subunit 2	PSMD2	16
P23276	Kell blood group glycoprotein	KEL	16
P34932	Heat shock 70 kDa protein 4	HSPA4	16
A0A0A0MSI0	Peroxiredoxin-1	PRDX1	16
Q9Y230	RuvB-like 2	RUVBL2	16
Q5XPI4	E3 ubiquitin-protein ligase RNF123	RNF123	16
P68871	Hemoglobin subunit beta; LW-hemorphin-7; Spinorphin	HBB	15
P60174	Triosephosphate isomerase	TPI1	15
P00491	Purine nucleoside phosphorylase	PNP	15
C9JIF9	Acylamino-acid-releasing enzyme	APEH	15
H7BYY1	Tropomyosin alpha-1 chain	TPM1	15
P35998	26S protease regulatory subunit 7	PSMC2	15
P17987	T-complex protein 1 subunit alpha	TCP1	15
P09543	2,3-cyclic-nucleotide 3-phosphodiesterase	CNP	15
Q99460	26S proteasome non-ATPase regulatory subunit 1	PSMD1	15
Q9Y4E8	Ubiquitin carboxyl-terminal hydrolase 15	USP15	15
Q9C0C9	E2/E3 hybrid ubiquitin-protein ligase UBE2O	UBE2O	15
P26038	Moesin	MSN	15
P04083	Annexin A1; Annexin	ANXA1	14
P30043	Flavin reductase (NADPH)	BLVRB	14
P11166	Solute carrier family 2, facilitated glucose transporter member 1	SLC2A1	14
P00918	Carbonic anhydrase 2	CA2	14
P06733	Alpha-enolase	ENO1	14
Q5TDH0	Protein DDI1 homolog 2	DDI2	14
B0QZ18	Copine-1	CPNE1	14
O75326	Semaphorin-7A	SEMA7A	14
P05164	Myeloperoxidase	MPO	14
Q9Y265	RuvB-like 1	RUVBL1	14
P29401	Transketolase	TKT	14
I3LON3	Vesicle-fusing ATPase	NSF	14
Q4VB86	Protein 4.1	EPB41	14
P11277	Spectrin beta chain, erythrocytic	SPTB	14
P13716	Delta-aminolevulinic acid dehydratase	ALAD	13
P07738	Bisphosphoglycerate mutase	BPGM	13
P48506	Glutamate-cysteine ligase catalytic subunit	GCLC	13

TABLE 4-continued

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Protein ID	Protein names	Gene names	Razor + unique peptides
Q99816	Tumor susceptibility gene 101 protein	TSG101	13
O14818	Proteasome subunit alpha type-7	PSMA7	13
P23526	Adenosylhomocysteinase	AHCY	13
P61225	Ras-related protein Rap-2b	RAP2B	13
O00231	26S proteasome non-ATPase regulatory subunit 11	PSMD11	13
P11413	Glucose-6-phosphate 1-dehydrogenase	G6PD	13
P00338	L-lactate dehydrogenase A chain	LDHA	12
Q99808	Equilibrative nucleoside transporter 1	SLC29A1	12
A6NJA2	Ubiquitin carboxyl-terminal hydrolase; Ubiquitin carboxyl-terminal hydrolase 14	USP14	12
Q06323	Proteasome activator complex subunit 1	PSME1	12
P28074	Proteasome subunit beta type-5	PSMB5	12
B3KQV6	Serine/threonine-protein phosphatase 2A 65 kDa regulatory subunit A alpha isoform	PPP2R1A	12
Q14974	Importin subunit beta-1	KPNB1	12
P25786	Proteasome subunit alpha type-1; Proteasome subunit alpha type	PSMA1	12
Q86X55	Histone-arginine methyltransferase CARM1	CARM1	12
A6NG10	WW domain-binding protein 2	WBP2	12
P63092	Guanine nucleotide-binding protein G(s) subunit alpha	GNAS	12
P31939	Bifunctional purine biosynthesis protein PURH; Phosphoribosylaminoimidazolecarboxamide formyltransferase; IMP cyclohydrolase	ATIC	12
P52209	6-phosphogluconate dehydrogenase, decarboxylating	PGD	12
A0A087X0C8	Calpain-5	CAPN5	12
F8W9S7	GTPase-activating protein and VPS9 domain-containing protein 1	GAPVD1	12
P60842	Eukaryotic initiation factor 4A-1	EIF4A1	12
P69905	Hemoglobin subunit alpha	HBA1	11
Q6XQN6	Nicotinate phosphoribosyltransferase	NAPRT	11
P48637	Glutathione synthetase	GSS	11
H7BZ94	Protein disulfide-isomerase	P4HB	11
P21980	Protein-glutamine gamma-glutamyltransferase 2	TGM2	11
P50895	Basal cell adhesion molecule	BCAM	11
A0A087X2I1	26S protease regulatory subunit 10B	PSMC6	11
G3V1D3	Dipeptidyl peptidase 3	DPP3	11
P05023	Sodium/potassium-transporting ATPase subunit alpha-1; Sodium/potassium-transporting ATPase subunit alpha-3	ATP1A1; ATP1A3	11
P40925	Malate dehydrogenase, cytoplasmic; Malate dehydrogenase	MDH1	11
Q9UKV8	Protein argonaute-2	AGO2	11
P30566	Adenylosuccinate lyase	ADSL	11
P20618	Proteasome subunit beta type-1	PSMB1	11
P17858	ATP-dependent 6-phosphofructokinase, liver type	PFKL	11
A0A087X253	AP-2 complex subunit beta	AP2B1	11
O95782	AP-2 complex subunit alpha-1	AP2A1	11
O00232	26S proteasome non-ATPase regulatory subunit 12	PSMD12	11
Q9BSL1	Ubiquitin-associated domain-containing protein 1	UBAC1	11
A0A087WUL0	Bifunctional ATP-dependent dihydroxyacetone kinase/FAD-AMP lyase (cyclizing); ATP-dependent dihydroxyacetone kinase; FAD-AMP lyase (cyclizing)	TKFC; DAK	11
P69891	Hemoglobin subunit gamma-1	HBG1	11
A0A087WZE4	Spectrin alpha chain, erythrocytic 1	SPTA1	11
P48426	Phosphatidylinositol 5-phosphate 4-kinase type-2 alpha	PIP4K2A	10
P10644	cAMP-dependent protein kinase type I-alpha regulatory subunit	PRKAR1A	10
Q9BWD1	Acetyl-CoA acetyltransferase, cytosolic	ACAT2	10
P62191	26S protease regulatory subunit 4	PSMC1	10
E9PBS1	Multifunctional protein ADE2; Phosphoribosylaminoimidazole-succinocarboxamide synthase; Phosphoribosylaminoimidazole carboxylase	PAICS	10
M0R0Y2	Alpha-soluble NSF attachment protein	NAPA	10
P78417	Glutathione S-transferase omega-1	GSTO1	10
P25789	Proteasome subunit alpha type-4; Proteasome subunit alpha type; Proteasome subunit beta type	PSMA4	10
Q9H0U4	Ras-related protein Rab-1B; Putative Ras-related protein Rab-1C	RAB1B; RAB1C	10
Q16401	26S proteasome non-ATPase regulatory subunit 5	PSMD5	10
P30101	Protein disulfide-isomerase A3	PDIA3	10
D6RAX7	COP9 signalosome complex subunit 4	COPS4	10
O75340	Programmed cell death protein 6	PDCD6	10
Q96P70	Importin-9	IPO9	10
P38606	V-type proton ATPase catalytic subunit A	ATP6V1A	10
A0A0G2JH68	Protein diaphanous homolog 1	DIAPH1	10
Q5T9B7	Adenylate kinase isoenzyme 1	AK1	10
O14980	Exportin-1	XPO1	10
P43686	26S protease regulatory subunit 6B	PSMC4	10
H0YH81	ATP synthase subunit beta; ATP synthase subunit beta, mitochondrial	ATP5B	10
Q04656	Copper-transporting ATPase 1	ATP7A	10
P16452	Erythrocyte membrane protein band 4.2	EPB42	10
Q9BY43	Charged multivesicular body protein 4a	CHMP4A	9

TABLE 4-continued

Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome <i>Homo sapiens</i> database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search.			
Protein ID	Protein names	Gene names	Razor + unique peptides
P51148	Ras-related protein Rab-5C	RAB5C	9
Q9NRV9	Heme-binding protein 1	HEBP1	9
Q9UNZ2	NSFL1 cofactor p47	NSFL1C	9
Q16851	UTP--glucose-1-phosphate uridylyltransferase	UGP2	9
P28066	Proteasome subunit alpha type-5	PSMA5	9
A0A0C4DGQ5	Calpain small subunit 1	CAPNS1	9
A0A087X1Z3	Proteasome activator complex subunit 2	PSME2	9
Q01518	Adenylyl cyclase-associated protein 1	CAP1	9
B1AKQ8	Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-1	GNB1	9
O75131	Copine-3	CPNE3	9
P54725	UV excision repair protein RAD23 homolog A	RAD23A	9
P11215	Integrin alpha-M	ITGAM	9
Q93008	Probable ubiquitin carboxyl-terminal hydrolase FAF-X	USP9X	9
Q96G03	Phosphoglucomutase-2	PGM2	9
P49721	Proteasome subunit beta type-2	PSMB2	9
Q15008	26S proteasome non-ATPase regulatory subunit 6	PSMD6	9
Q9UNQ0	ATP-binding cassette sub-family G member 2	ABCG2	9
P22303	Acetylcholinesterase; Carboxylic ester hydrolase	ACHE	9
G3V5Z7	Proteasome subunit alpha type; Proteasome subunit alpha type-6	PSMA6	9
O15439	Multidrug resistance-associated protein 4	ABCC4	9
P37837	Transaldolase	TALDO1	9
O14744	Protein arginine N-methyltransferase 5	PRMT5	9
P02042	Hemoglobin subunit delta	HBD	9
P25788	Proteasome subunit alpha type-3	PSMA3	8
J3QS39	Ubiquitin-60S ribosomal protein L40; Ubiquitin; 60S ribosomal protein L40; Ubiquitin-40S ribosomal protein S27a; Ubiquitin; 40S ribosomal protein S27a; Polyubiquitin-B; Ubiquitin; Polyubiquitin-C; Ubiquitin	UBB; RPS27A; UBC; UBA52; UBBP4	8
O94919	Endonuclease domain-containing 1 protein	ENDOD1	8
P31946	14-3-3 protein beta/alpha	YWHA8	8
P60891	Ribose-phosphate pyrophosphokinase 1; Ribose-phosphate pyrophosphokinase 2; Ribose-phosphate pyrophosphokinase 3	PRPS1; PRPS2; PRPS1L1	8
P62195	26S protease regulatory subunit 8	PSMC5	8
A0A024RA52	Proteasome subunit alpha type; Proteasome subunit alpha type-2	PSMA2	8
P04899	Guanine nucleotide-binding protein G(i) subunit alpha-2	GNAI2	8
Q13561	Dynactin subunit 2	DCTN2	8
P53004	Biliverdin reductase A	BLVRA	8
P00387	NADH-cytochrome b5 reductase 3	CYB5R3	8
P16152	Carbonyl reductase [NADPH] 1	CBR1	8
P06744	Glucose-6-phosphate isomerase	GPI	8
Q99733	Nucleosome assembly protein 1-like 4	NAP1L4	8
P04792	Heat shock protein beta-1	HSPB1	8
P17612	cAMP-dependent protein kinase catalytic subunit alpha; cAMP-dependent protein kinase catalytic subunit beta	PRKACA; KIN27; PRKACB	8
P20340	Ras-related protein Rab-6A	RAB6A	8
P13796	Plastin-2	LCP1	8
P52907	F-actin-capping protein subunit alpha-1	CAPZA1	8
Q14697	Neutral alpha-glucosidase AB	GANAB	8
P08514	Integrin alpha-IIb; Integrin alpha-IIb heavy chain; Integrin alpha-IIb light chain, form 1; Integrin ITGA2B alpha-IIb light chain, form 2	ITGA2B	8
P26641	Elongation factor 1-gamma	EEF1G	8
Q9UQ80	Proliferation-associated protein 2G4	PA2G4	8
Q9Y4D1	Disheveled-associated activator of morphogenesis 1	DAAM1	8
P11166	Solute carrier family 2, facilitated glucose transporter member 1	SLC2A1	8
P63261	Actin, cytoplasmic 2; Actin, cytoplasmic 2, N-terminally processed; Actin, cytoplasmic 1; Actin, cytoplasmic 1, N-terminally processed; Actin, gamma-enteric smooth muscle; Actin, alpha skeletal muscle; Actin, alpha cardiac muscle 1; Actin, aortic smooth muscle	ACTG1; ACTB; ACTG2; ACTA1; ACTC1; ACTA2	8
P02008	Hemoglobin subunit zeta	HBZ	8
P20073	Annexin A7	ANXA7	8
P35613	Basigin	BSG	7
P49720	Proteasome subunit beta type-3	PSMB3	7
P17931	Galectin-3; Galectin	LGALS3	7
P63104	14-3-3 protein zeta/delta	YWHAZ	7
Q9H444	Charged multivesicular body protein 4b	CHMP4B	7
O43396	Thioredoxin-like protein 1	TXNL1	7
P50502	Hsc70-interacting protein; Putative protein FAM10A4; Putative protein FAM10A5	ST13; ST13P4; ST13P5	7
E7EQ12	Calpastatin	CAST	7
P49189	4-trimethylaminobutyraldehyde dehydrogenase	ALDH9A1	7
Q9Y3I1	F-box only protein 7	FBXO7	7
P07954	Fumarate hydratase, mitochondrial	FH	7

TABLE 4-continued

Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome <i>Homo sapiens</i> database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search.			
Protein ID	Protein names	Gene names	Razor + unique peptides
F6S8N6	Protein-L-isoaspartate O-methyltransferase; Protein-L-isoaspartate(D-aspartate) O-methyltransferase	PCMT1	7
P49247	Ribose-5-phosphate isomerase	RPIA	7
P62834	Ras-related protein Rap-1A	RAP1A	7
A0A087WUQ6	Glutathione peroxidase; Glutathione peroxidase 1	GPX1	7
O60256	Phosphoribosyl pyrophosphate synthase-associated protein 2	PRPSAP2	7
Q00796	Sorbitol dehydrogenase	SORD	7
Q00299	Chloride intracellular channel protein 1	CLIC1	7
Q99497	Protein deglycase DJ-1	PARK7	7
P17174	Aspartate aminotransferase, cytoplasmic	GOT1	7
H7BXD5	Grancalcin	GCA	7
E9PGT1	Translin	TSN	7
C9J7K9	Phospholipid scramblase 1	PLSCR1	7
Q13618	Cullin-3	CUL3	7
O75695	Protein XRP2	RP2	7
P09960	Leukotriene A-4 hydrolase	LTA4H	7
E9PLK3	Puromycin-sensitive aminopeptidase	NPEPPS	7
P00492	Hypoxanthine-guanine phosphoribosyltransferase	HPRT1	7
F5H4B6	Aldehyde dehydrogenase family 16 member A1	ALDH16A1	7
D6RA82	Annexin; Annexin A3	ANXA3	7
P61106	Ras-related protein Rab-14	RAB14	7
Q16775	Hydroxyacylglutathione hydrolase, mitochondrial	HAGH	7
P07355	Annexin A2; Annexin; Putative annexin A2-like protein	ANXA2; ANXA2P2	7
A0A087WX08	Gamma-adducin	ADD3	7
P08238	Heat shock protein HSP 90-beta	HSP90AB1	7
Q8LZY2	ATP-binding cassette sub-family A member 7	ABCA7	7
P14780	Matrix metalloproteinase-9	MMP9	7
P12955	Xaa-Pro dipeptidase	PEPD	7
P68371	Tubulin beta-4B chain; Tubulin beta-4A chain; Tubulin beta chain	TUBB4B; TUBB4A; TUBB	7
O15067	Phosphoribosylformylglycinamide synthase	PFAS	7
P35241	Radixin	RDX	7
O60488	Long-chain-fatty-acid--CoA ligase 4	ACSL4	7
A0A0C4DGX4	Cullin-1	CUL1	7
P50148	Guanine nucleotide-binding protein G(q) subunit alpha	GNAQ	7
Q15907	Ras-related protein Rab-11B; Ras-related protein Rab-11A	RAB11B; RAB11A	7
P30086	Phosphatidylethanolamine-binding protein 1; Hippocampal cholinergic neurostimulating peptide	PEBP1	6
P55036	26S proteasome non-ATPase regulatory subunit 4	PSMD4	6
P28070	Proteasome subunit beta type-4	PSMB4	6
Q9UN52	COP9 signalosome complex subunit 3	COPS3	6
P08754	Guanine nucleotide-binding protein G(k) subunit alpha	GNAI3	6
P61006	Ras-related protein Rab-8A	RAB8A	6
P61019	Ras-related protein Rab-2A	RAB2A	6
H0Y8C6	Importin-5	IPO5	6
Q00577	Transcriptional activator protein Pur-alpha	PURA	6
P52565	Rho GDP-dissociation inhibitor 1	ARHGDI1A	6
Q9Y5Z4	Heme-binding protein 2	HEBP2	6
J3KNF4	Copper chaperone for superoxide dismutase; Superoxide dismutase [Cu—Zn]	CCS	6
A0A087WXS7	ATPase ASNA1	ASNA1	6
P05089	Arginase-1	ARG1	6
O95336	6-phosphogluconolactonase	PGLS	6
Q92508	Piezo-type mechanosensitive ion channel component 1	PIEZO1	6
P84077	ADP-ribosylation factor 1; ADP-ribosylation factor 3	ARF1; ARF3	6
P21281	V-type proton ATPase subunit B, brain isoform; V-type proton ATPase subunit B, kidney isoform	ATP6V1B2; ATP6V1B1	6
P04259	Keratin, type II cytoskeletal 6B	KRT6B	6
Q92905	COP9 signalosome complex subunit 5	COPS5	6
P61163	Alpha-centractin	ACTR1A	6
O95373	Importin-7	IPO7	6
C9JD73	Protein phosphatase 1 regulatory subunit 7	PPP1R7	6
Q99536	Synaptic vesicle membrane protein VAT-1 homolog	VAT1	6
Q86UX7	Fermitin family homolog 3	FERMT3	6
C9JFE4	COP9 signalosome complex subunit 1	GPS1	6
P36959	GMP reductase 1	GMPR	6
B5MDF5	GTP-binding nuclear protein Ran	RAN	6
F5GY90	Porphobilinogen deaminase	HMBS	6
E7EX90	Dynactin subunit 1	DCTN1	6

TABLE 4-continued

Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome <i>Homo sapiens</i> database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search.			
Protein ID	Protein names	Gene names	Razor + unique peptides
H0Y512	Adipocyte plasma membrane-associated protein	APMAP	6
Q32Q12	Nucleoside diphosphate kinase; Nucleoside diphosphate kinase B; Putative nucleoside diphosphate kinase	NME1-NME2; NME2; NME1; NME2P1	6
P13807	Glycogen [starch] synthase, muscle	GYS1	6
K7ES02	Bleomycin hydrolase	BLMH	6
P17213	Bactericidal permeability-increasing protein	BPI	6
Q9UNM6	26S proteasome non-ATPase regulatory subunit 13	PSMD13	6
J3KQ32	Obg-like ATPase 1	OLA1	6
Q9Y490	Talin-1	TLN1	6
H0YD13	CD44 antigen	CD44	6
P18669	Phosphoglycerate mutase 1; Phosphoglycerate mutase 2; Probable phosphoglycerate mutase 4	PGAM1; PGAM2; PGAM4	6
P23528	Cofilin-1	CFL1	6
Q5SR44	Complement receptor type 1	CR1	6
Q99436	Proteasome subunit beta type-7	PSMB7	6
P47756	F-actin-capping protein subunit beta	CAPZB	6
P30740	Leukocyte elastase inhibitor	SERPINB1	6
A0A024R571	EH domain-containing protein 1	EHD1	6
P30043	Flavin reductase (NADPH)	BLVRB	6
P32119	Peroxisomal oxidin-2	PRDX2	6
C9J0K6	Sorcin	SRI	6
P27105	Erythrocyte band 7 integral membrane protein	STOM	6
P84077	ADP-ribosylation factor 1; ADP-ribosylation factor 3; ADP-ribosylation factor 5; ADP-ribosylation factor 4	ARF1; ARF3; ARF5; ARF4	6
P61981	14-3-3 protein gamma; 14-3-3 protein gamma, N-terminally processed	YWHAQ	5
P09211	Glutathione S-transferase P	GSTP1	5
P13489	Ribonuclease inhibitor	RNH1	5
Q96PU5	E3 ubiquitin-protein ligase NEDD4-like	NEDD4L	5
Q5SRN7	HLA class I histocompatibility antigen, A; HLA class I histocompatibility antigen, B; HLA class I histocompatibility antigen, Cw	HLA-A; HLA-C; HLA-B	5
P02042	Hemoglobin subunit delta	HBD	5
A0A087WU29	Glycophorin-A	GYPA	5
P00390	Glutathione reductase, mitochondrial	GSR	5
Q9UBV8	Peflin	PEF1	5
Q8WVM8	Sec1 family domain-containing protein 1	SCFD1	5
F6TLX2	Glyoxalase domain-containing protein 4	GLOD4	5
H3BQF1	Adenine phosphoribosyltransferase	APRT	5
Q07960	Rho GTPase-activating protein 1	ARHGAP1	5
F6XSS0	Blood group Rh(CE) polypeptide; Blood group Rh(D) polypeptide	RHCE; RHD	5
A0A087WY55	Vacuolar protein sorting-associated protein VTA1 homolog	VTA1	5
G5E9W8	Glycogenin-1	GYG1	5
P07451	Carbonic anhydrase 3	CA3	5
Q08722	Leukocyte surface antigen CD47	CD47	5
X6RA14	S-formylglutathione hydrolase	ESD	5
Q8IU18	Cytokine receptor-like factor 3	CRLF3	5
Q5VW32	BRO1 domain-containing protein BROX	BROX	5
P61026	Ras-related protein Rab-10	RAB10	5
A0A087WWY3	Filamin-A	FLNA	5
H0YGX7	Rho GDP-dissociation inhibitor 2	ARHGDIB	5
Q92783	Signal transducing adapter molecule 1	STAM	5
Q7Z6Z7	E3 ubiquitin-protein ligase HUWE1	HUWE1	5
H0YHC3	Nucleosome assembly protein 1-like 1	NAP1L1	5
Q5QPM7	Proteasome inhibitor PI31 subunit	PSMF1	5
P09104	Gamma-enolase; Enolase	ENO2	5
Q5T2B5	Cullin-2	CUL2	5
Q8WW22	DnaJ homolog subfamily A member 4	DNAJA4	5
P61201	COP9 signalosome complex subunit 2	COPS2	5
X6R433	Protein-tyrosine-phosphatase; Receptor-type tyrosine-protein phosphatase C	PTPRC	5
P63000	Ras-related C3 botulinum toxin substrate 1	RAC1	5
F5GXM3	IST1 homolog	IST1	5
H3BLU7	Aflatoxin B1 aldehyde reductase member 2	AKR7A2	5
P25325	3-mercaptopyruvate sulfurtransferase; Sulfurtransferase	MPST	5
A0A087X0K1	Calcium-binding protein 39	CAB39	5
P23381	Tryptophan--tRNA ligase, cytoplasmic; T1-TrpRS; T2-TrpRS	WARS	5
P01116	GTPase KRas; GTPase KRas, N-terminally processed	KRAS	5
P30040	Endoplasmic reticulum resident protein 29	ERP29	5
P05198	Eukaryotic translation initiation factor 2 subunit 1	EIF2S1	5
Q6UX06	Olfactomedin-4	OLFM4	5
Q96KP4	Cytosolic non-specific dipeptidase	CNDP2	5
Q04760	Lactoylglutathione lyase	GLO1	5

TABLE 4-continued

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Protein ID	Protein names	Gene names	Razor + unique peptides
J3QS39	Ubiquitin-60S ribosomal protein L40; Ubiquitin; 60S ribosomal protein L40; Ubiquitin-40S ribosomal protein S27a; Ubiquitin; 40S ribosomal protein S27a; Polyubiquitin-B; Ubiquitin; Polyubiquitin-C; Ubiquitin	UBB; RPS27A; UBC; UBA52; UBBP4	5
P06702	Protein S100-A9	S100A9	5
P15531	Nucleoside diphosphate kinase A; Nucleoside diphosphate kinase; Nucleoside diphosphate kinase B	NME1; NME2; NME1-NME2	5
E7EV99	Alpha-adducin	ADD1	5
P51149	Ras-related protein Rab-7a	RAB7A	4
K7N7A8	Aquaporin-1	AQP1	4
P61020	Ras-related protein Rab-5B	RAB5B	4
X6R4N5	Erythroid membrane-associated protein	ERMAP	4
E5RJR5	S-phase kinase-associated protein 1	SKP1	4
Q9Y315	Deoxyribose-phosphate aldolase	DERA	4
X6R8F3	Neutrophil gelatinase-associated lipocalin	LCN2	4
O75396	Vesicle-trafficking protein SEC22b	SEC22B	4
Q15102	Platelet-activating factor acetylhydrolase IB subunit gamma	PAFAH1B3	4
P51665	26S proteasome non-ATPase regulatory subunit 7	PSMD7	4
Q96FZ7	Charged multivesicular body protein 6	CHMP6	4
Q9NRQ2	Phospholipid scramblase 4	PLSCR4	4
F8VWS0	60S acidic ribosomal protein P0; 60S acidic ribosomal protein P0-like	RPLP0; RPLP0P6	4
O14964	Hepatocyte growth factor-regulated tyrosine kinase substrate	HGS	4
A0A0A0MTJ9	Neutral cholesterol ester hydrolase 1	NCEH1	4
A0A087WY82	Junctional adhesion molecule A	F11R	4
J3QSB7	Purine nucleoside phosphorylase; S-methyl-5-thioadenosine phosphorylase	MTAP	4
Q5WQ6	Ubiquitin thioesterase OTU1	YOD1	4
P36543	V-type proton ATPase subunit E 1	ATP6V1E1	4
Q9BS40	Latexin	LXN	4
P47755	F-actin-capping protein subunit alpha-2	CAPZA2	4
Q9GZT8	NIF3-like protein 1	NIF3L1	4
H6UYS7	Alpha-synuclein	SNCA	4
P60953	Cell division control protein 42 homolog	CDC42	4
Q14773	Intercellular adhesion molecule 4	ICAM4	4
H0Y6T7	Nicastrin	NCSTN	4
P69891	Hemoglobin subunit gamma-1	HBG1	4
P14625	Endoplasmic	HSP90B1	4
Q96GD0	Pyridoxal phosphate phosphatase	PDXP	4
P08311	Cathepsin G	CTSG	4
Q9H9Q2	COP9 signalosome complex subunit 7b	COPS7B	4
P09417	Dihydropteridine reductase	QDPR	4
F8WE6	Peptidyl-prolyl cis-trans isomerase; Peptidyl-prolyl cis-trans isomerase A	PPIA	4
O15173	Membrane-associated progesterone receptor component 2	PGRMC2	4
P28072	Proteasome subunit beta type-6; Proteasome subunit beta type	PSMB6	4
H3BSW0	Leucine-rich repeat-containing protein 57	LRRCS7	4
Q9UBW8	COP9 signalosome complex subunit 7a	COPS7A	4
O00560	Syntenin-1	SDCBP	4
F5H157	Ras-related protein Rab-35	RAB35	4
Q9H479	Fructosamine-3-kinase	FN3K	4
Q04917	14-3-3 protein eta	YWHAH	4
C9IJ47	AP-2 complex subunit mu	AP2M1	4
C9JIG9	Serine/threonine-protein kinase OSR1	OSR1	4
Q13336	Urea transporter 1	SLC14A1	4
P46926	Glucosamine-6-phosphate isomerase 1; Glucosamine-6-phosphate isomerase; Glucosamine-6-phosphate isomerase 2	GNPDA1; GNPDA2	4
O43633	Charged multivesicular body protein 2a	CHMP2A	4
F8VVB9	Tubulin alpha-1A chain; Tubulin alpha-1C chain; Tubulin alpha-1B chain; Tubulin alpha-3C/D chain; Tubulin alpha-3E chain	TUBA1B; TUBA1C; TUBA1A; TUBA3C; TUBA3E	4
A6PVN5	Serine/threonine-protein phosphatase 2A activator	PPP2R4	4
B8ZZB8	CB1 cannabinoid receptor-interacting protein 1	CNRIP1	4
Q9P2R3	Rabankyrin-5	ANKFY1	4
Q86Y87	C2 domain-containing protein 5	C2CD5	4
R4GMR5	26S proteasome non-ATPase regulatory subunit 8	PSMD8	4
E7EM64	COP9 signalosome complex subunit 6	COPS6	4
J3KNI6	Integrin beta; Integrin beta-2	ITGB2	4
H0Y5R6	Uroporphyrinogen decarboxylase	UROD	4
M0R165	Epidermal growth factor receptor substrate 15-like 1	EPS15L1	4
O00487	26S proteasome non-ATPase regulatory subunit 14	PSMD14	4
Q13630	GDP-L-fucose synthase	TSTA3	4
P55060	Exportin-2	CSE1L	4
P20020	Plasma membrane calcium-transporting ATPase 1; Calcium-transporting ATPase	ATP2B1	4

TABLE 4-continued

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Protein ID	Protein names	Gene names	Razor + unique peptides
Q9NYU2	UDP-glucose: glycoprotein glucosyltransferase 1	UGGT1	4
H3BND8	Ubiquitin carboxyl-terminal hydrolase; Ubiquitin carboxyl-terminal hydrolase 7	USP7	4
Q9GZP4	PITH domain-containing protein 1	PITHD1	4
D6RD66	WD repeat-containing protein 1	WDR1	4
P48729	Casein kinase I isoform alpha	CSNK1A1	4
P25685	DnaJ homolog subfamily B member 1	DNAJB1	4
P14550	Alcohol dehydrogenase [NADP(+)]	AKR1A1	4
Q6PCE3	Glucose 1,6-bisphosphate synthase	PGM2L1	4
Q9UPN7	Serine/threonine-protein phosphatase 6 regulatory subunit 1	PPP6R1	4
P62805	Histone H4	HIST1H4A	4
P62937	Peptidyl-prolyl cis-trans isomerase A	PPIA	4
P30046	D-dopachrome decarboxylase; D-dopachrome decarboxylase-like protein	DDT; DDTL	4
Q08495	Dematin	DMTN	4
K7EIJ0	WW domain-binding protein 2	WBP2	4
E5RHP7	Carbonic anhydrase 1	CA1	4
P09105	Hemoglobin subunit theta-1	HBQ1	4
E7ESC6	Exportin-7	XPO7	4
P61225	Ras-related protein Rap-2b; Ras-related protein Rap-2c; Ras-related protein Rap-2a	RAP2B; RAP2A; RAP2C	4
O15400	Syntaxin-7	STX7	3
Q5VZR0	Golgi-associated plant pathogenesis-related protein 1	GLIPR2	3
P40199	Carcinoembryonic antigen-related cell adhesion molecule 6	CEACAM6	3
P27348	14-3-3 protein theta	YWHAQ	3
P62820	Ras-related protein Rab-1A	RAB1A	3
C9JEN3	Protein lifeguard 3	TMBIM1	3
Q9UDX3	SEC14-like protein 4	SEC14L4	3
Q9Y570	Protein phosphatase methyltransferase 1	PPME1	3
Q96GG9	DCN1-like protein 1; DCN1-like protein	DCUN1D1	3
B0YJC4	Vimentin	VIM	3
Q86VN1	Vacuolar protein-sorting-associated protein 36	VPS36	3
F8WFB9	Endophilin-B2	SH3GLB2	3
A0A087WVQ9	Elongation factor 1-alpha 1; Putative elongation factor 1-alpha-like 3	EEF1A1; EEF1A1P5	3
Q53TN4	Cytochrome b reductase 1	CYBRD1	3
P10809	60 kDa heat shock protein, mitochondrial	HSPD1	3
F5H7X1	26S proteasome non-ATPase regulatory subunit 9	PSMD9	3
E9PNW4	CD59 glycoprotein	CD59	3
E9PS74	Solute carrier family 43 member 3	SLC43A3	3
P53985	Monocarboxylate transporter 1	SLC16A1	3
Q8WWI5	Choline transporter-like protein 1	SLC44A1	3
E9PIR7	Thioredoxin reductase 1, cytoplasmic	TXNRD1	3
Q5TD07	Ribosyl-dihydroxynicotinamide dehydrogenase [quinone]	NQO2	3
Q9UL25	Ras-related protein Rab-21	RAB21	3
P27824	Calnexin	CANX	3
U3KPS2	Myeloblastin	PRTN3	3
Q9UKU0	Long-chain-fatty-acid--CoA ligase 6	ACSL6	3
P24666	Low molecular weight phosphotyrosine protein phosphatase	ACP1	3
H7C2G2	NAD(P)(+)-arginine ADP-ribosyltransferase; Ecto-ADP-ribosyltransferase 4	ART4	3
I3L1K6	Myosin light chain 4	MYL4	3
Q9UK41	Vacuolar protein sorting-associated protein 28 homolog	VPS28	3
Q9NUQ9	Protein FAM49B	FAM49B	3
Q9UBQ7	Glyoxylate reductase/hydroxypyruvate reductase	GRHPR	3
B8ZZG1	MAGUK p55 subfamily member 6	MPP6	3
Q10567	AP-1 complex subunit beta-1	AP1B1	3
O75387	Large neutral amino acids transporter small subunit 3	SLC43A1	3
Q9BTU6	Phosphatidylinositol 4-kinase type 2-alpha	PI4K2A	3
J3KS22	L-xylulose reductase	DCXR	3
O15498	Synaptobrevin homolog YKT6	YKT6	3
Q08211	ATP-dependent RNA helicase A	DHX9	3
P50416	Carnitine O-palmitoyltransferase 1, liver isoform	CPT1A	3
P08237	ATP-dependent 6-phosphofructokinase, muscle type	PFKM	3
P14735	Insulin-degrading enzyme	IDE	3
Q9H0R3	Transmembrane protein 222	TMEM222	3
Q14166	Tubulin-tyrosine ligase-like protein 12	TLL12	3
Q14558	Phosphoribosyl pyrophosphate synthase-associated protein 1	PRPSAP1	3
A0A087WTB8	Ubiquitin carboxyl-terminal hydrolase; Ubiquitin carboxyl-terminal hydrolase isozyme L3	UCHL3	3
Q9NPQ8	Synembryn-A	RIC8A	3
F6WQW2	Ran-specific GTPase-activating protein	RANBP1	3
Q15691	Microtubule-associated protein RP/EB family member 1	MAPRE1	3
A0A0A0MR50	Cullin-4A	CUL4A	3
Q96IU4	Alpha/beta hydrolase domain-containing protein 14B	ABHD14B	3

TABLE 4-continued

Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome <i>Homo sapiens</i> database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search.			
Protein ID	Protein names	Gene names	Razor + unique peptides
Q9P0L0	Vesicle-associated membrane protein-associated protein A	VAPA	3
E9PRY8	Elongation factor 1-delta	EEF1D	3
Q16543	Hsp90 co-chaperone Cdc37	CDC37	3
P06702	Protein S100-A9	S100A9	3
P13639	Elongation factor 2	EEF2	3
E9PJC7	Tetraspanin; CD82 antigen	CD82	3
Q01432	AMP deaminase 3	AMPD3	3
B1AUU8	Epidermal growth factor receptor substrate 15	EPS15	3
P54709	Sodium/potassium-transporting ATPase subunit beta-3	ATP1B3	3
P54727	UV excision repair protein RAD23 homolog B	RAD23B	3
P36507	Dual specificity mitogen-activated protein kinase kinase 2	MAP2K2	3
Q9Y2V2	Calcium-regulated heat stable protein 1	CARHSP1	3
P31146	Coronin-1A; Coronin	CORO1A	3
Q8IY17	Neuropathy target esterase	PNPLA6	3
E7EQR4	Ezrin	EZR	3
F8WDS9	LanC-like protein 1	LANCL1	3
X6RJP6	Transgelin-2	TAGLN2	3
A0A0J9YXM6	WD repeat-containing protein 81	WDR81	3
P62140	Serine/threonine-protein phosphatase PP1-beta catalytic subunit; Serine/threonine-protein phosphatase	PPP1CB	3
H0YLJ3	Mortality factor 4-like protein 1	MORF4L1	3
E7EPV7	Alpha-synuclein	SNCA	3
K7EK07	Histone H3	H3F3B	3
P05109	Protein S100-A8; Protein S100-A8, N-terminally processed	S100A8	3
I3L0A0	Ubiquitin-conjugating enzyme E2 variant 1; Ubiquitin-conjugating enzyme E2 variant 2	TMEM189-UBE2V1; UBE2V1; UBE2V2	3
Q6B0K9	Hemoglobin subunit mu	HBM	3
O75368	SH3 domain-binding glutamic acid-rich-like protein	SH3BGRL	3
A0A0J9YXB3	Ras-related protein Rap-1b; Ras-related protein Rap-1A; Ras-related protein Rap-1b-like protein	RAP1B; RAP1A	3
P60953	Cell division control protein 42 homolog	CDC42	3
A0A087WTI1	Ras-related protein Rab-1B; Ras-related protein Rab-1A	RAB1B; RAB1A	3
B5MDF5	GTP-binding nuclear protein Ran	RAN	3
P61088	Ubiquitin-conjugating enzyme E2 N	UBE2N	3
Q00013	55 kDa erythrocyte membrane protein	MPP1	3
B1AKQ8	Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-1; Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-3; Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-2	GNB1; GNB2; GNB3	3
P04406	Glyceraldehyde-3-phosphate dehydrogenase	GAPDH	3
Q9UJC5	SH3 domain-binding glutamic acid-rich-like protein 2	SH3BGRL2	3
P04921	Glycophorin-C	GYPC	2
Q9NP59	Solute carrier family 40 member 1	SLC40A1	2
K7EKH5	Fructose-bisphosphate aldolase; Fructose-bisphosphate aldolase C	ALDOC	2
P11233	Ras-related protein Ral-A	RALA	2
F8WBR5	Calmodulin	CALM2; CALM3; CALM1	2
P00441	Superoxide dismutase [Cu—Zn]	SOD1	2
H0YDI1	Lymphocyte function-associated antigen 3	CD58	2
H0YNE9	Ras-related protein Rab-8B	RAB8B	2
J3KN67		TPM3	2
P20160	Azurocidin	AZU1	2
A0A087WZZ4	Ammonium transporter Rh type A	RHAG	2
F5GXS0	Complement C4-A; Complement C4-B	C4B; C4A	2
Q8ND76	Cyclin-Y	CCNY	2
Q14739	Lamin-B receptor	LBR	2
Q15181	Inorganic pyrophosphatase	PPA1	2
J3QS92	Galectin-9	LGALS9	2
I3L471	Phosphatidylinositol transfer protein alpha isoform	PITPNA	2
P01111	GTPase NRas	NRAS; KRAS	2
F5H4Q5	Vacuolar protein sorting-associated protein 37C	VPS37C	2
A0A0A0MSW4	Phosphatidylinositol transfer protein beta isoform	PITPNB	2
P67775	Serine/threonine-protein phosphatase 2A catalytic subunit alpha isoform; Serine/threonine-protein phosphatase	PPP2CA; PPP2CB	2
P41091	Eukaryotic translation initiation factor 2 subunit 3; Putative eukaryotic translation initiation factor 2 subunit 3-like protein	EIF2S3; EIF2S3L	2
A0A087WWS7	Syntaxin-binding protein 2	STXBP2	2
F8VQX6	Methyltransferase-like protein 7A	METTL7A	2
B3KT28	FAS-associated factor 1	FAF1	2
K7EIJ8	Katanin p60 ATPase-containing subunit A-like 2	KATNAL2	2

TABLE 4-continued

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Protein ID	Protein names	Gene names	Razor + unique peptides
P20339	Ras-related protein Rab-5A	RAB5A	2
O95456	Proteasome assembly chaperone 1	PSMG1	2
K7EK45	Polypyrimidine tract-binding protein 1	PTBP1	2
Q15365	Poly(rC)-binding protein 1; Poly(rC)-binding protein 3	PCBP1; PCBP3	2
Q9BSJ8	Extended synaptotagmin-1	ESYT1	2
H3BP35	Diphosphomevalonate decarboxylase	MVD	2
H0Y8C4	Serine/threonine-protein phosphatase 2A 56 kDa regulatory subunit delta isoform	PPP2R5D	2
K7EP09	Bifunctional coenzyme A synthase; Phosphopantetheine adenylyltransferase; Dephospho-CoA kinase	COASY	2
F8VNT9	Tetraspanin; CD63 antigen	CD63	2
Q8NEV1	Casein kinase II subunit alpha 3; Casein kinase II subunit alpha	CSNK2A3; CSNK2A1	2
F8VTQ5	Heterogeneous nuclear ribonucleoprotein A1	HNRNPA1	2
A0A087X2E2	Carcinoembryonic antigen-related cell adhesion molecule 8; Carcinoembryonic antigen-related cell adhesion molecule 1; Carcinoembryonic antigen-related cell adhesion molecule 5	CEACAM8; CEACAM5; CEACAM1	2
H0YAS8	Clusterin; Clusterin beta chain; Clusterin alpha chain; Clusterin	CLU	2
E9PP54	Tubulin-specific chaperone cofactor E-like protein	TBCEL	2
P43034	Platelet-activating factor acetylhydrolase IB subunit alpha	PAFAH1B1	2
Q9NT62	Ubiquitin-like-conjugating enzyme ATG3	ATG3	2
K7ERZ3	Perilipin-3	PLIN3	2
Q6DD88	Atlastin-3	ATL3	2
Q15084-3	Protein disulfide-isomerase A6	PDIA6	2
K7EQH4	ATP synthase subunit alpha, mitochondrial	ATP5A1	2
G3V1U5	Vesicle transport protein GOT1B	GOLT1B	2
O00186	Syntaxin-binding protein 3	STXBP3	2
Q5T6W2	Heterogeneous nuclear ribonucleoprotein K	HNRNPK	2
H0YFY4	ADP-sugar pyrophosphatase	NUDT5	2
P35580	Myosin-10	MYH10	2
F5H081	Solute carrier family 2, facilitated glucose transporter member 4	SLC2A4	2
Q9H1C7	Cysteine-rich and transmembrane domain-containing protein 1	CYSTM1	2
K7EJ83	Cyclin-dependent kinase 2; Cyclin-dependent kinase 3	CDK3; CDK2	2
C9J352	Importin subunit alpha-5; Importin subunit alpha-5, N-terminally processed	KPNA1	2
H9KV75	Alpha-actinin-1; Alpha-actinin-4; Alpha-actinin-2; Alpha-actinin-3	ACTN1; ACTN4; ACTN3; ACTN2	2
A0A0A0MQS1	Pyrroline-5-carboxylate reductase; Pyrroline-5-carboxylate reductase 3	PYCRL	2
D6RBY0	Rieske domain-containing protein	RFESD	2
C9JC71	Low affinity immunoglobulin gamma Fc region receptor III-A	FCGR3A	2
P51811	Membrane transport protein XK	XK	2
C9JIG2	DnaJ homolog subfamily B member 2	DNAJB2	2
Q9UN37	Vacuolar protein sorting-associated protein 4A; Vacuolar protein sorting-associated protein 4B; Fidgetin-like protein 1	VPS4A; VPS4B; FIGNL1	2
P61160	Actin-related protein 2	ACTR2	2
P16930	Fumarylacetoacetase	FAH	2
A0A087X0K4	CUB and sushi domain-containing protein 2	CSMD2	2
P10746	Uroporphyrinogen-III synthase	UROS	2
K7EQ02	DAZ-associated protein 1	DAZAP1	2
F5GYN4	Ubiquitin thioesterase OTUB1	OTUB1	2
P10599	Thioredoxin	TXN	2
O95197	Reticulon-3	RTN3	2
E7ETB3	Aspartyl aminopeptidase	DNPEP	2
P14868	Aspartate--tRNA ligase, cytoplasmic	DARS	2
Q08AM6	Protein VAC14 homolog	VAC14	2
Q93034	Cullin-5	CUL5	2
Q9BQA1	Methylosome protein 50	WDR77	2
A0A0B4J2G9	Ubiquitin-conjugating enzyme E2 L3	UBE2L3	2
Q04446	1,4-alpha-glucan-branching enzyme	GBE1	2
Q96NA2	Rab-interacting lysosomal protein	RILP	2
Q92539	Phosphatidate phosphatase LPIN2	LPIN2	2
P28482	Mitogen-activated protein kinase 1	MAPK1	2
Q7Z406	Myosin-14; Myosin-11	MYH14; 11	2
Q8IU68	Transmembrane channel-like protein 8	TMC8	2
Q96BJ3	Axin interactor, dorsalization-associated protein	AIDA	2
P20042	Eukaryotic translation initiation factor 2 subunit 2	EIF2S2	2
A0A0A0MQR0	Docosahexaenoic acid omega-hydroxylase CYP4F3; Phylloquinone omega-hydroxylase CYP4F2; Cytochrome P450 4F12; Phylloquinone omega-hydroxylase CYP4F11	CYP4F2; CYP4F3; CYP4F11; CYP4F12	2
Q9BS26	Endoplasmic reticulum resident protein 44	ERP44	2
O00178	GTP-binding protein 1	GTPBP1	2

TABLE 4-continued

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Protein ID	Protein names	Gene names	Razor + unique peptides
P27797	Calreticulin	CALR	2
G3V0E5	Transferrin receptor protein 1; Transferrin receptor protein 1, serum form	TFRC	2
P48147	Prolyl endopeptidase	PREP	2
H7BZC1	Hippocalcin-like protein 1; Neuron-specific calcium-binding protein hippocalcin; Neurocalcin-delta	HPCAL1; NCALD; HPCA	2
F8VPD4	CAD protein; Glutamine-dependent carbamoyl-phosphate synthase; Aspartate carbamoyltransferase; Dihydroorotase	CAD	2
Q9NTJ5	Phosphatidylinositol phosphatase SAC1	SACM1L	2
P26447	Protein S100-A4	S100A4	2
Q9UIW2	Plexin-A1	PLXNA1	2
H0YJS0	V-type proton ATPase subunit D	ATP6V1D	2
P25445	Tumor necrosis factor receptor superfamily member 6	FAS	2
C9JEU5	Fibrinogen gamma chain	FGG	2
F5H562	Copper-transporting ATPase 2; WND/140 kDa	ATP7B	2
B5MCF3	Protein GUCD1	GUCD1	2
Q96TA1	Niban-like protein 1	FAM129B	2
F5GWT4	Serine/threonine-protein kinase WNK1	WNK1	2
E9PLT1	Platelet glycoprotein 4	CD36	2
Q8TDB8	Solute carrier family 2, facilitated glucose transporter member 14; Solute carrier family 2, facilitated glucose transporter member 3	SLC2A14; SLC2A3	2
B4DDD6	Drebrin-like protein	DBNL	2
E9PII3	Band 4.1-like protein 2	EPB41L2	2
E9PNF5	Glutamine-dependent NAD(+) synthetase	NADSYN1	2
K7ELL7	Glucosidase 2 subunit beta	PRKCSH	2
Q9BV20	Methylthioribose-1 -phosphate isomerase	MR11	2
J3KNB4	Cathelicidin antimicrobial peptide; Antibacterial protein FALL-39; Antibacterial protein LL-37	CAMP	2
Q5T1Z0	Phospholysine phosphohistidine inorganic pyrophosphate phosphatase	LHPP	2
H3BLV0	Complement decay-accelerating factor	CD55	2
Q9Y3E7	Charged multivesicular body protein 3	CHMP3	2
Q5T6H7	Xaa-Pro aminopeptidase 1	XPNPEP1	2
O94779	Contactin-5	CNTN5	2
E9PNR2	Ras and Rab interactor 1	RIN1	2
E7ESJ7	Protein FAM114A2	FAM114A2	2
A0A0A0MS99	Multidrug resistance-associated protein 1	ABCC1	2
A0A0G2JM15	Large neutral amino acids transporter small subunit 4	SLC43A2	2
H0YBF7	Arf-GAP with SH3 domain, ANK repeat and PH domain-containing protein 1; Arf-GAP with SH3 domain, ANK repeat and PH domain-containing protein 2	ASAP1; ASAP2	2
P29992	Guanine nucleotide-binding protein subunit alpha-11	GNA11	2
F8WF69	Clathrin light chain A	CLTA	2
B5MCN0	Atlastin-2	ATL2	2
F8W9F9	Serine/threonine-protein kinase WNK2; Serine/threonine-protein kinase WNK3	WNK2; WNK3	2
A2A3F3	Transient receptor potential cation channel subfamily M member 3	TRPM3	2
Q5KU26	Collectin-12	COLEC12	2
O14523	C2 domain-containing protein 2-like	C2CD2L	2
Q96DG6	Carboxymethylglutaminyl hydrolase homolog	CMBL	2
Q5THJ4	Vacuolar protein sorting-associated protein 13D	VPS13D	2
Q02790	Peptidyl-prolyl cis-trans isomerase FKBP4; Peptidyl-prolyl cis-trans isomerase FKBP4, N-terminally processed	FKBP4	2
A0A0G2JQD2	Glutathione S-transferase theta-1	GSTT1	2
Q9NV96	Cell cycle control protein 50A; Cell cycle control protein 50B	TMEM30A; TMEM30B	2
P62330	ADP-ribosylation factor 6	ARF6	2
Q9UPU5	Ubiquitin carboxyl-terminal hydrolase 24	USP24	2
B2R4S9	Histone H2B	HIST1H2BI	2
P10599	Thioredoxin	TXN	2
R4GN98	Protein S100; Protein S100-A6	S100A6	2
H3BS66	Small integral membrane protein 1	SMIM1	2
Q8NHG7	Small VCP/p97-interacting protein	SVIP	2
C9JIG9	Serine/threonine-protein kinase OSR1; STE20/SPS1-related proline-alanine-rich protein kinase	OXSRI; STK39	2
Q5VZR0	Golgi-associated plant pathogenesis-related protein 1	GLIPR2	2
F8WD49	Anion exchange protein 3; Anion exchange protein; Anion exchange protein 2	SLC4A3; SLC4A2	2
P61626	Lysozyme C; Lysozyme	LYZ	2
Q5T123	SH3 domain-binding glutamic acid-rich-like protein 3	SH3BGR1	2
F5H571	Ubiquitin carboxyl-terminal hydrolase 5	USP5	2
Q9H3K6	Bola-like protein 2	BOLA2; BOLA2B	2
P05164	Myeloperoxidase; Myeloperoxidase; 89 kDa myeloperoxidase; 84 kDa myeloperoxidase; Myeloperoxidase light chain; Myeloperoxidase heavy chain	MPO	2

TABLE 4-continued

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Protein ID	Protein names	Gene names	Razor + unique peptides
P50395	Rab GDP dissociation inhibitor beta	GDI2	2
G3V2U7	Acylphosphatase; Acylphosphatase-1	ACYP1	2
J3KNB4	Cathelicidin antimicrobial peptide; Antibacterial protein FALL-39; Antibacterial protein LL-37	CAMP	2
H0Y9X3	Programmed cell death protein 6	PDCD6	2
A6NJA2	Ubiquitin carboxyl-terminal hydrolase; Ubiquitin carboxyl-terminal hydrolase 14	USP14	2
P35998	26S protease regulatory subunit 7	PSMC2	2
P30041	Peroxisome protein-6	PRDX6	2
P35612	Beta-adducin	ADD2	2
P00918	Carbonic anhydrase 2	CA2	2
A0A0C4DGH5	Cullin-associated NEDD8-dissociated protein 1	CAND1	2
C9JZN1	Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-2; Guanine nucleotide-binding protein subunit beta-4	GNB2; GNB4	1
I3L4X8	Integrin beta; Integrin beta-3	ITGB3	1
H7BZJ3		PDIA3	1
P17066	Heat shock 70 kDa protein 6; Putative heat shock 70 kDa protein 7	HSPA6; HSPA7	1
J3KT70	Arf-GAP with dual PH domain-containing protein 2	ADAP2; CENTA2	1
Q9Y6M5	Zinc transporter 1	SLC30A1	1
P54725	UV excision repair protein RAD23 homolog A	RAD23A	1
F8WD59	40S ribosomal protein SA	RPSA; RPSAP58	1
P06753			1
E9PMI6	Methylome subunit pICln	CLNS1A	1
Q9BVK6	Transmembrane emp24 domain-containing protein 9	TMED9	1
C9JIV6	Myeloid-associated differentiation marker	MYADM	1
Q8IXQ3	Uncharacterized protein C9orf40	C9orf40	1
Q8NDC0	MAPK-interacting and spindle-stabilizing protein-like	MAPK11P1L	1
Q99747	Gamma-soluble NSF attachment protein	NAPG	1
E7EWE1	Ubiquitin-like modifier-activating enzyme 5	UBA5	1
J3KNE3	Platelet-activating factor acetylhydrolase IB subunit beta	PAFAH1B2	1
Q8IUI8	Cytokine receptor-like factor 3	CRLF3	1
Q9BTX7	Alpha-tocopherol transfer protein-like	TTPAL	1
P63261	Actin, cytoplasmic 2; Actin, cytoplasmic 2, N-terminally processed	ACTG1	1
P31153	S-adenosylmethionine synthase isoform type-2	MAT2A	1
O00560	Syntenin-1	SDCBP	1
Q4VB86	Protein 4.1	EPB41	1
Q8WUD1	Ras-related protein Rab-2B	RAB2B; DKFZp313C1541	1
Q9NWX4	UPF0587 protein C1orf123	C1orf123	1
Q9Y2Z0	Suppressor of G2 allele of SKP1 homolog	SUGT1	1
H0YK48	Tropomyosin alpha-1 chain	TPM1	1
H3BS66	Small integral membrane protein 1	SMIM1	1
Q5VTS0	Neurensin-1	NRSN1	1
H0Y904	Multidrug resistance-associated protein 7	ABCC10	1
P08F94	Fibrocystin	PKHD1	1
G3V5X4	Nesprin-2	SYNE2	1
P48507	Glutamate--cysteine ligase regulatory subunit	GCLM	1
H3BUF4	Cyclin-D1-binding protein 1	CCNDBP1	1
F8WB30	Target of Myb protein 1	TOM1	1
E5RJ18		CA1	1
S4R3E5	Importin subunit alpha-7	KPNA6	1
E7EVS6		ACTB	1
P04206	Ig kappa chain V-III region GOL; Ig kappa chain V-III region WOL; Ig kappa chain V-III region 1		
Q5SSV3	Ti; Ig kappa chain V-III region SIE		
H0YG54	N(G),N(G)-dimethylarginine dimethylaminohydrolase 2	DDAH2	1
P08246	Oligoribonuclease, mitochondrial	REXO2	1
P40926	Neutrophil elastase	ELANE	1
P98172	Malate dehydrogenase, mitochondrial	MDH2	1
Q96DD7	Ephrin-B1	EFNB1	1
Q71RC9	Protein shisa-4	SHISA4	1
P68133	Small integral membrane protein 5	SMIM5	1
H0Y9Q6	Actin, alpha skeletal muscle; Actin, alpha cardiac muscle 1; Actin, gamma-enteric smooth muscle; Actin, aortic smooth muscle	ACTA1; ACTC1; ACTG2; ACTA2	1
G3V5P0	Clathrin light chain B	CLTB	1
Q9NWX6	Probable tRNA(His) guanylyltransferase	KTN1	1
Q15404	Ras suppressor protein 1	THG1L	1
B2R4S9	Histone H2B	RSU1	1
K7EK06	Phenylalanine--tRNA ligase alpha subunit	HIST1H2B	1
H7C3S9	COP9 signalosome complex subunit 8	FARSA	1
		COPS8	1

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Protein ID	Protein names	Gene names	Razor + unique peptides
J3KRV4	Dual specificity mitogen-activated protein kinase kinase 3	MAP2K3	1
Q15042	Rab3 GTPase-activating protein catalytic subunit	RAB3GAP1	1
P01893	Putative HLA class I histocompatibility antigen, alpha chain H; HLA class I histocompatibility antigen, Cw-6 alpha chain; HLA class I histocompatibility antigen, B-38 alpha chain; HLA class I histocompatibility antigen, B-67 alpha chain; HLA class I histocompatibility antigen, B-82 alpha chain; HLA class I histocompatibility antigen, B-39 alpha chain; HLA class I histocompatibility antigen, Cw-18 alpha chain; HLA class I histocompatibility antigen, Cw-7 alpha chain; HLA class I histocompatibility antigen, B-42 alpha chain; HLA class I histocompatibility antigen, B-14 alpha chain; HLA class I histocompatibility antigen, B-8 alpha chain; HLA class I histocompatibility antigen, B-7 alpha chain	HLA-H; HLA-C; HLA-B	1
Q9UNW1	Multiple inositol polyphosphate phosphatase 1	MINPP1	1
Q6B0K9	Hemoglobin subunit mu	HBM	1
O75915	PRA1 family protein 3	ARL6IP5	1
O95376	E3 ubiquitin-protein ligase ARIH2	ARIH2	1
E9PBW4	Hemoglobin subunit gamma-2	HBG2	1
C9J1X0	WD repeat-containing protein 91	WDR91	1
Q8NCV1			1
Q9Y4P8	WD repeat domain phosphoinositide-interacting protein 2	WIPI2	1
A0A087WUX6	Proteasomal ubiquitin receptor ADRM1	ADRM1	1
Q9NQS7	Inner centromere protein	INCENP	1
P62805	Histone H4	HIST1H4A	1
Q8TDY2	RB1-inducible coiled-coil protein 1	RB1CC1	1
C9JFM5	Syntaxin-4	STX4	1
B0QZ43	Erlin-2; Erlin-1	ERLIN1; ERLIN2	1
C9J8T0	Selenocysteine-specific elongation factor	EEFSEC	1
P61224	Ras-related protein Rap-1b; Ras-related protein Rap-1b-like protein	RAP1B	1
Q99828	Calcium and integrin-binding protein 1	CIB1	1
Q53GQ0	Very-long-chain 3-oxoacyl-CoA reductase	HSD17B12	1
H0Y9Q9	ADP-ribosyl cyclase/cyclic ADP-ribose hydrolase 2	BST1	1
R4GN98	Protein S100; Protein S100-A6	S100A6	1
Q8TB73	Protein NDNF	NDNF	1
Q6P1A2	Lysophospholipid acyltransferase 5	LPCAT3	1
F2Z2Y4	Pyridoxal kinase	PDXK	1
Q9HA65	TBC1 domain family member 17	TBC1D17	1
G3V126	V-type proton ATPase subunit H	ATP6V1H	1
D6RGE2	Isochorismatase domain-containing protein 1	ISOC1	1
A0A0B4J222	ADP-ribosylation factor-like protein 15	ARL15	1
F8W7W4	Androglobin	ADGB	1
Q9GZR7			1
A0A087WVC4	cAMP-dependent protein kinase catalytic subunit beta	PRKACB	1
A8MU39	Serine/threonine-protein phosphatase; Serine/threonine-protein phosphatase 5	PPP5C	1
G3V1N2		HBA2	1
P69892	Hemoglobin subunit gamma-2	HBG2	1
F8WEZ0	1-phosphatidylinositol 3-phosphate 5-kinase	PIKFYVE	1
A0A087WZZ4	Ammonium transporter Rh type A	RHAG	1
Q1JUQ3	Peptidyl-prolyl cis-trans isomerase; Peptidyl-prolyl cis-trans isomerase FKBP1A	FKBP12-Exin; FKBP1A	1
F8VWZ5	H2.0-like homeobox protein	HLX	1
P62877	E3 ubiquitin-protein ligase RBX1; E3 ubiquitin-protein ligase RBX1, N-terminally processed	RBX1	1
Q7Z5P9	Mucin-19	MUC19	1
F8VPB3		TPK1	1
F8WBF4	Transmembrane protein 50B	TMEM50B	1
C9JL85	Myotrophin	MTPN	1
Q53EQ6	Tigger transposable element-derived protein 5	TIGD5	1
E3W974		DNPEP	1
H7BYV1	Interferon-induced transmembrane protein 1; Interferon-induced transmembrane protein 2; Interferon-induced transmembrane protein 3	IFITM2; IFITM3; IFITM1	1
D6RC06	Histidine triad nucleotide-binding protein 1	HINT1	1
E9PNW4	CD59 glycoprotein	CD59	1
H7C2Z6	Grancalcin	GCA	1
Q9H1C7	Cysteine-rich and transmembrane domain-containing protein 1	CYSTM1	1
H0YK07	ATP-dependent Clp protease ATP-binding subunit clpX-like, mitochondrial	CLPX	1
E9PG15	14-3-3 protein theta	YWHAQ	1
K7ENK9	Vesicle-associated membrane protein 3; Vesicle-associated membrane protein 2	VAMP2; VAMP3	1
F8WDD6	Cation channel sperm-associated protein subunit gamma	CATSPERG	1
P35754	Glutaredoxin-1	GLRX	1
Q5THJ4	Vacuolar protein sorting-associated protein 13D	VPS13D	1
B4E3H6	Transforming acidic coiled-coil-containing protein 1	TACC1	1
B0YJC4	Vimentin	VIM	1
H7C5R6	AT-rich interactive domain-containing protein 4B	ARID4B	1

TABLE 4-continued

Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome <i>Homo sapiens</i> database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search.			
Protein ID	Protein names	Gene names	Razor + unique peptides
P49755	Transmembrane emp24 domain-containing protein 10	TMED10	1
E9PMJ3	Ribonuclease inhibitor	RNH1	1
B1AHA9	DNA replication licensing factor MCM5	MCM5	1
P23634	Plasma membrane calcium-transporting ATPase 4	ATP2B4	1
Q14152	Eukaryotic translation initiation factor 3 subunit A	EIF3A	1
C9IY70	60S ribosomal export protein NMD3	NMD3	1
A0A087WUS7	Ig delta chain C region	IGHD	1
P07384	Calpain-1 catalytic subunit	CAPN1	1
G3V2C9	Guanine nucleotide-binding protein subunit gamma; Guanine nucleotide-binding protein G(I)/G(S)/G(O) subunit gamma-2	GNG2	1
Q9Y2Y8	Proteoglycan 3	PRG3	1
V9GY70		DNAJB12	1
U5GXS0	MAM and LDL-receptor class A domain-containing protein 1	MALRD1	1
Q92625	Ankyrin repeat and SAM domain-containing protein 1A	ANKS1A	1

The present description refers to a number of documents, the contents of which are herein incorporated by reference in their entirety.

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note = MISC_FEATURE - AIDA
source 1..306
mol_type = protein
organism = Homo sapiens
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KTIGKIATCL ELRSAALQST QSQEEFKLED LKKLEPILKN ILTYNKEPFP DVQVPPLRRI 120
LAPGEEENLE FEEDEEEGGA GAGSPDSFPA RVPGTLLPRL PSEPGMTLLT IRIEKIGLKD 180
AGQCIDPYIT VSVKDLNGID LTPVQDTPVA SRKEDTYVHF NVDIELQKHV EKLTKGAAIF 240
FEFKHYKPKK RFTSTKCFAP MEMDEIKPGP IVIELYKKPT DPKRKKLQLL TKKPLYLHLH 300
QTLHKE 306
SEQ ID NO: 2 moltype = AA length = 210
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source 1..210
mol_type = protein
organism = Homo sapiens
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GYRAVAIDL PGLGHSKEAAA PAPIGELAPG SFLAAVVDAL ELGPPVVIS PSLSGMYSLPF 120
LTAPGSQLPG FVPVAPICD KINAANYASV KTPALIVYGD QDPMGQTSFE HLKQLPNHRV 180
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IEWKYHSPEE EISLGPACWL WDFLRRSQQA GFLPLSGGV DSAATACLIY SMCCQVCEAV 120
RSGNEEVLD VRTIVNQISY TPQDPRDLCC RILTTCYMAS KNSSQETCTR ARELAQQIGS 180
HHISLNIDPA VKAVMGIFSL VTGKSPLFAA HGGSSRENLA LQNVQARIM VLAYLFAQLS 240
LWSRGVHGGL LVLGSANVDE SLLGYLTKYD CSSADINPIG GISKDRLRAF VQFCIQRFQL 300
PALQSILLAP ATAELPLAD GQVSQTDEED MGMTYAELSV YGKLRKVAKM GPYSMFCKLL 360
GMWRHICTPR QVADKVKRFF SKYSMNRHKM TTLTPAYHAE NYSPEDNRFD LRPFLYNTSW 420
PWQFRCIENQ VLQLERAEPQ SLDGVD 446

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GGNAKSKSLF KNCCLMWKQS IWTSTISSHL ATKHLKEGGL LTLGAKAAL DGTGPMIGYG 120
MAKGAVHQLC QSLAGKNSGM PPGAAAIAVL PVTLDTPMNR KSMPEADFSS WTPLEFLVET 180
FHDWITGKNR PSSGSLIQVV TTEGRTELTP AYP 213

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KEDVGPGKAV PREELFVTSK LWNTKHHPED VEPALRKTLA DLQLEYLDLY LMHWPYAFER 120
GDNPPFPKNAD GTICYDSTHY KETWKALEAL VAKGLVQALG LSNFNSRQID DILSVASVRP 180
AVLQVECHPY LAQNELIAHC QARGLEVTAI SPLGSSDRAW RDPDEPVLE EPVVLALAEK 240
YGRSPAQILL RWQQRKVIC IPKSITPSRI LQNIKVFDFI FSPPEMKQLN ALNKNWRYIV 300
PMLTVDGKRV PRDAGHPLYP FNDPY 325

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NISIGGLVLP LELKSKPEDG DRVVYTGTYD TEGVTPTKSG ERQPIQITMP VRLACGYRSM 120
VPWKMETTEK KTTITSL 137

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                 note = MISC_FEATURE - USP24
source           1..2620
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IEFPTTNLYE LESRVLDTHW SIPYKREESL GKCLLASTYL ARLGLSEDE NCRRFMDRCM 180
PEAFKLLTS SAVHKWGTET HEGIYNMLML LIELVAERIK QDPIPTGLLG VLTMAFNPDN 240
EYHFKNRMKV SQRNWAEVFG EGNMFAVSPV STFQKEPHGW VVDLVNKFGE LGGFAAIQAK 300
LHSEDIELGA VSALIQLGV CAEYLNSSVV QPMLDPVILT TIQDVRSVEE KDLKDKRLVS 360
IPELLSAVKL LCMRFQPDIV TIVDDLRLDI LRLMLKSPHF SAKMNSLKEV TKLIEDSTLS 420
KSVKNAIDTD RLDDWLVSNS VLSIALEGNI DQAQYCDRIK GIIELLGSKL SLDELTKIWK 480
IQSGQSSTVI ENIHITIIAAA AVKFNSDQLN HLFVLIQKSW ETESDRVRQK LLSLIGRIGR 540
EARFETTSKG VLDVLWELAH LPTLPSSLIQ QALEEHLTIL SDAVAVKEAI KRSYIICKIE 600
DTRKPGESG LEKNKKGDFK SSQNLNPPQFV WVVPALRQLH EITRSFIQT YQKQDKSIIQ 660
DLKKNFEIVK LVTGSLIACH RLAAAVAGPG GLSGSTLVDG RYTYREYLEA HLKFLAFFLQ 720
EATLYLGWNR AKEIWECLVT GQDVCELDRE MCFEWFTEGQ HDLESVDVQQQ LFKEKILKLE 780
SYEITMNGFN LFKTFEFENVN LCDHRLKRQG AQLYVEKLEL IGMDFIWKIA MESPDEEIAN 840
EAIQLIINYS YINLNPRLLK DSVSLHKKFI ADCYTRLEAA SSALGGPTLT HAVTRATKML 900
TATAMPTVAT SVQSPYRSTK LVIIERLLLL AERYVITIED FYSVPRTILP HGASFHGLL 960
TLNVTYESTK DTFTVEAHNS ETIGSVRWKI AKQLCSPVDN IQIFTNDSLL TVNKDQKLLH 1020

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QLGFSDEQIL	TVKTSGSGTP	SGSSADSSTS	SSSSSSGVFS	SSYAMEQES	LPGVVMALVC	1080
NVFDMLYQLA	NLEEPRLTLR	VRKLLLLIPT	DPAIQEALDQ	LDLGRKKTLL	LSESSSQSSK	1140
SPSLSSKQOH	QPSASSILES	LFRSFAPGMS	TFRVLYNLEV	LSSKLMPTAD	DDMARSCAKS	1200
FCENFLKAGG	LSLVVNVQR	DSIPSEVDYE	TRQGVYSICL	QLARFLLVGQ	TMPTLLDEDL	1260
TKDGIEALSS	RPFNRVSRQT	SRQMSLCGTP	EKSSYRQLSV	SDRSSIRVEE	IIPAARVAIQ	1320
TMEVSDFTST	VACFMRLSWA	AAAGRLDLVG	SSQPIKESNS	LCPAGIRNRL	SSSGSNCSG	1380
SEGEFVALHA	GICVRQQSVS	TKDSLIAGEA	LSLLVTCLQL	RSQQLASFYN	LPCVADFIID	1440
ILLGSPSAEI	RRVACDQLYT	LSQTDTSAPH	DVQKPNQPLL	GVILTAQLPL	WSPTSIMRGV	1500
NQRLLSQCME	YFDLRCQLLD	DLTTSEMEQL	RISPATMLED	EITWLDNFEP	NRTAECETSE	1560
ADNILLAGHL	RLIKTLLSLC	GAEKMLGSS	LIKPLDDDFL	FRASRIILNS	HSPAGSAAIS	1620
QQDFHPKCST	ANSRLAAEYV	LVMLADSSPS	NLQIIKELL	SMHHQDPDAL	TKEFDYLPVP	1680
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TFQGIYSQDK	ICKDCPHRYE	REEAFMALNL	GVTSCQSLEI	SLDQFVRGEV	LEGSNAYYCE	1860
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CGKGKWKYFN	DTVIEFDLNL	DETLEYECFG	GEYRPKVYDQ	TNPYTDVRRR	YWNAYMLFYQ	2040
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HNTVALLVLH	SDVSSQRNVA	PGIFKQRPP	SIAPSSPLLP	LHEEVEALLF	MSEGKPYLLE	2400
VMFALRELTV	SLLLALIMV	YCCFCNEHFS	FTMLHFIKNQ	LETAPPHELK	NTFQLLHEIL	2460
VIEDPIQVER	VKFPVETENG	LLALMHHSNH	VDSSRCYQCV	KFLVTLAQKC	PAAKEYFKEN	2520
SHHWSWAVQW	LQKKMSEHYW	TPQSNVSNET	STGKTFQRTI	SAQDTLAYAT	ALLNEKEQSG	2580
SSNGSESSPA	NENGDRHLQQ	GSESPMMIGE	LRSDDLDDVP			2620

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 GADTSVDLEE TGRVLSIGDG IARVHGLRNV QAEEMVEFSS GLKGMSLNLE PDNVGVVVF 120
 NDKLIKEGDI VKRTGAIVDV PVGEELLGRV VDALGNAIDG KGPIGSKTRR RVGLKAPGII 180
 PRISVREPMQ TGIIKAVDSL PIGRGQRELI IGDRQTGNTG W 221

What is claimed is:

1. An in vitro method for preparing a clinical human blood sample, the method comprising

(a) obtaining a preparation of isolated erythrocyte-derived extracellular vesicles (EEV) from a blood sample of a subject having or suspected of having Parkinson's disease;

(b) removing hemoglobin from the preparation of isolated EEV, thereby obtaining a processed blood sample depleted of hemoglobin; and

(c) quantifying the expression levels of one or more proteins in the processed blood sample of (b).

2. The method of claim 1, wherein the quantification in (c) is performed on EEVs having a diameter of between 100 nm and 1000 nm.

3. The method of claim 1, wherein the EEV are CD235a+ extracellular vesicles.

4. The method of claim 1, wherein the EEV are TSG101+, Rab5+, CD9+, CD63+, CD81+, or any combination thereof.

5. The method of claim 1, wherein said blood sample is platelet-free plasma.

6. The method of claim 1, further comprising, prior to (a), inducing calcium-dependent production of EEV from activated erythrocytes in the blood sample of the subject using a calcium ionophore, cold-storage, or ATP-depletion.

7. The method of claim 1, wherein the preparation of isolated EEV is obtained by separating the EEV by flow cytometry, differential centrifugation, nanomembrane ultra-filtration, immunoabsorbent capture, size-exclusion chromatography, ultracentrifugation, magnetic activated cell sorting (MACS), nanoparticle tracking analysis, light scattering, electrophoretic light scattering, dynamic light scattering, electron microscopy, or any combination thereof.

8. The method of claim 1, wherein the quantification in (c) comprises mass spectrometry.

9. The method of claim 1, wherein the quantification in (c) comprises nano liquid chromatography tandem mass spectrometry (nanoLC MS/MS).

10. An in vitro method for preparing a clinical human blood sample, the method comprising

(a) obtaining a preparation of isolated erythrocyte-derived extracellular vesicles (EEV) depleted of hemoglobin from a blood sample of a subject having or suspected of having Parkinson's disease; and

(b) quantifying the expression levels of one or more proteins biomarkers, in the preparation of isolated EEV depleted of hemoglobin.

* * * * *